SMRTTECH 4SC3

Smart Cities and Communities Lab (Rail/Roadway)

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Lab 4:
Digital Twin in Smart Cities II

Fall 2026

McMaster University
Engineering
W Booth School of Engineering Practice and Technology



Agenda

- ❑ Explore a Pre-Built Unity Project
- ❑ Build The Unity Project
- ❑ Milestone 1
- ❑ Deliverable1
- ❑ Milestone 2
- ❑ Deliverable2
- ❑ Deliverable3



Game Engine Summary (1/4)

1 Game Engine vs 3D Modeling Software

3D Modeling Software (Blender/Maya)	Game Engine (Unity/Unreal)
Create asset (geometry)	Import & use in a scene



Model Output

FBX / OBJ

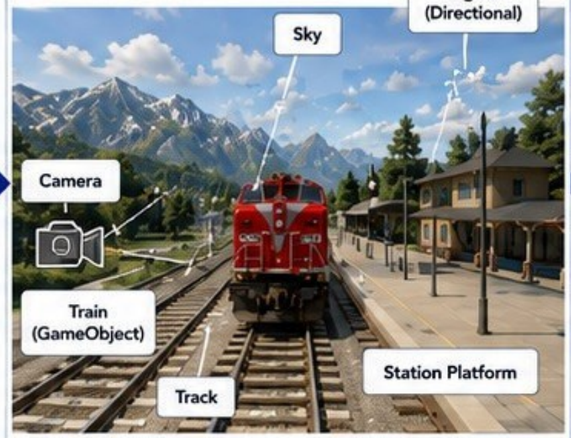
Import Asset

- [x] Mesh
- [x] Materials
- [x] Textures
- [x] Prefabs

3D tools create assets; the game engine brings them to life.

2 The Scene Concept

A scene is a self-contained railway world.



A scene contains GameObjects (objects), lights, cameras, and environments.

3 The GameObject Concept

The locomotive is a GameObject.



GameObject: Locomotive

A GameObject is an entity in the scene that can hold components.

4 Component-Based Design

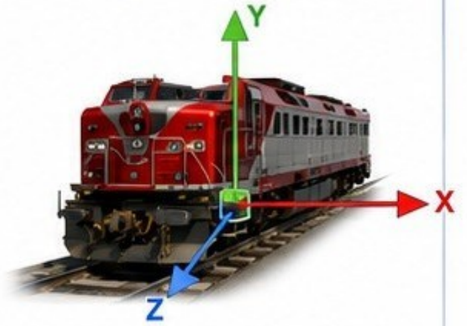
GameObjects are made of components.

- Transform
- Mesh Renderer
- Material
- Collider
- Rigidbody
- Script (C#)
- Light (optional)
- Audio (optional)

Components define how the object looks, behaves, and interacts.

5 Transform in a Game Engine

Position, rotation, scale (local or world).



Translation (Position)		Rotation (Euler °)		Scale	
X	10.0	X	0°	X	1.0
Y	0.0	Y	45°	Y	1.0
Z	25.0	Z	0°	Z	1.0

Transform controls where and how an object is placed and oriented.

6 The Camera Concept

A camera defines what the player sees.



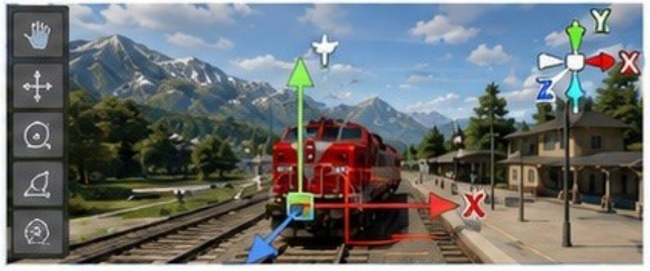
Camera Properties

- Field of View (FOV)
- Aspect Ratio
- Near / Far Clip
- Position & Rotation

The camera acts like the player's eyes, capturing a view of the scene.

7 Scene View vs Game View

Scene View (Editor)
Edit the scene with gizmos.



Game View (Camera Output)
What the player sees.



Scene View is for editing and placement; Game View is the final camera output.

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From asset creation to in-game view: the foundation of every railway game.

Game Engine Summary (2/4)

1 Rendering Pipeline Overview

Transforms scene data into the final image displayed on screen.

Scene (Models, Lights, Cameras) → Input Materials & Textures → Lighting (Per Pixel) → Post-Processing (Effects)

RENDERING PIPELINE

Culling → Lighting → Shading → Post-Processing

Final Image

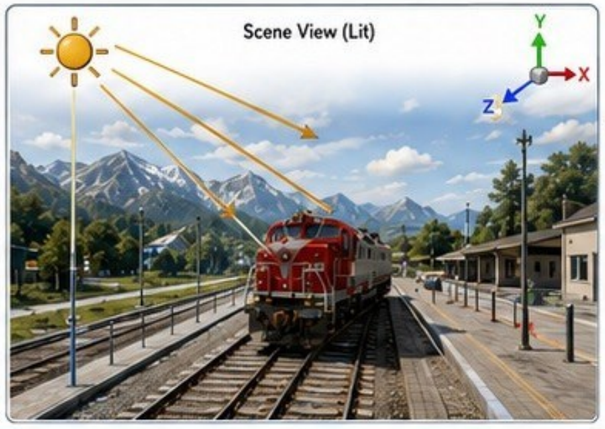


2 Lighting in Game Engines

Lights illuminate the scene.

Light Types

- Directional (Sun)
- Point
- Spot
- Area



3 Directional Light as the Sun

Simulates sunlight coming from infinity.

Direction	
X	-0.45
Y	1.00
Z	-0.25

Directional lights affect all objects equally and cast shadows.



4 Skybox

A sky surrounds the scene.



Six-sided texture (Cubemap) creates an immersive background.

Sky Faces

- Top
- Bottom
- Front
- Back
- Left
- Right

5 Shadows

Real-time shadows add realism.

Shadow Comparison

Shadows Off → Shadows On



Shadows are calculated from lights (Directional, Spot, Point, Area). They ground objects and add depth.

6 Materials and Shaders

Materials = parameters fed to the shader.

Material Properties

Base Color	
Metallic	0.0 – 1.0
Smoothness	0.0 – 1.0
Normal Scale	1.0
Emission	
Tile (Tiling)	(X, Y)

Shader (Standard)

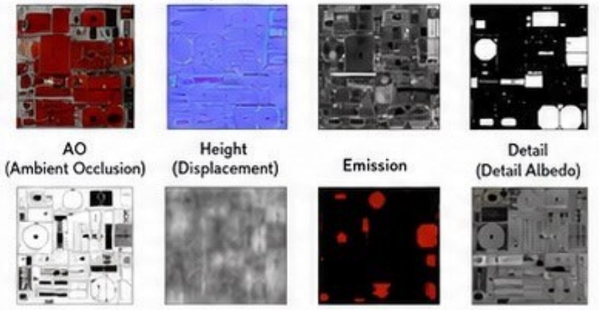
Surface Response

Diffuse · Specular · Normal · Emission

Shaders define how materials react to lights and how surfaces are rendered.

7 Texture Maps

Textures provide surface detail.



Combined, these maps create rich, realistic surfaces with high visual fidelity.

Key Takeaway: The rendering pipeline uses lights, environment (skybox), shadows, materials, shaders, and textures to produce realistic frames in real time.



Game Engine Summary (3/4)

1 Animation Basics

Animate an object properties over time.



Position, rotation, scale, material, door state, etc.

2 Keyframe Animation

Define important poses (keyframes) and interpolate between them.

Door Open Animation (Example)



Keyframes

Wheel Rotation (Loop)



Interpolation
- Linear, Slerp, Bezier

Looping
- Loop, Ping-Pong

3 Procedural Animation

Rules or code drive animation based on state or time.

- Wheel rotation from speed
- Signal lights cycle (e.g., red/yellow/green)
- Camera shake on bumps
- Train doors open/close based on station events
- Coupler compression based on forces

Examples

```
wheelAngle += speed * dt;
signalState = (t / cycle) % 3;
if (bump) camera.Shake();
```

Sources
- Sensors (speed, force)
- Game state (station, AI)
- Time / Random / Noise

4 The Physics Engine

Simulates forces and resolves collisions.



Calculates response to forces, applies constraints, and updates motion each frame.

5 Rigid Body

Physical properties of an object.

- **Mass:** 80,000 kg
- **Velocity:** 20 m/s
- **Gravity:** On
- **Is Kinematic:** Off

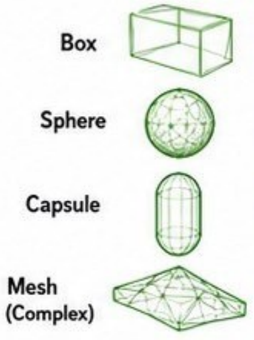


A rigid body moves and rotates in response to forces and collisions.

6 Collider

Defines the shape used for collision detection.





Colliders don't affect visuals—only collision detection. Choose simple shapes for performance.

7 Mesh Collider vs Simple Collider

Accuracy vs performance.

Mesh Collider (Accurate)



High detail, matches geometry.
More accurate collisions.
Higher cost.

VS

Simple Collider (Efficient)



Low detail, fewer primitives.
Faster and more efficient.
Lower cost.

1 — 2 — 3 — 4 — 5 — 6 — 7

From animation to simulation: building believable and interactive railway worlds.

Game Engine Summary (4/4)

1 Scripting and Behaviour

Code controls trains, signals, cameras, and gameplay.

```
// Accelerate to target speed
if (speed < targetSpeed)
    throttle += accel * dt;
else
    throttle = idle;

if (signal == RED)
    Brake();

if (distanceToStation < 50)
    Announce("Next station");
```

Signals Cameras Announcements

Behaviours drive the rules and reactions in the world.

2 Input and Interaction

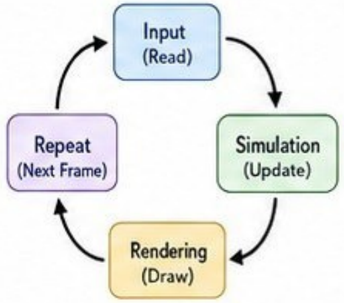
User input drives control and interaction.



Players and systems send commands to the game.

3 The Game Loop

The engine runs a continuous loop.



Read input → Update simulation → Render → Repeat.

4 Time in Simulation

Maintain consistent motion.



Fixed Time Step	1	2	3	4	5	...
Δt (ms)	16.2	17.0	15.4	18.1	16.6	...

✓ Smooth appearance on frame rate changes
✓ Stable physics & deterministic results

Use a fixed time step for physics and consistent simulation.

5 Why Optimization Matters

More performance = better visuals, more features, and smoother gameplay.



Optimization balances quality and performance.

6 FPS and Frame Time

Higher FPS = smoother.

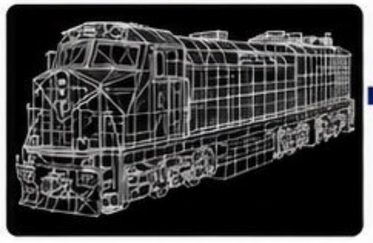


16.7 ms per frame @ 60 FPS

FPS = 1 / Frame Time
Target: 16.7 ms for 60 FPS

7 Triangles and Vertices

Mesh complexity impacts performance.

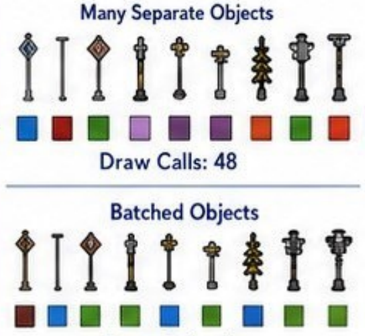


Vertices: 48,172
Triangles: 96,404

Fewer triangles = faster rendering.

8 Draw Calls and Batches

Batching reduces draw calls.



Fewer draw calls = better performance.

9 Texture Resolution

Higher resolution = more detail, higher memory cost.



Low 256×256 Medium 1024×1024 High 2048×2048

Use the right resolution for the need.

10 Real-Time Shadows (Cost)

Shadows add realism but are expensive.



Cost increases with number of lights and shadow resolution.

11 Level of Detail (LOD)

Use simpler models when objects are far away.

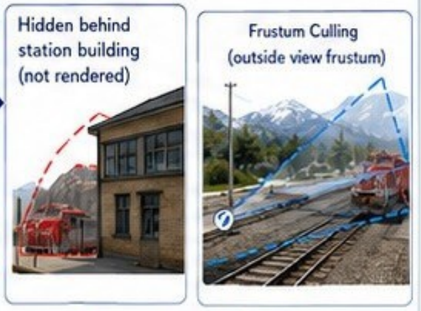


Near (LOD0) High detail ~ 50k tris
Mid (LOD1) Medium detail ~ 10k tris
Far (LOD2) Low detail ~ 1k tris

Reduces triangles and draw calls at distance.

12 Occlusion and Culling

Don't render what the camera can't see.



Render only visible objects.
Outdoor frustum (far) culled.

Logic (1–4) drives the simulation. Optimization (5–12) ensures a smooth, high-quality experience.

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Virtual Reality Summary

1 What Is Virtual Reality?



- Computer-generated immersive environment
- User perceives being inside the scene
- Supports viewing and interaction in 3D space

2 Immersion vs Presence

Immersion

Immersion = system capability

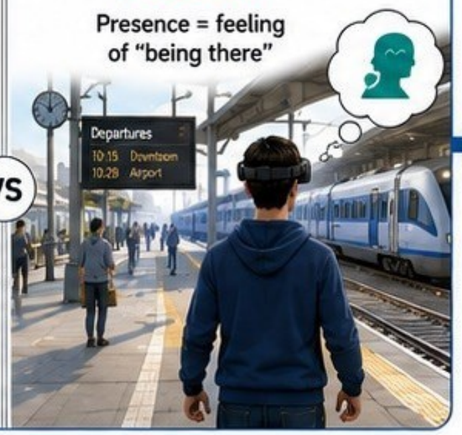
- Headset / Stereo display
- Spatial audio
- Tracking
- Wide field of view



VS

Presence

Presence = feeling of "being there"



3 3DoF vs 6DoF

3DoF: rotation only



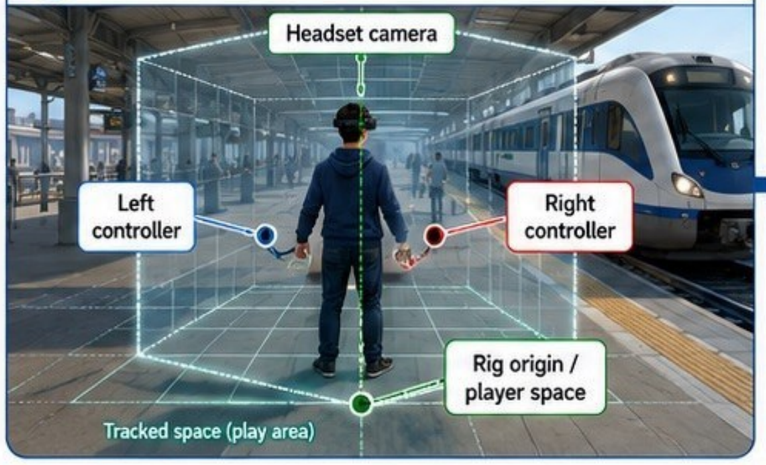
Look around only (X, Y, Z rotation)

6DoF: rotation + position



Look around and move (rotate + move in 3D space)

4 The VR Camera Rig



5 360° Video vs Real-Time VR

360° Video



- Recorded panorama
- Mostly 3DoF viewing
- Not fully interactive

Real-Time VR



- Live 3D scene
- 6DoF movement
- Interactive

6 Optimization for VR

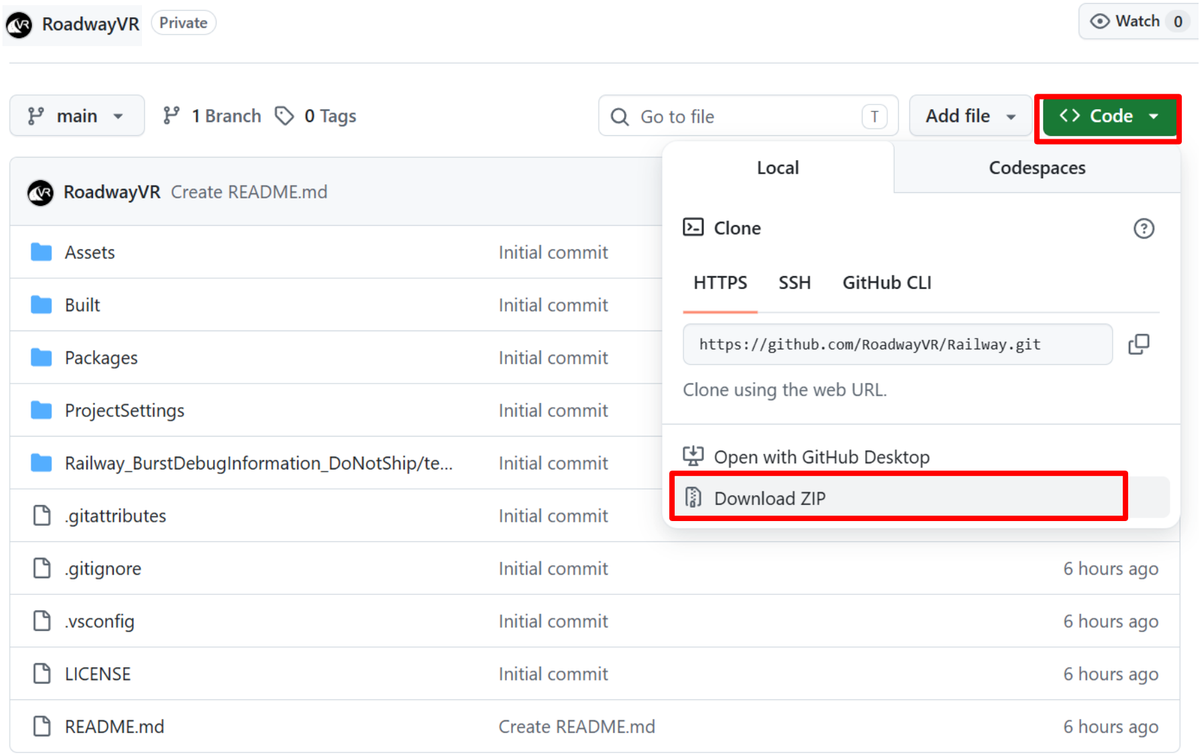
- High FPS needed (90 FPS+)
- Low latency (<20 ms)
- Efficient meshes & textures
- Reduce draw calls and shadow cost
- Comfort and responsiveness

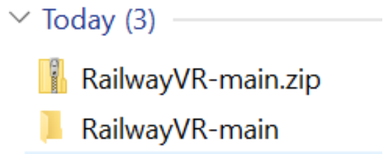


In short: VR combines immersive display, tracking, and real-time rendering to let users experience and interact with a virtual railway environment.

Download and Run Project

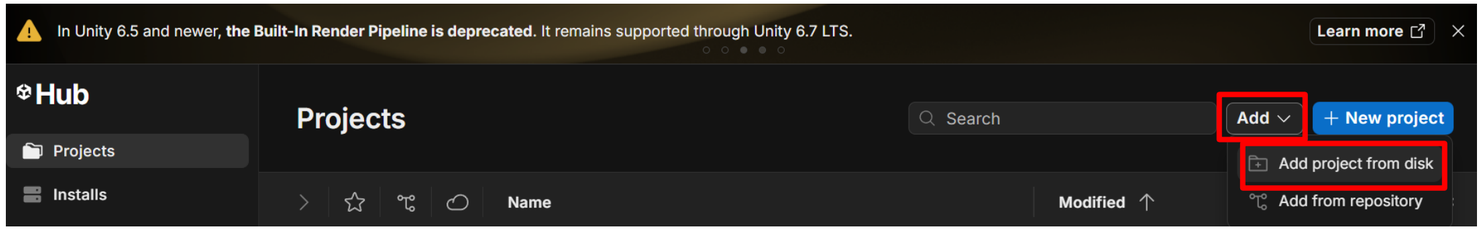
- ❑ Download Unity source file “RailwayVR” from Github or Avenue
- ❑ https://github.com/RoadwayVR/RailwayVR
- ❑ Extract the zip file

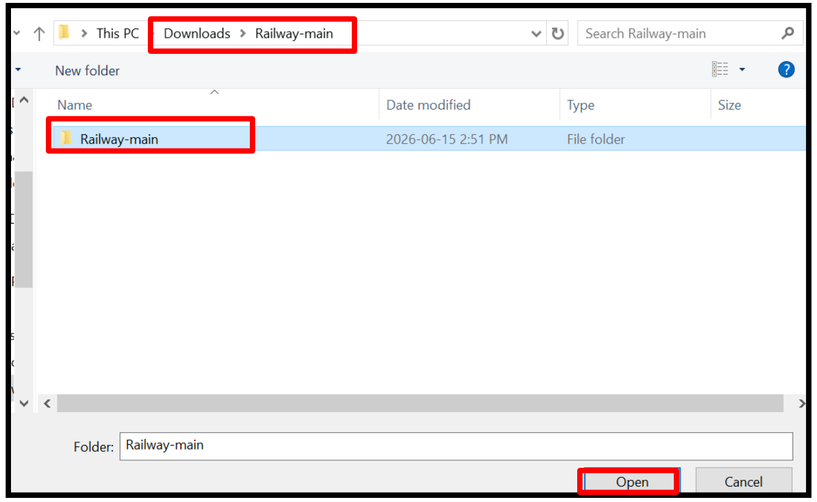


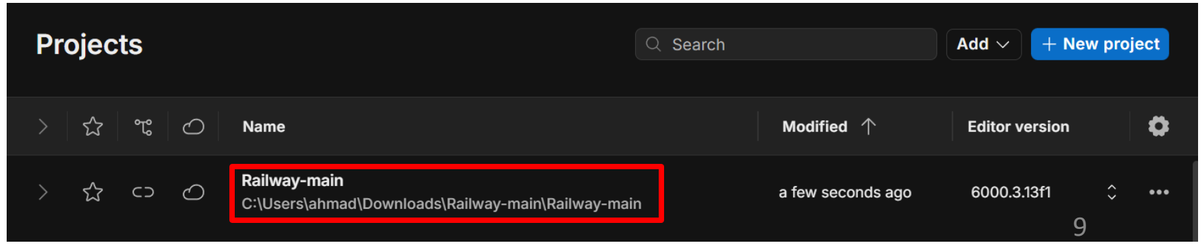


Download and Run Project

- ❑ Unity Hub → Create an Account & Sign in
- ❑ Add → Add project from disk
- ❑ Open Folder RailwayVR-main and Select folder “RailwayVR-main” → Open
- ❑ Unity Hub → Click on the red box to open the project

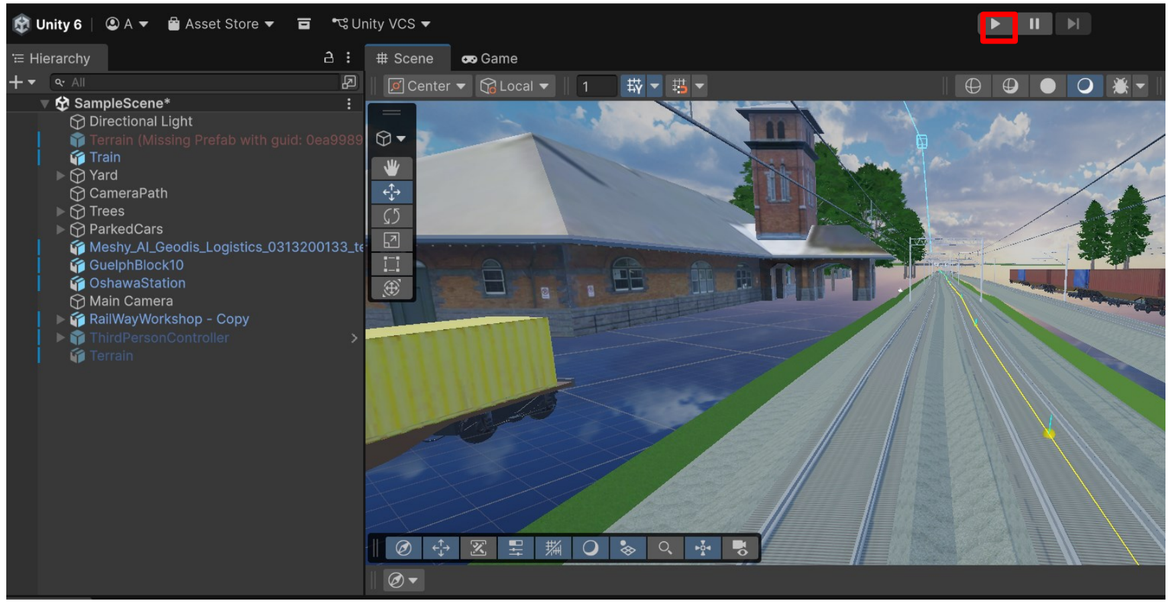


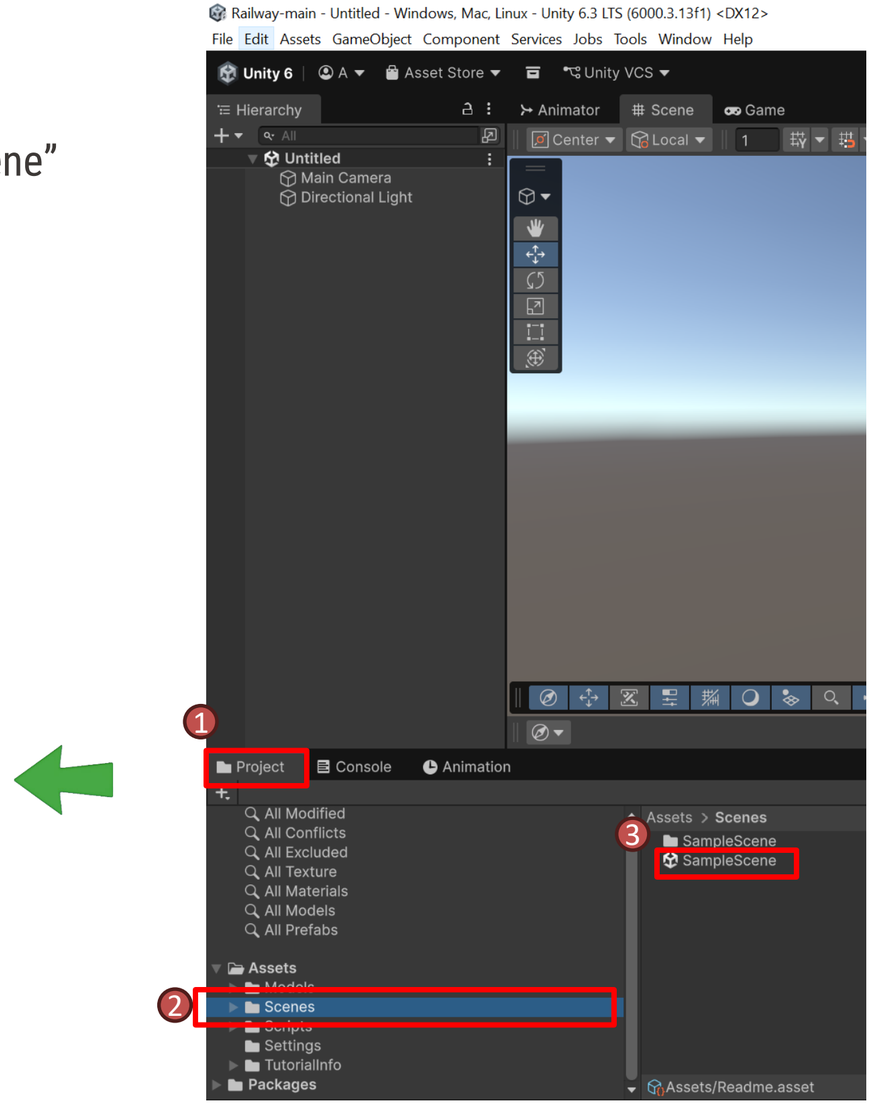




Download and Run Project

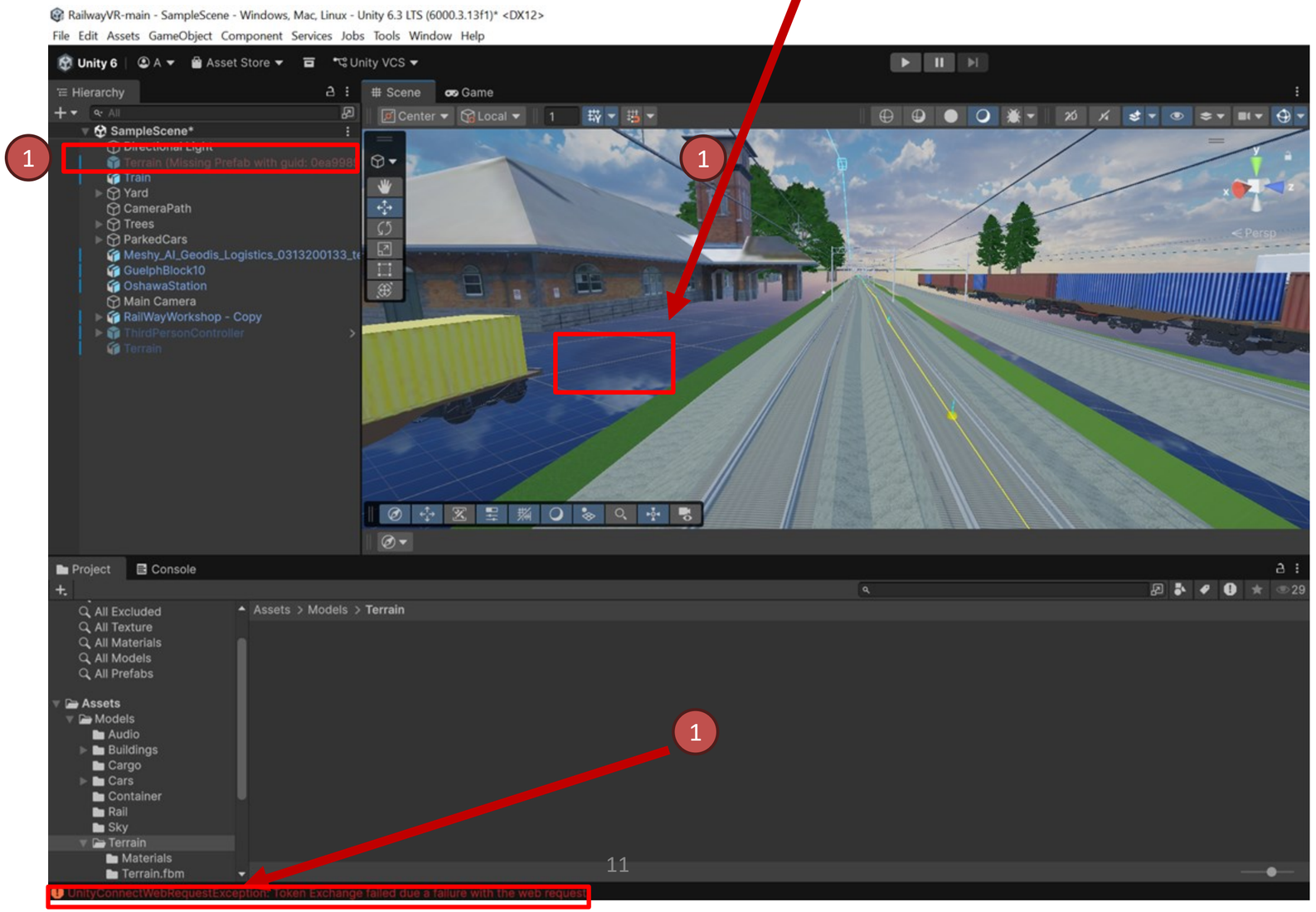
- ❑ Open Unity project → Select folder “Scenes” → Double click on “SampleScene”
- ❑ Select “Play Button”





What Problem Do You Notice in This Unity Scene?

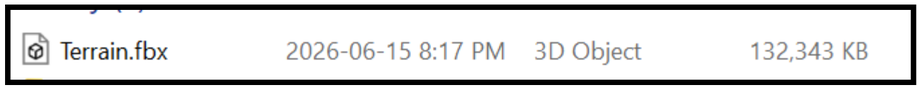
Terrain is Missing



Fixing Problem: Importing the Terrain FBX Object

❑ Download the Terrain FBX file from below link or Avenue

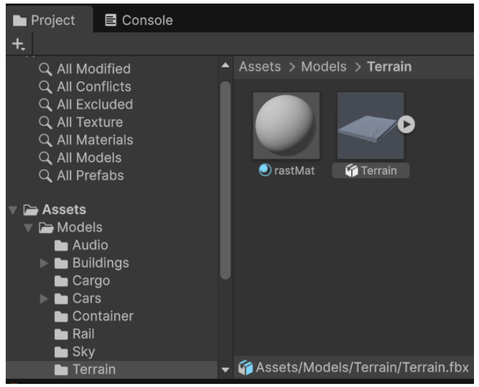
https://drive.google.com/file/d/19zJZRr2IyJx1lnjTV8NYgTTyMgy-3hUL/view?usp=drive_link

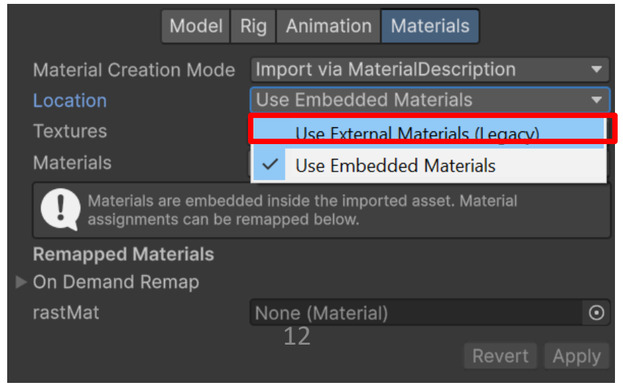


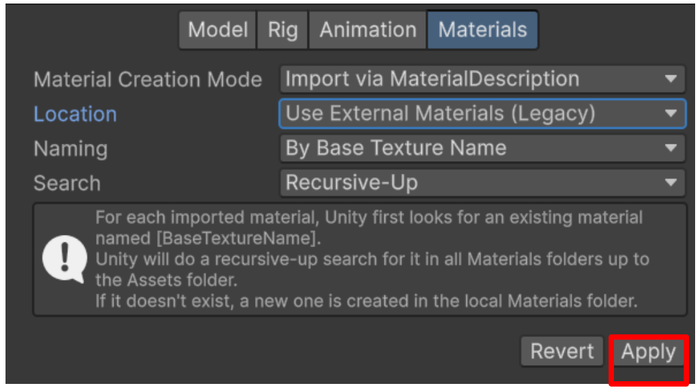
❑ What is FBX? FBX stands for Filmbox. It is a common 3D file format used to transfer 3D models

❑ Move the FBX file into the Unity project folder: RailwayVR-main\Assets\Models\Terrain\

❑ Configure the material import settings: Select the Terrain FBX → go to Materials tab (Right Side Top)→ Change the Location to “Use External Materials (Legacy) → click Apply.





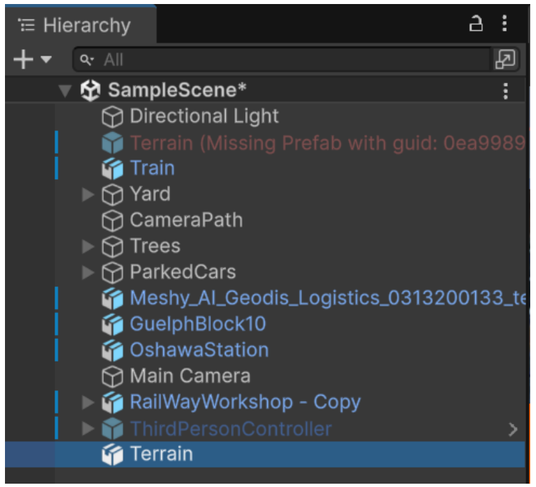


Fixing Problem 1: Adding the Terrain 3D Object to the Scene

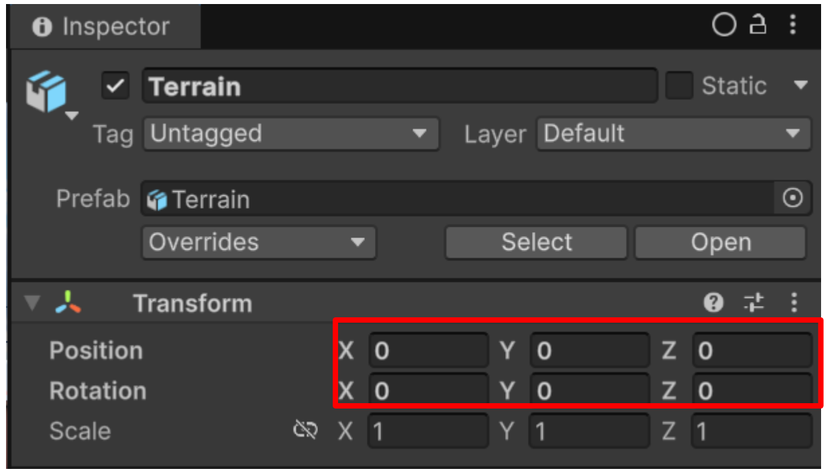
Drag and drop the Terrain prefab into the Hierarchy window



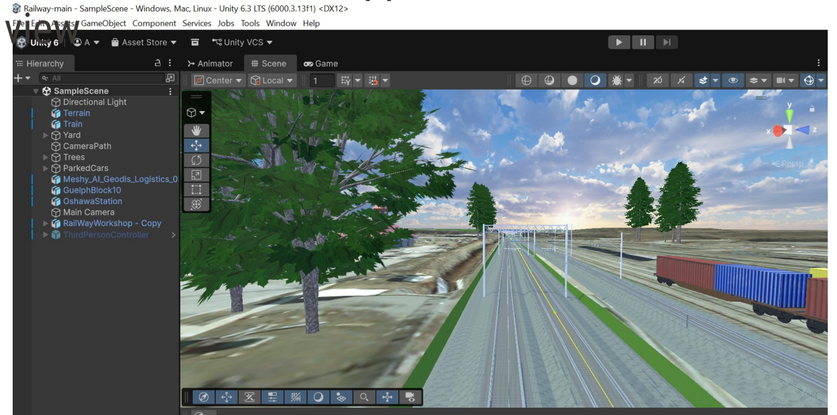
Select the Terrain object



Set Position and Rotation values to 0

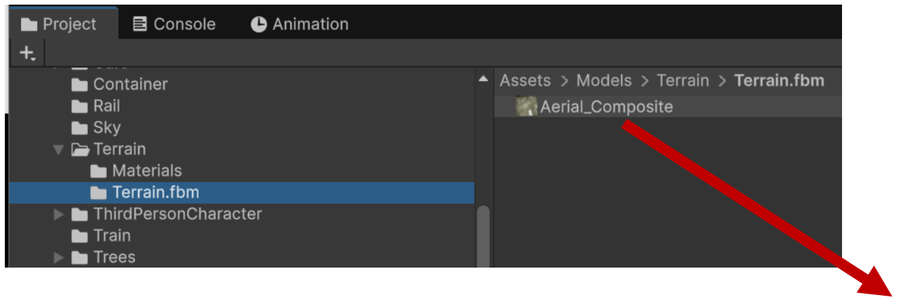


Confirm the Terrain appears in the Scene view

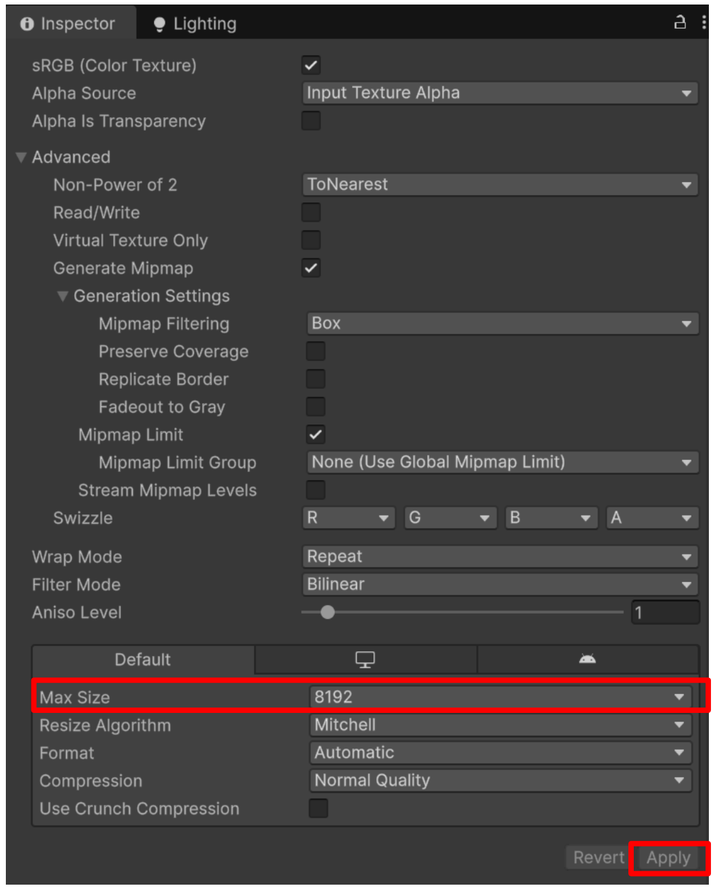


Fixing Problem: Adding the Terrain 3D Object to the Scene

Select Terrain.fbm → Select Texture (Aerial_Composite.tif)



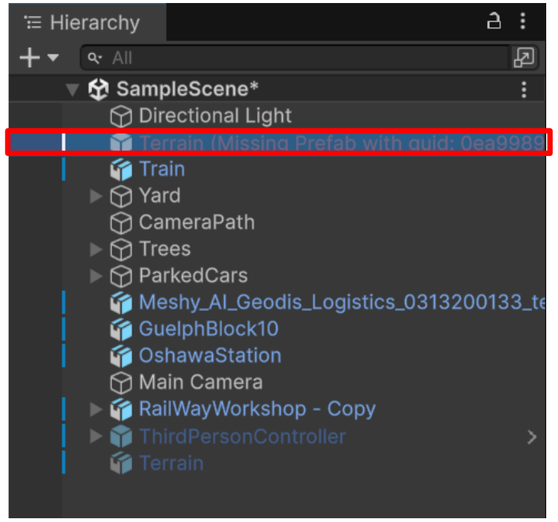
Inspector → Change Image Resolution



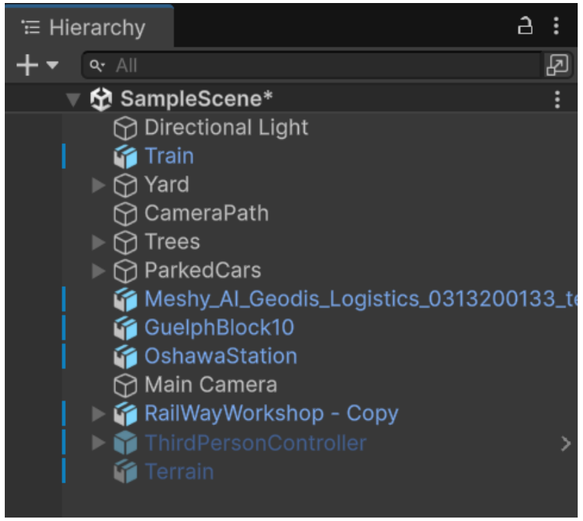
This enhance the texture resolution of the Terrain

Fixing Problem: Removing the Missing Object Reference

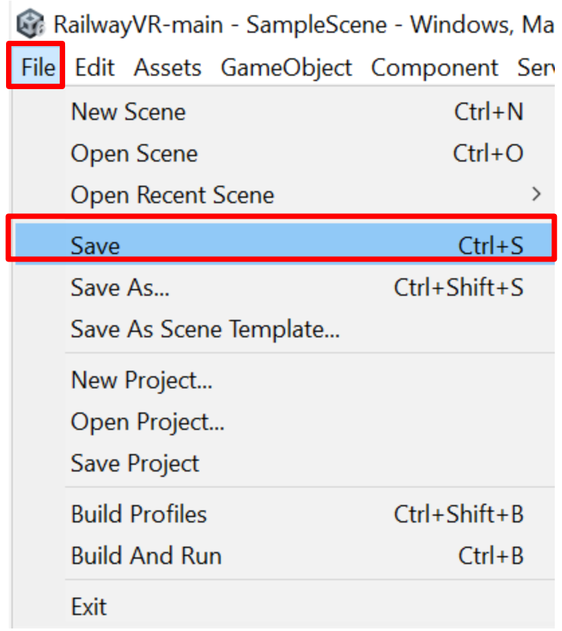
Select the Missing Prefab Object



Press Delete on the Keyboard

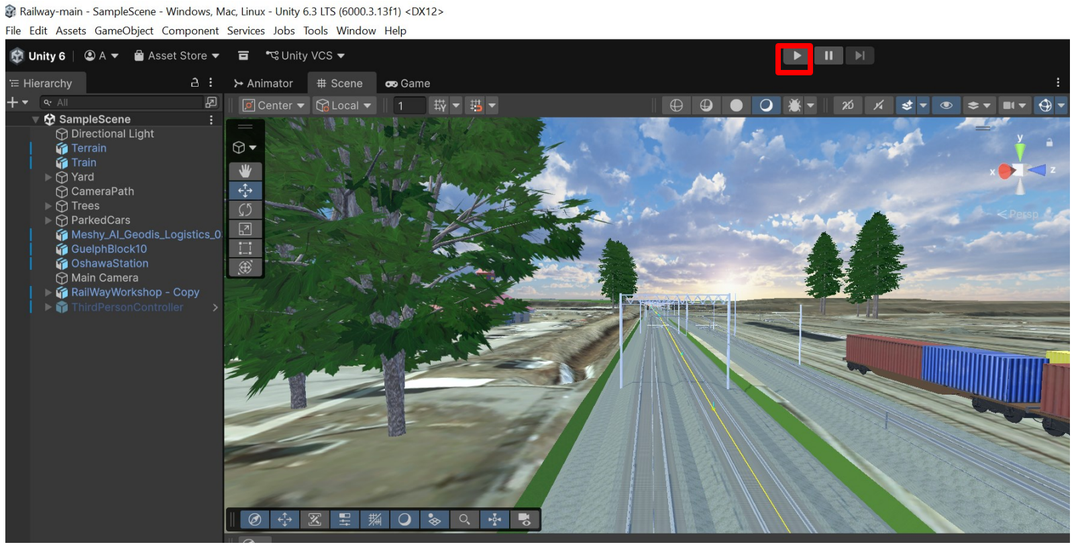


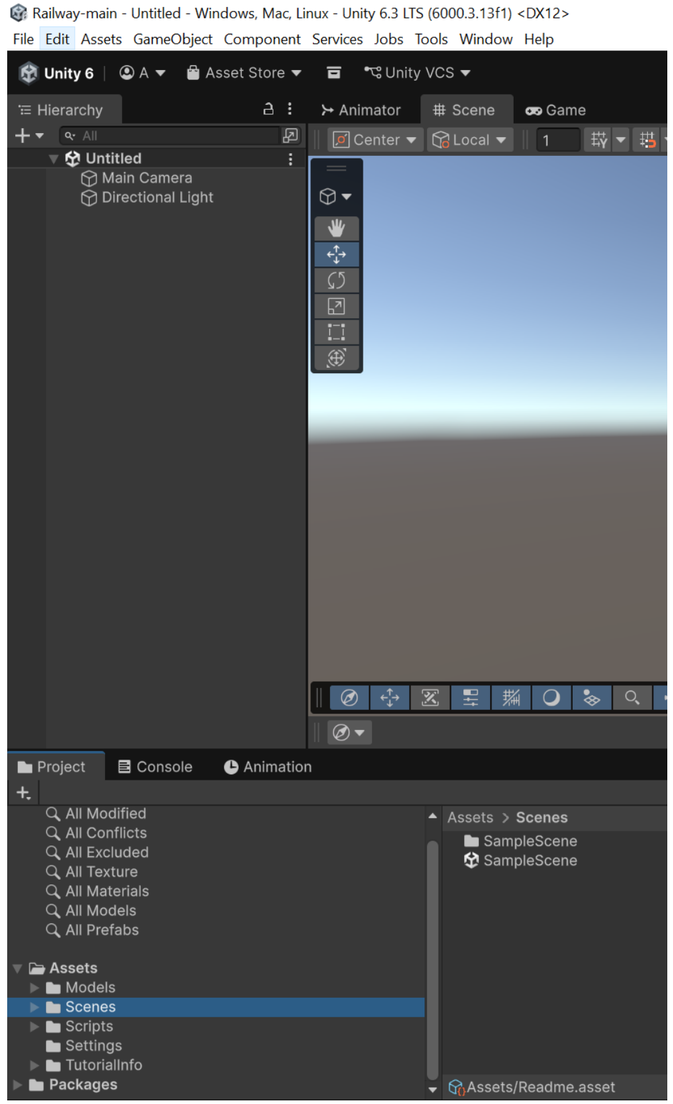
Always Save The Changes



Run the Unity Project

- ❑ Click the Play button to preview the scene.
- ❑ In this workshop, we will develop the railway environment, animation, and 360-degree VR view step by step.





Welcome to the Game Engine World

All your favorite games or animations are developed using **game engines.**

Unity is one of the most popular game engines. There are other game engines like **Unreal Engine**, etc.

Unity
- Widely used
- User-friendly
- Cross-platform

Unreal Engine
- High-fidelity graphics
- Real-time rendering
- Used in many AAA games



Examples of Games Built with Different Engines


Assassin's Creed (Ubisoft Anvil)


Cities: Skylines (Unity)



The Witcher 4 (Unreal Engine 5)



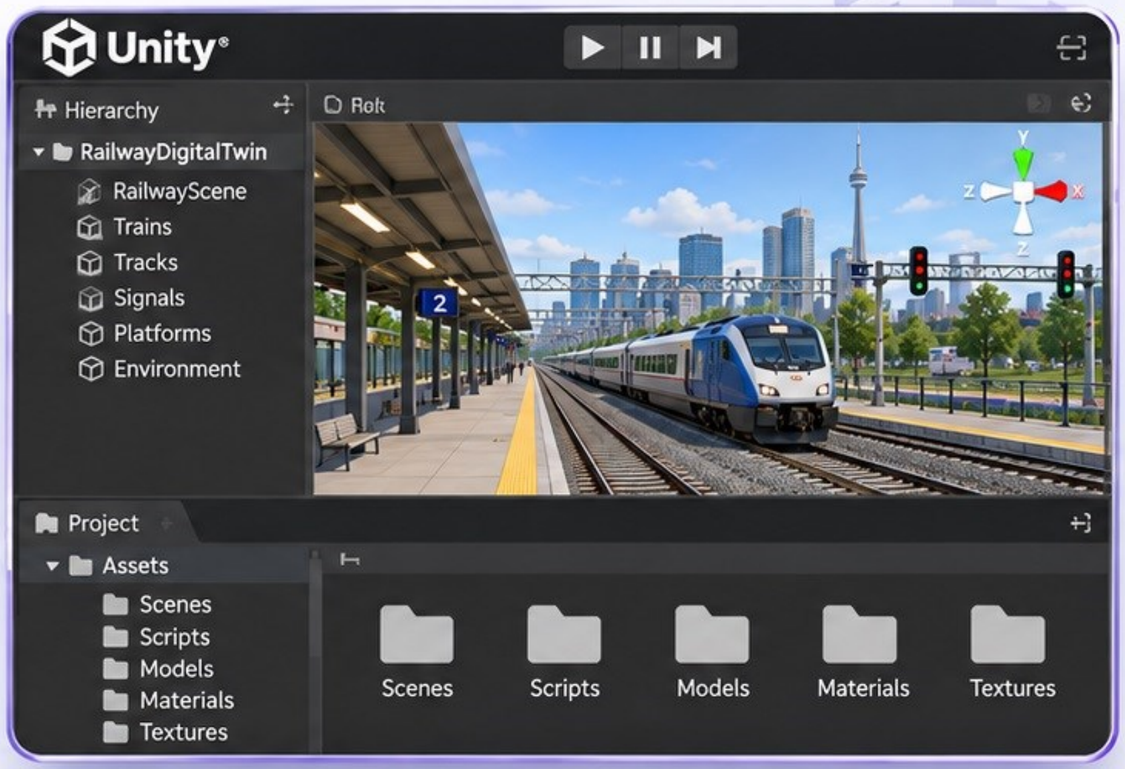
Fortnite (Unreal Engine)

Digital Twins in Railway Industry

Digital twins are often developed using game engines such as **Unity.**

In **North America,** these platforms support many railway applications and real-world projects.

Unity® | **Game engines help create interactive, visual, and data-driven railway environments for analysis, training, and planning.**



- **Training Simulators**
- **Operations & Control**
- **Infrastructure Planning**
- **Safety & Maintenance**

Sample North American Railway Use Cases



Station Simulation


Rail Yard Visualization


Driver Training


Asset Monitoring

Geographic Classification of Railway VR, AR, and Immersive Visualization Projects

North American industry case studies grouped by country and province / state / jurisdiction

Canada

4 projects | **2** jurisdictions

United States

6 projects | **4** jurisdictions

Canada

Ontario — 2 projects

- TTC Virtual Reality Station Models
- TTC Subway Familiarization VR Game

National / Multi-province — 2 projects

- CN Rail Safety Pledge immersive campaign
- CN Campus VR/AR railway training

United States

California — 1 project

- BART SFO Extension design visualization

Texas — 1 project

- DART asset management and digital condition assessment

Washington, DC — 2 projects

- WMATA Wayfinding Guidelines VR testing
- WMATA Metro Training Center AR/VR

National / Multi-state — 2 projects

- FRA Grade-Crossing Treatments in VR
- FRA CTIL virtual cab human-factors research

Takeaway: U.S. examples are more geographically diverse, while Canadian cases cluster around CN's safety/training initiatives and TTC's passenger-facing immersive applications.

Classification of Railway VR, AR & Immersive Visualization Projects

Grouped by application area and primary immersive use



1 Safety & Workforce Training

How can immersive tools improve rail safety readiness and workforce preparation?



Representative Projects

- CN Campus VR/AR railway training (2025)
- CN Rail Safety Pledge immersive campaign (2016)
- FRA CTIL virtual cab human-factors research (2017)
- WMATA Metro Training Center AR/VR (2026)

Main Immersive Focus

Scenario-based workforce training, near-miss and live-track awareness, cab simulation, operational decision-making, emergency preparedness, and repeatable safety exercises away from the active railway.

2 Infrastructure Inspection & Digital Twins

How can immersive models support asset inspection and railway digital twins?



Representative Projects

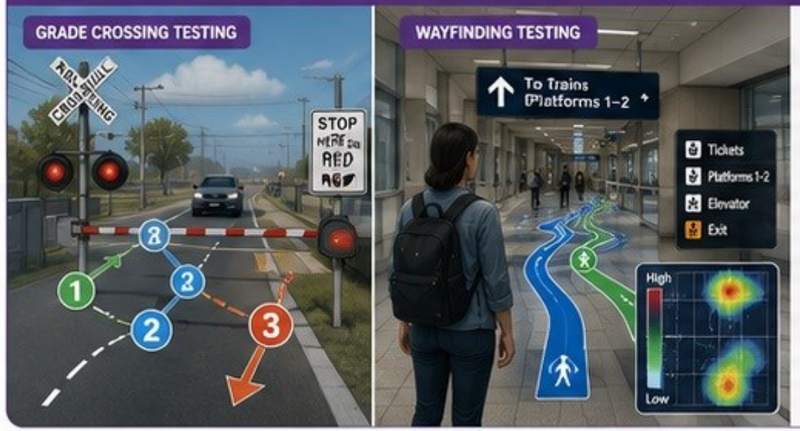
- DART real-time asset management & digital condition assessment (2025)

Main Immersive Focus

LiDAR, photogrammetry, AI-supported condition assessment, spatially accurate asset visualization, remote inspection, and navigation of digital railway environments for maintenance and asset-management decisions.

3 Human Behaviour & Design Evaluation

How can immersive environments test behaviour and design before field deployment?



Representative Projects

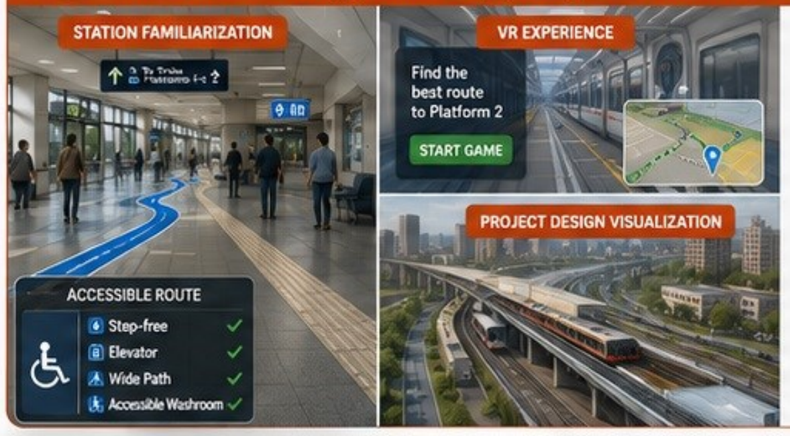
- FRA Grade-Crossing Treatments in VR (2023)
- WMATA Wayfinding Guidelines VR testing (2026)

Main Immersive Focus

Pre-implementation testing of crossing treatments, signs, passenger navigation, decision points, and human responses in controlled virtual environments before physical installation.

4 Passenger Information, Accessibility & Project Visualization

How can immersive applications help passengers and communicate projects?



Representative Projects

- TTC Virtual Reality Station Models (2021)
- TTC Subway Familiarization VR Game (2023)
- BART SFO Extension design visualization (1997)

Main Immersive Focus

Station familiarization, accessibility support, anxiety reduction for unfamiliar trips, passenger journey preview, and communication of project alternatives and design concepts through immersive visualization.

IMMERSIVE FOUNDATION (across all categories): Simulation • Spatial Context • Human Factors • Visualization • Digital Models • Accessibility

Outcome:
Safer training, smarter inspection, better design testing, and more user-friendly railway systems.

Project References

Key reports and sources supporting the integrated Railway Track-Design project set

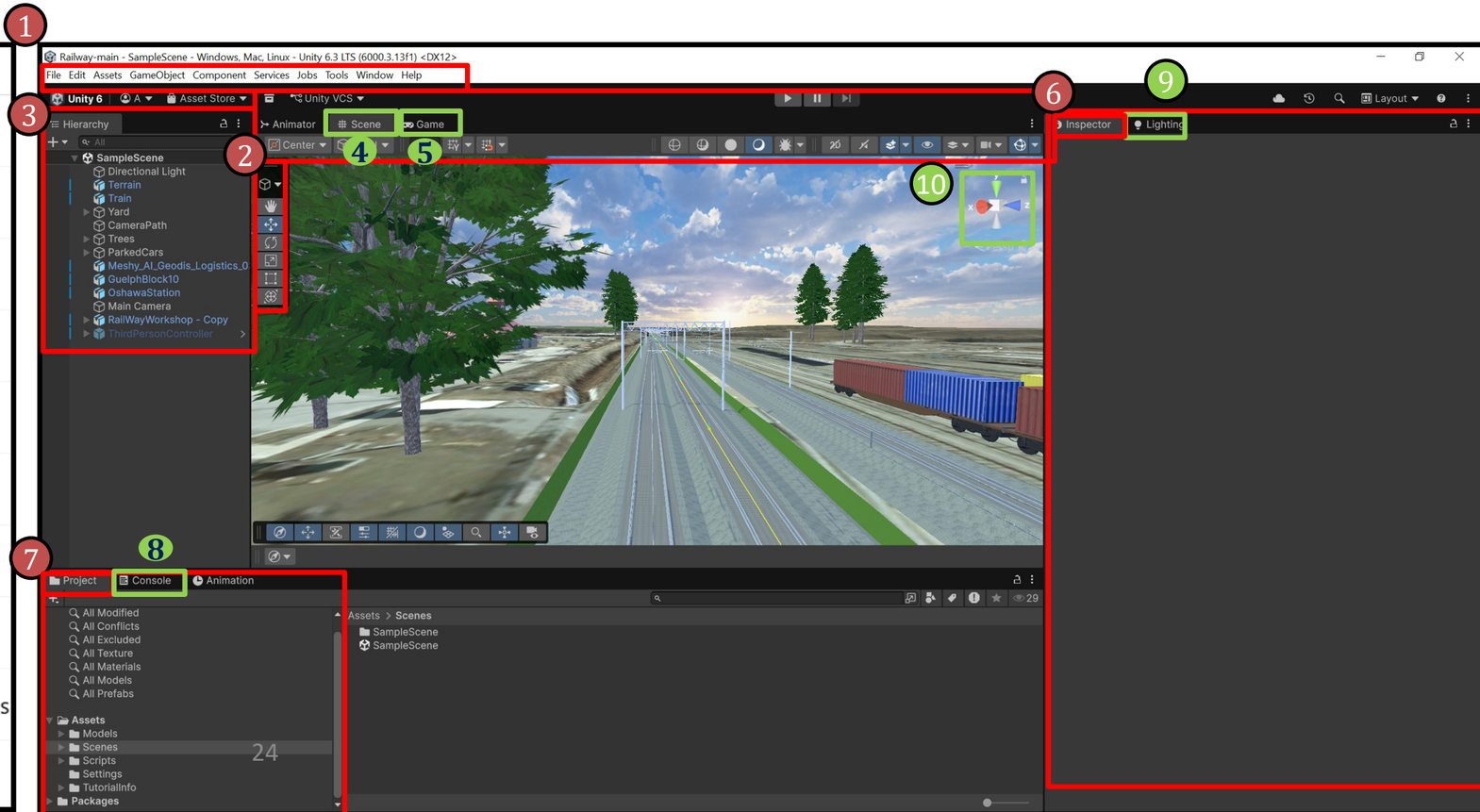
Project	Province / State	Country	Year (Report)	Reference (Short APA Style)	Source / PDF Link
1. **Hazel McCallion Line / Hurontario LRT**	Ontario	Canada	2023	Arcadis. (2023). *Hurontario Light-Rail Project overview.*	thepwi.org/.../Hurontario-Light-Rail-Project-overview.pdf
2. **La Grande Alliance - Phase I Rail Corridors**	Québec	Canada	2024	La Grande Alliance. (2024). *Technical report, Volume 2.*	lagrandealliance.quebec/.../LGATechnical-LOCAL_2024-03-19.pdf
3. **Eglinton Crosstown LRT**	Ontario	Canada	2022	Metrolinx. (2022). *Eglinton Crosstown LRT project overview.*	metrolinx.com/.../eglinton-crosstown-lrt
4. **GO Expansion Corridor Packages**	Ontario	Canada	2021	Metrolinx. (2021). *GO Expansion full business case.*	metrolinx.com/.../go-expansion-full-business-case.pdf
5. **Scarborough Subway Extension - Reference Concept**	Ontario	Canada	2020	Metrolinx. (2020). *Scarborough Subway Extension preliminary design business case.*	metrolinx.com/../scarborough-subway-extension-business-case.pdf
6. **Yonge North Subway Extension - Reference Concept**	Ontario	Canada	2021	Metrolinx. (2021). *Yonge North Subway Extension initial business case.*	metrolinx.com/.../yonge-north-subway-extension-initial-business-case.pdf
7. **Richmond-to-Raleigh S-Line Preliminary Engineering**	Virginia / North Carolina	USA	2023	Virginia Passenger Rail Authority. (2023). *Preliminary engineering RFP: Richmond to Raleigh.*	vapassengerrailauthority.org/.../RFP-1-006-23-002.pdf
8. **CTA Red and Purple Modernization Program**	Illinois	USA	2019	Chicago Transit Authority. (2019). *Bentley software scope of work for RPM program.*	transitchicago.com/.../Bentley_SOW_10-29-19.pdf
9. **California High-Speed Rail - Southern California Corridors**	California	USA	2018	California High-Speed Rail Authority. (2018). *Southern California project section materials.*	hsr.ca.gov/.../southern-california-project-sections.pdf

Note: Years refer to major public project / evidence years. Links are abbreviated for slide readability.

Beginning of Milestones 1

Unity Editor Interface Overview

Unity Area	What It Does
1 Menu Bar	Access project, assets, build, and editor tools
2 Toolbar	Play, pause, transform tools, layout controls
3 Hierarchy	Lists all objects in the current scene
4 Scene View	Main 3D editing workspace
5 Game View	Shows what the player/camera sees
6 Inspector	Shows and edits selected object properties
7 Project Window	Shows project files and folders
8 Console	Shows errors, warnings, and debug logs
9 Lighting Window	Controls sky, environment, and lighting settings
10 Scene Gizmo	Shows X/Y/Z direction and view orientation

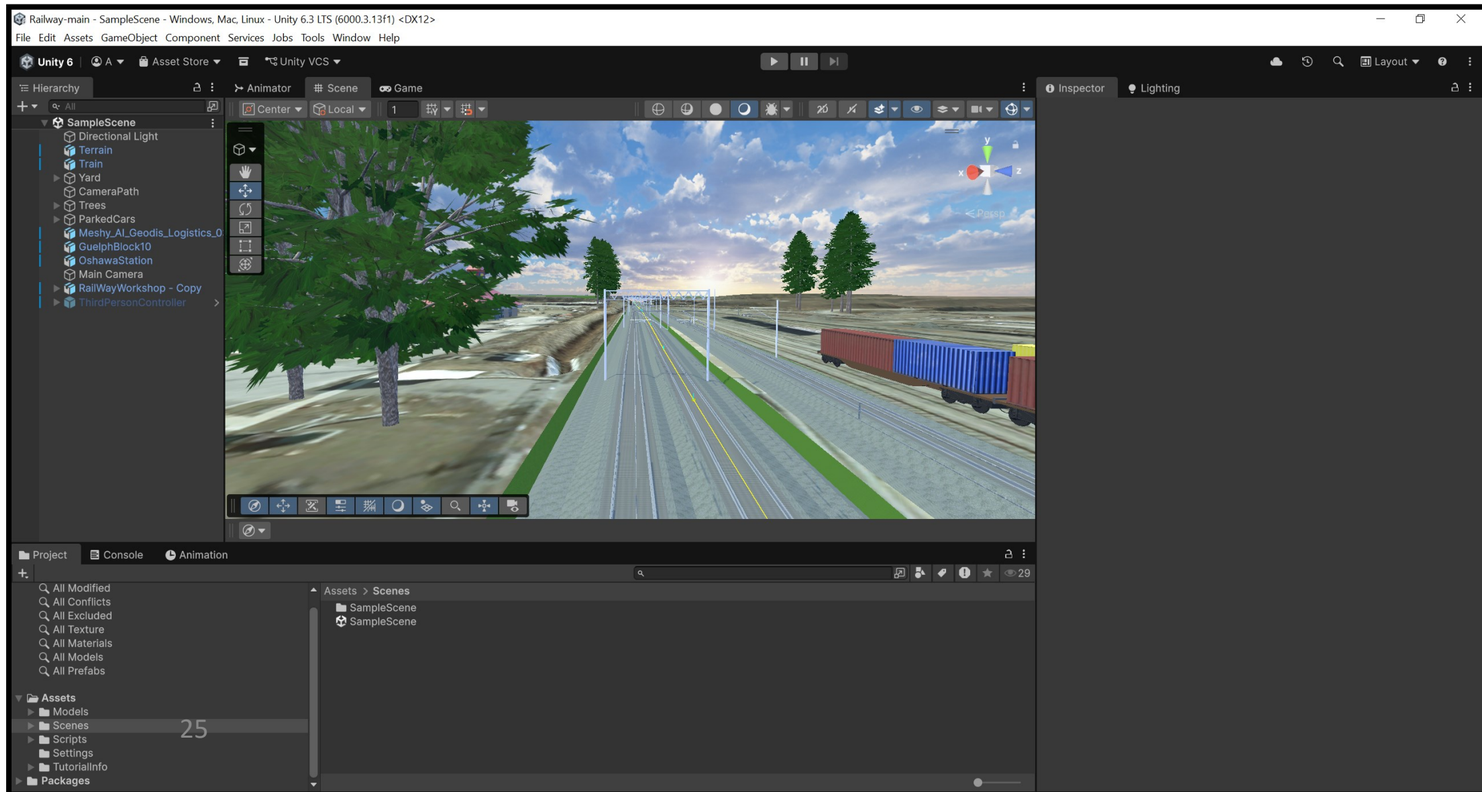


Unity Navigation Control

- ❑ Practice these Scene View controls for few minutes.
- ❑ Move around the railway scene and inspect the station, tracks, trees, and trains.

Unity Scene View Navigation Controls

Action	Control
Move Forward	Right Mouse Button + W
Move Backward	Right Mouse Button + S
Move Left	Right Mouse Button + A
Move Right	Right Mouse Button + D
Move Up	Right Mouse Button + E
Move Down	Right Mouse Button + Q
Move Faster	Right Mouse Button + Shift + W/A/S/D/Q/E
Look Around	Hold Right Mouse Button + Move Mouse
Zoom In/Out	Scroll Wheel
Pan View	Middle Mouse Button + Drag



Create a New Scene

Task: Create a new Scene, for that:

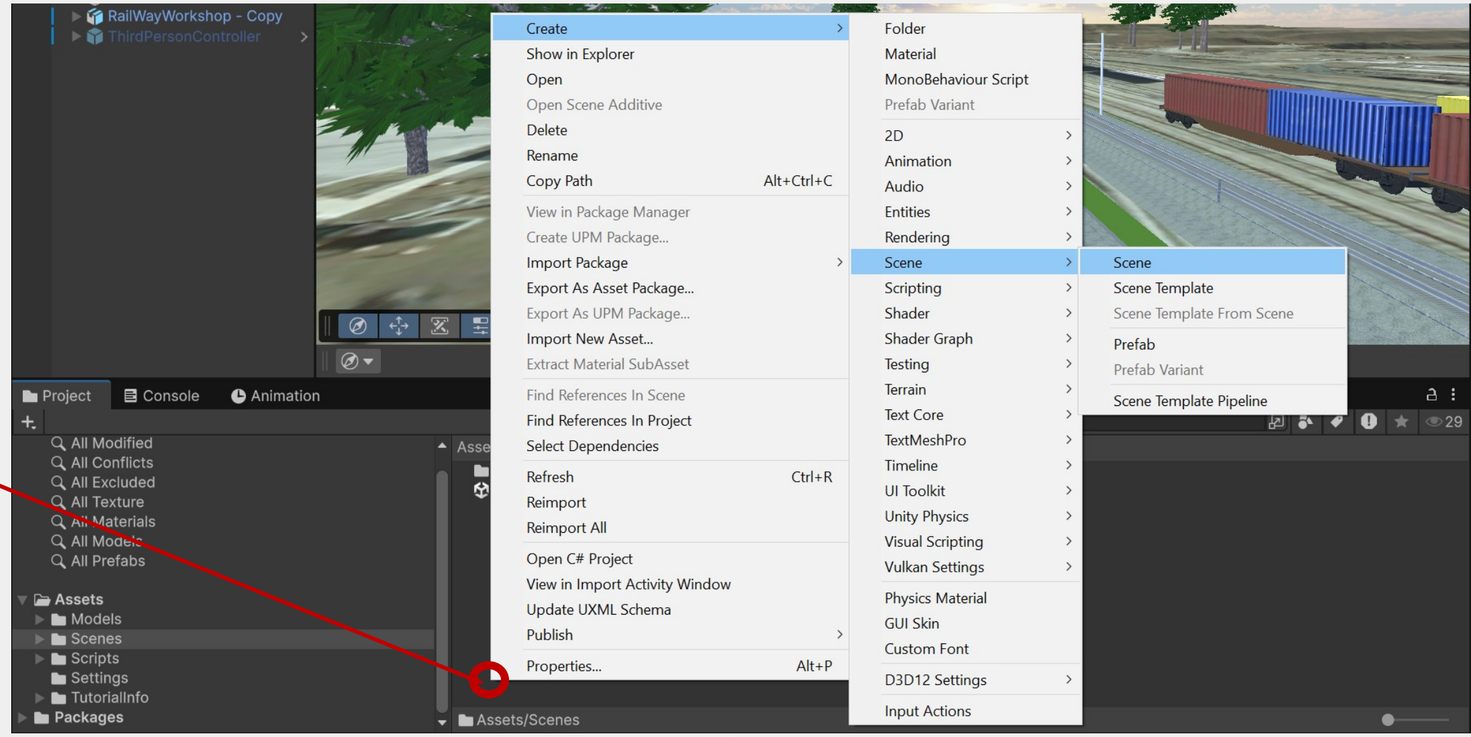
❑ Project Window → Select Folder “Scenes” →

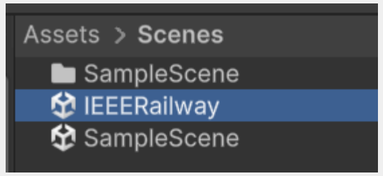
Right Click here → Create → Scene → Scene

❑Rename the new scene to IEEERailway

❑Double-click the scene to open it

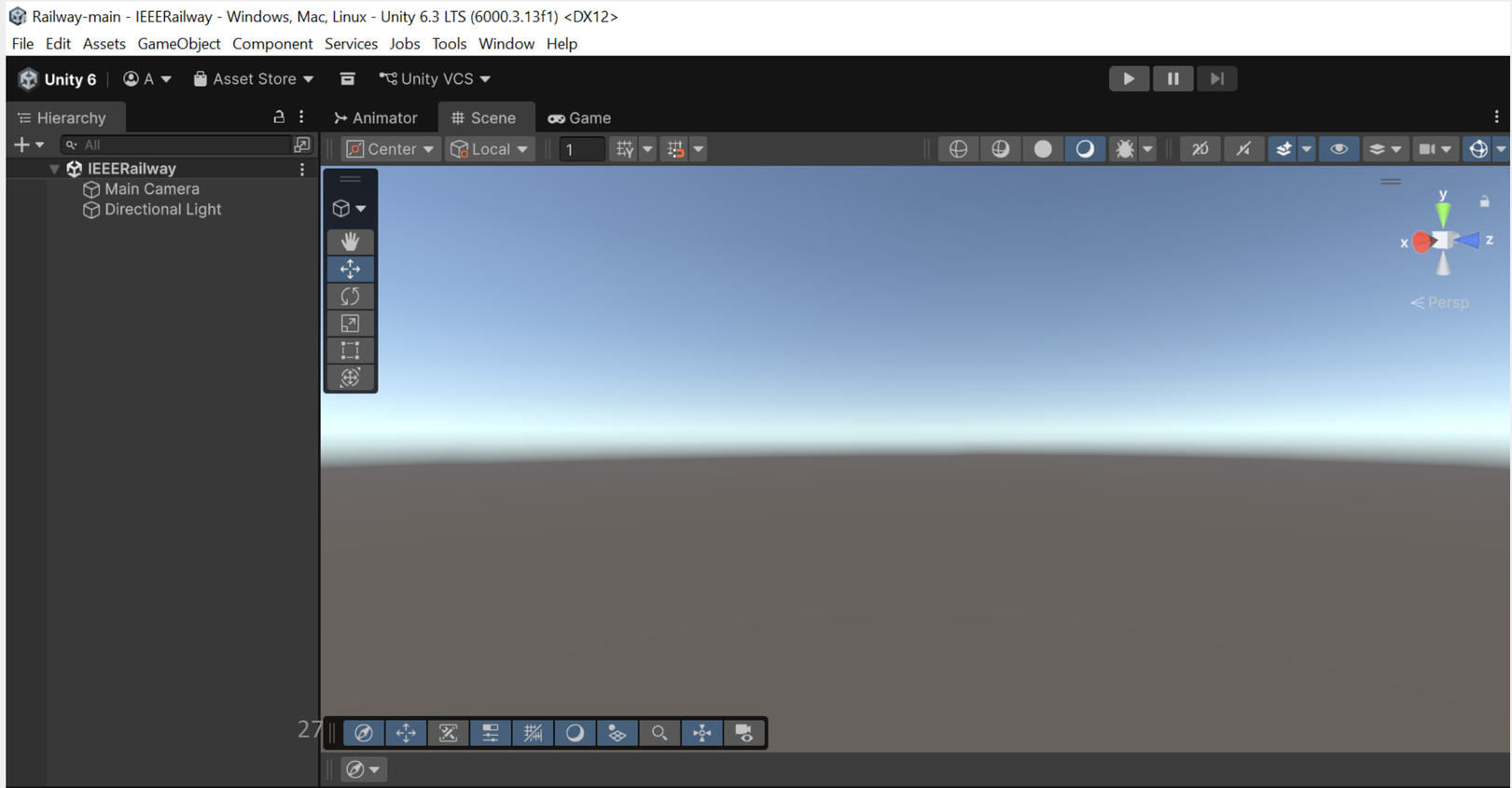
Question: What objects are included in the new scene?





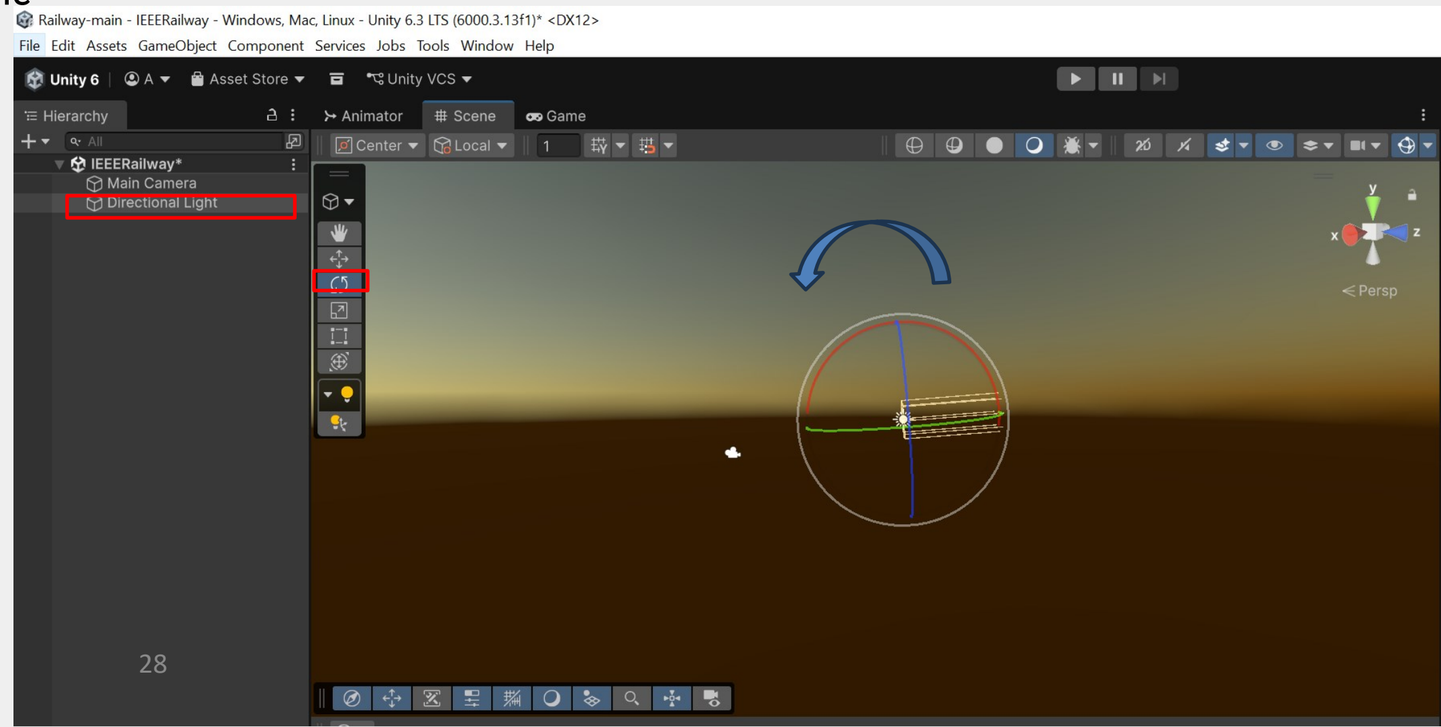
Create a new Scene

- ❑ A new default scene is created.
- ❑ The scene includes only the basic Unity objects: Main Camera and Directional Light.
- ❑ We will learn the fundamentals of the Unity game engine by creating new elements step by step.



Simulating Sunlight with Directional Light

- ❑ Select the Directional Light in the Hierarchy.
- ❑ Double-click the Directional Light to focus on it in the Scene view.
- ❑ Rotate the Directional Light and observe how the scene changes.
- ❑ Try different lighting conditions: Sunrise, sunset, daytime, and nighttime.
- ❑ This is how the games/digital twins simulated different time

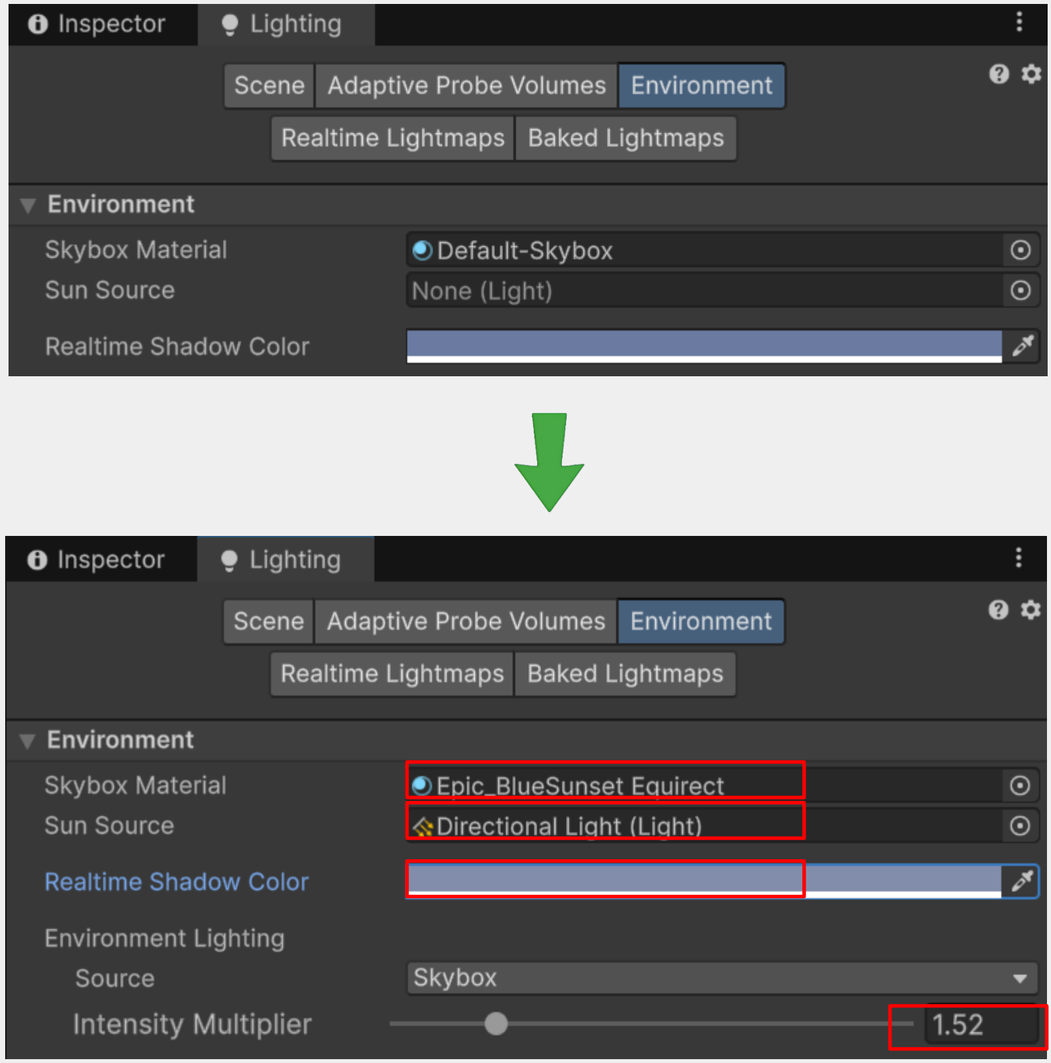


Changing the Skybox

- ❑ Open the Lighting window:

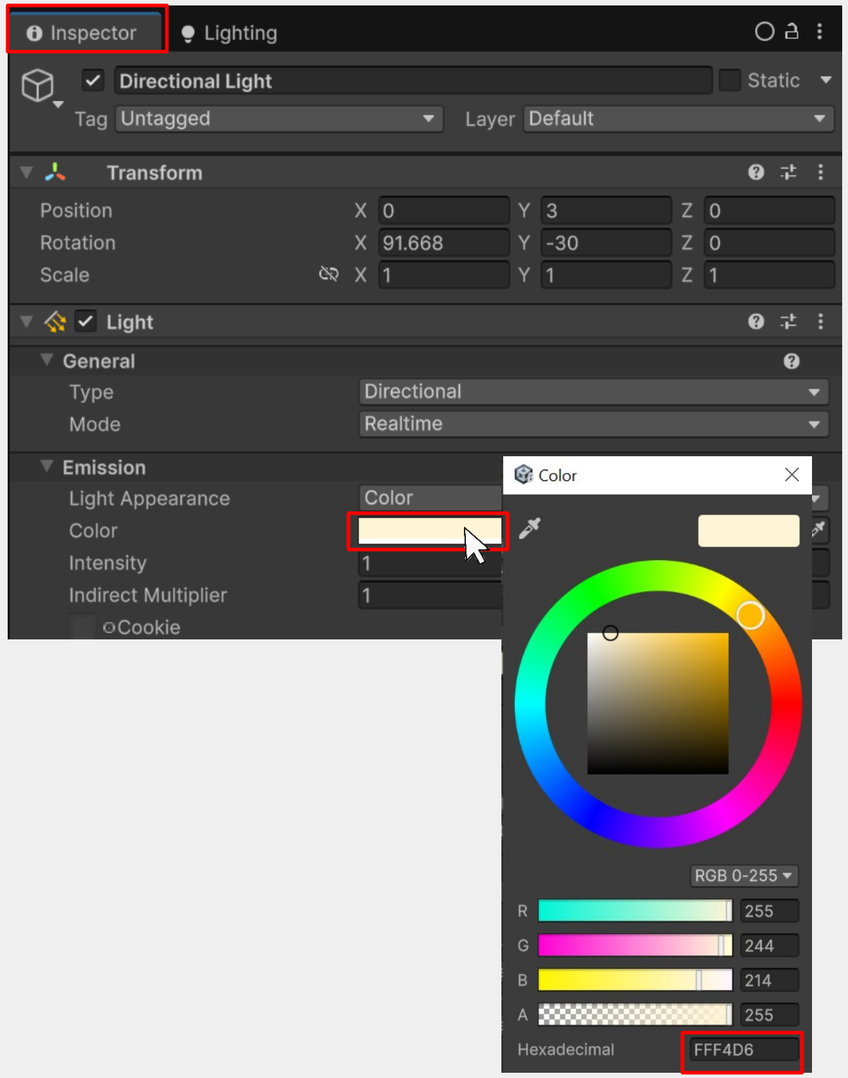
Window → Rendering → Lighting → Environment

- ❑ Set Skybox Material to: Epic_BlueSunset_Equirect
- ❑ Set Sun Source to: Directional Light
- ❑ Set Realtime Shadow Color to: #828DAB
- ❑ Set Intensity Multiplier to:1.52
- ❑ File → save
- ❑ Rotate to see the sky, clouds and sun
- ❑ This is one way that games/digital twins change the sky (cloudy, sunny etc)



Directional Light

- Select Directional Light in the Hierarchy
- In the Inspector window, go to: Light → Emission → Color
- Set the Hexadecimal color code to:#FFF4D6
- This creates a warmer sunlight effect in the scene.

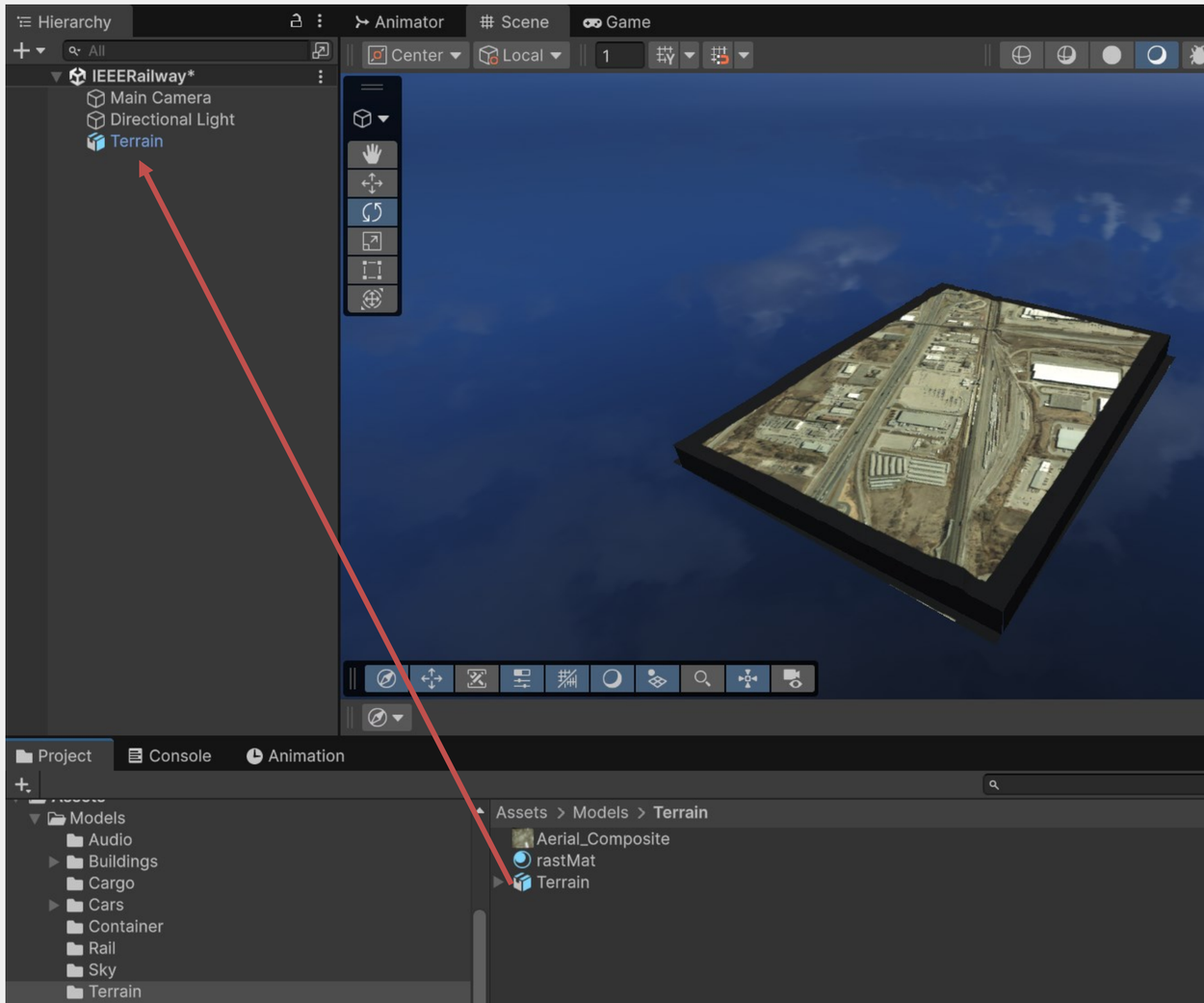


Adding Terrain and Trees to the Scene

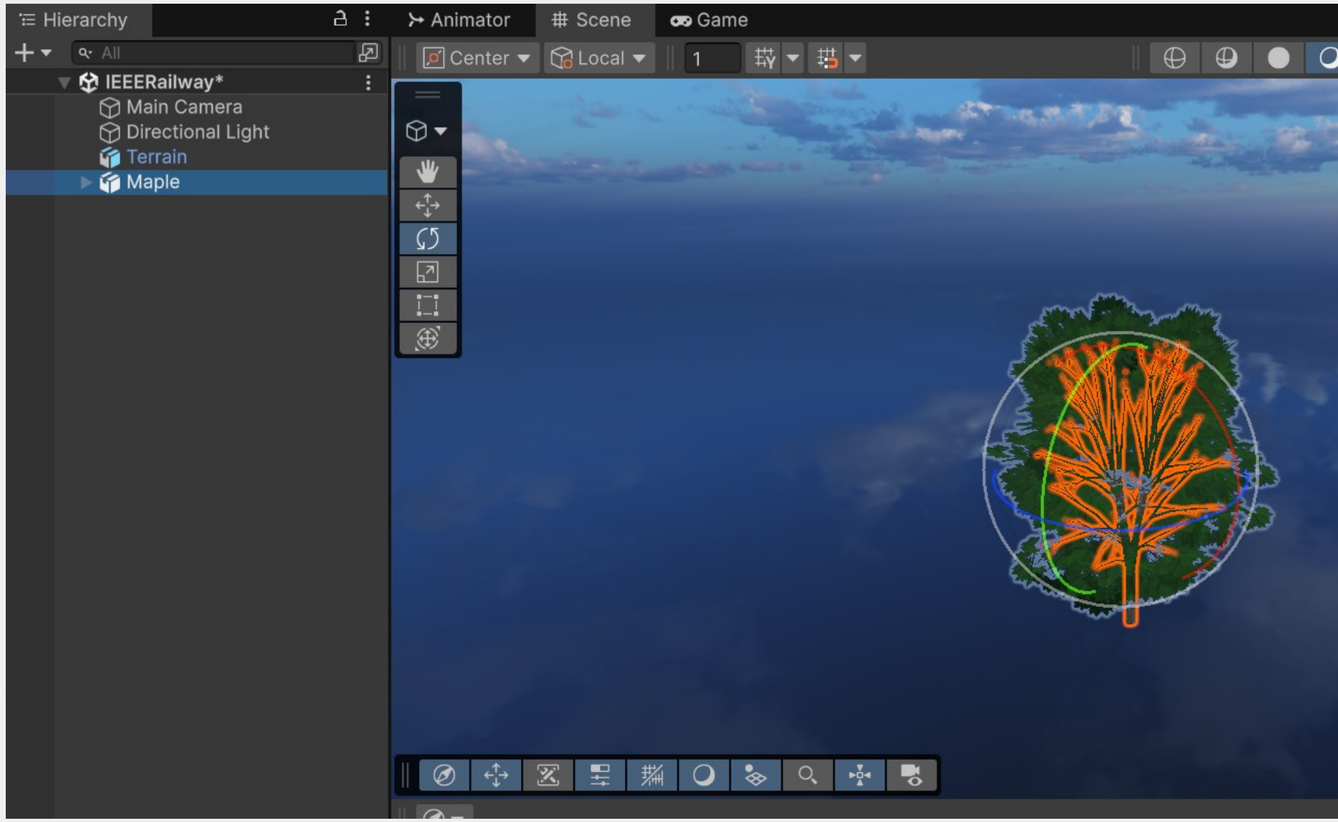
- ❑ Project Window → Models→ Terrain → Drag and Drop Terrain.fbx into Hierarchy window
- ❑ Hold Right Click and E to move vertically to observe the terrain
- ❑ The terrain object will appear in the Scene view.
- ❑ Now, similarly try to drag and drop a tree

Models → Trees → Maple → Maple.fbx

- ❑ Drag and drop Maple.fbx into the Hierarchy window.



Object Position and Coordinates

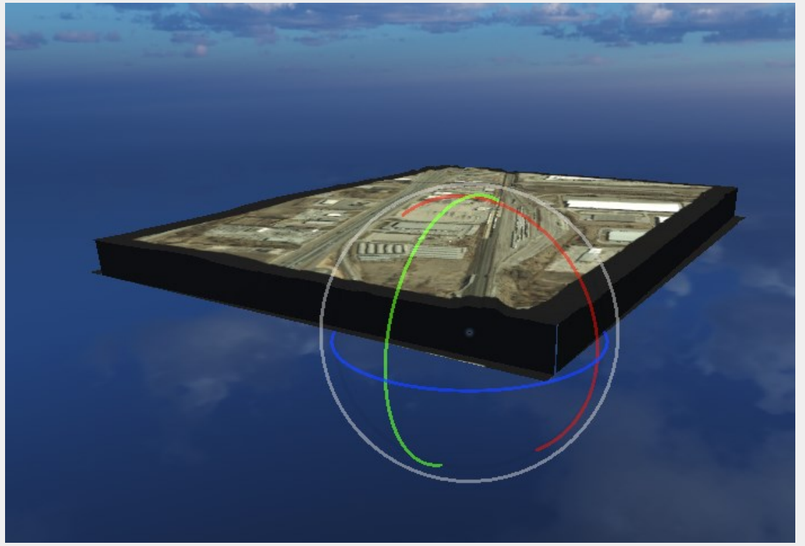


- ❑ Select the Maple object in the Hierarchy.
- ❑ Double-click the Maple object to focus on it in the Scene view.
- ❑ Zoom out (Hold Right Click and Press S) and

observe the tree position relative to the terrain.

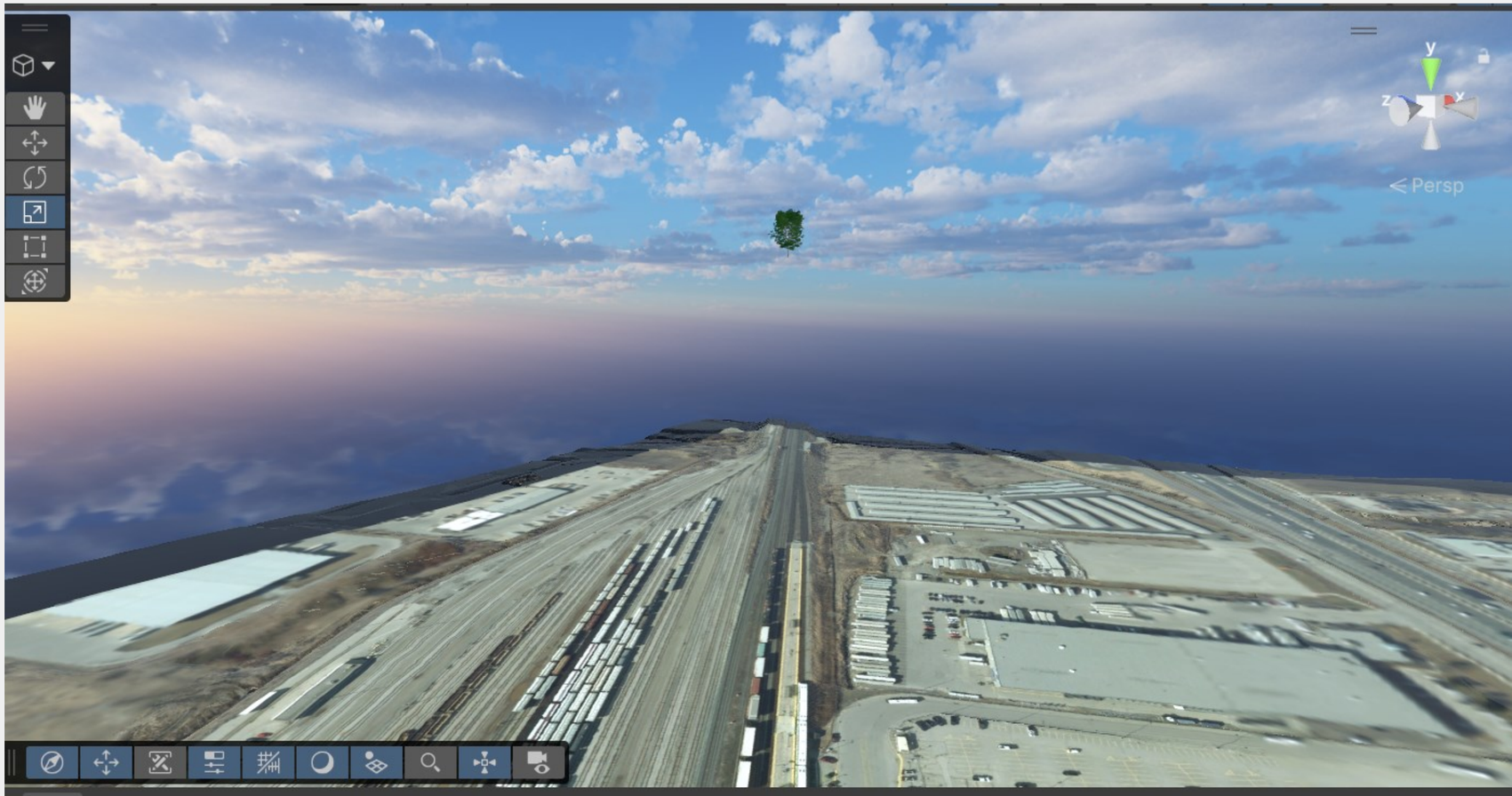
- ❑ Question:

Is the tree placed on the terrain, or is it far away?



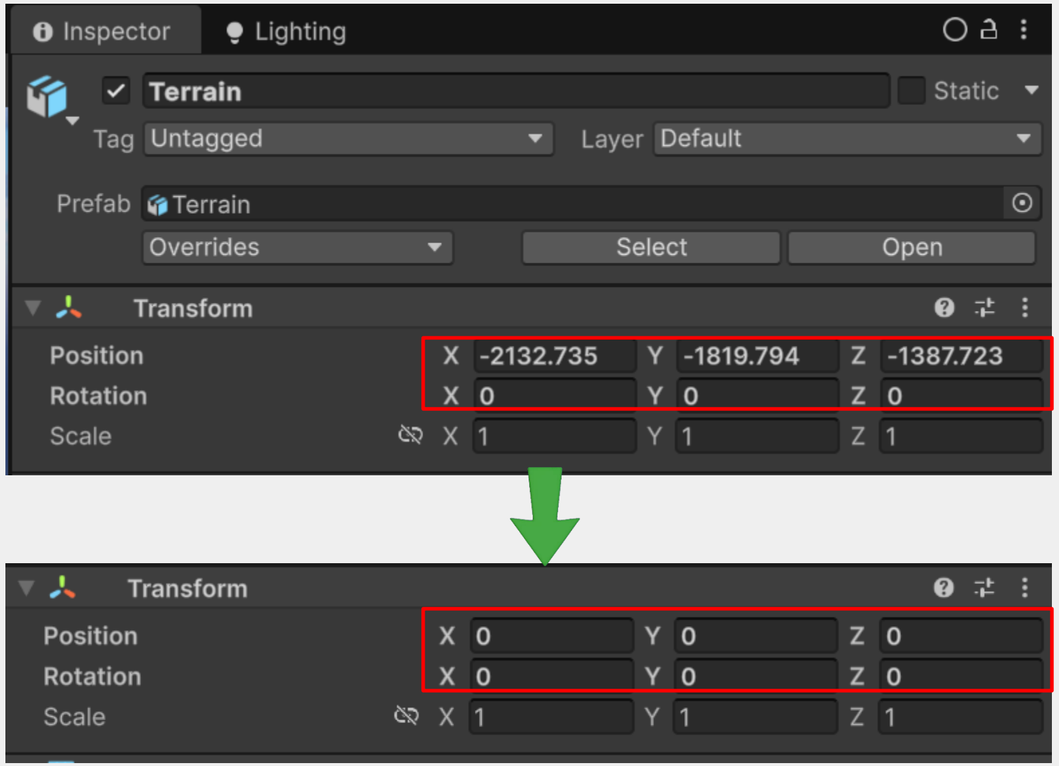
Object Position and Coordinates

- For example, I am seeing the tree there in the middle of nowhere



Object Position and Coordinates

- ❑ Hierarchy → Select Terrain
- ❑ In the Inspector window → Change Position and Rotation to (0,0,0)
- ❑ Repeat the same process for Maple
- ❑ Double-click Maple to focus on it in the Scene view.



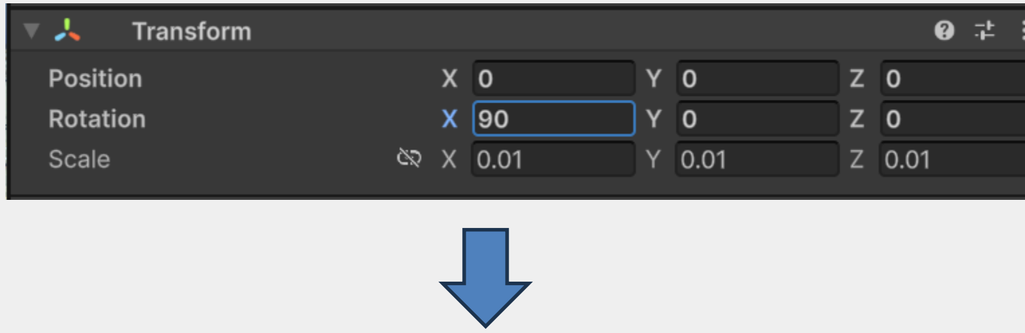
- ❑ **Think & Observe - No submission required:**

What do you see now? Left Side or Right Side



Object Position and Coordinates

- Rotate the maple properly to stand vertically





- Another observation here is about scale, right now it is (0.01,0.01,0.01)
- If you remember, the scale should always be (1,1,1)
- In Game Engine, we cannot do anything, we need to import this in Blender to fix it and Export as fbx again → **We skip the process for this course and lab.**

Object Position and Coordinates

- ❑ Now you understand the importance of Milestone 2 of the Lab 3 (previous lab: Blender) which was about fixing all of these issues before importing a digtal asset to Unity; for example:
- ❑ Set realistic scale
- ❑ Align the asset
- ❑ Apply the transforms

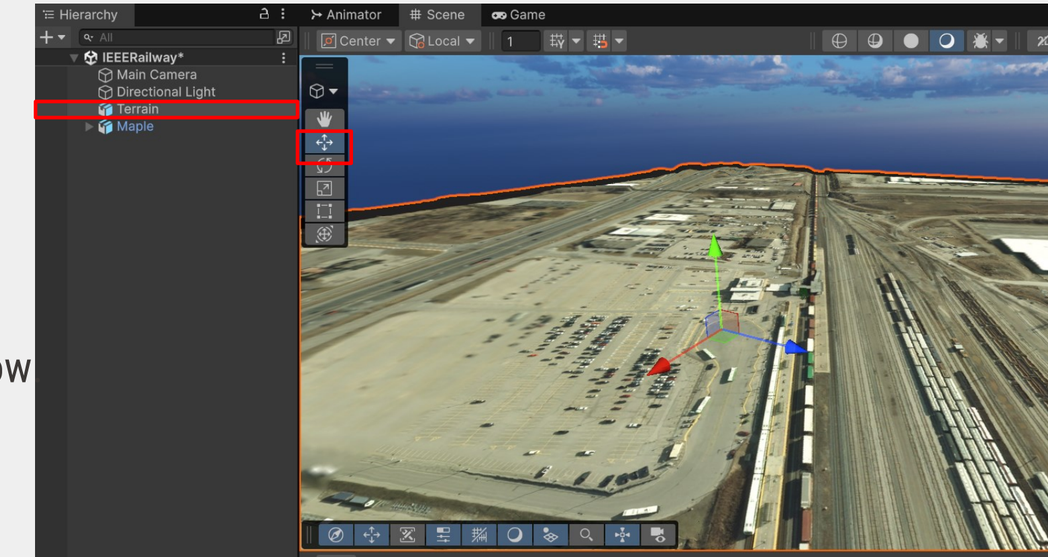
Moving Objects Using Coordinates

- ❑ Hierarchy → Select Terrain → Double Click on It → Zoom a bit → Hold Left Click + Press “W” + Hold Shift to speed up
- ❑ Alternatively, you can Press “F” to Focus
- ❑ Select the “Move” Tool



- ❑ The coordinate arrows appear in the Scene view:
- ❑ Y = Up/Down, X = Left/Right, Z = Forward/Backward
- ❑ Move the terrain upward by dragging the green Y arrow.
- ❑ Use Ctrl + Z to undo before the next exercise
- ❑ Move the terrain downward by dragging the green Y arrow.
- ❑ Use Ctrl + Z to undo before the next exercise
- ❑ Move the object along the X axis.
- ❑ Move the object along the Z axis.

- In each exercise, observe how the Position values change in the Inspector window
- Now, move the terrain by directly changing the Position values in the Inspector.



Moving Objects Using Coordinates

- Make Sure the Terrain Position/Rotaiton Values are all 0 after previous exercises



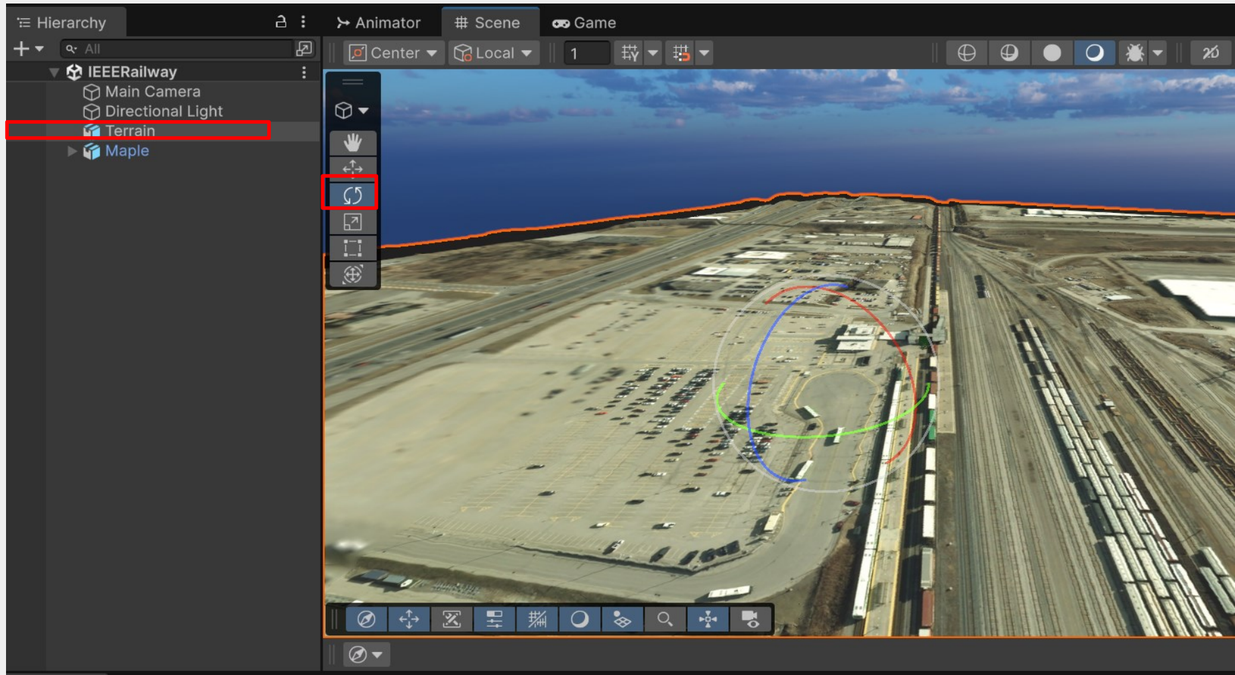
Rotating Objects in Unity

- ❑ Hierarchy → Select Terrain
- ❑ Select the “Rotate” Tool



- ❑ The rotation handles appear in the Scene view.
- ❑ Rotate the object around the Z axis.
- ❑ Rotate the object around the Y axis.
- ❑ Rotate the object around the X axis.
- ❑ In each exercise, observe how the Rotation values change in the Inspector window.

- Now, rotate the terrain by directly changing the Rotation values in the Inspector.



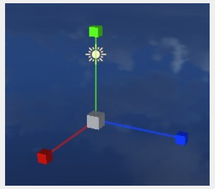
Moving Objects Using Coordinates

- Make Sure the Terrain Position/Rotaiton Values are all 0 after previous exercises



Scaling Objects in Unity

- ❑ Hierarchy → Select Terrain
- ❑ Select “Scale” Tool

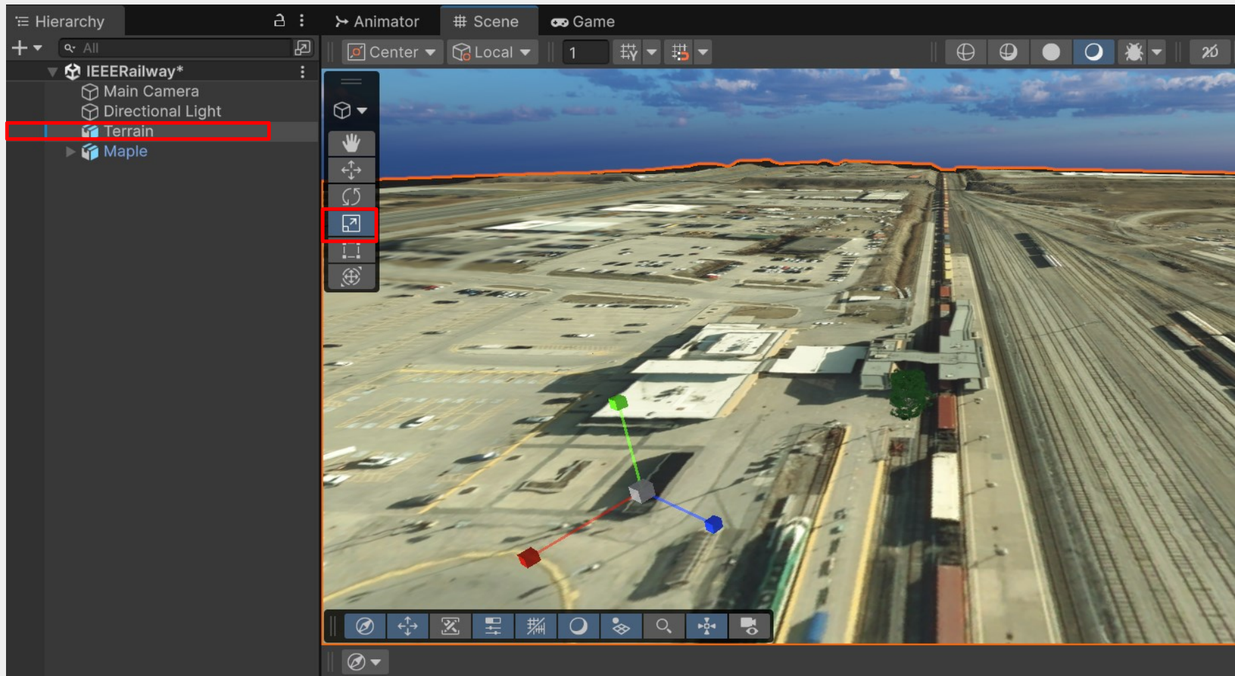


- ❑ The scale handles appear in the Scene view.

Practice the following exercises. (Use Ctrl + Z to undo before the next exercise.)

- ❑ Scale the object along the Z axis.
- ❑ Scale the object along the Y axis.
- ❑ Scale the object along the X axis.

- In each exercise, observe how the Scale values change in the Inspector window.
- Now, scale the terrain by directly changing the Scale values in the Inspector.



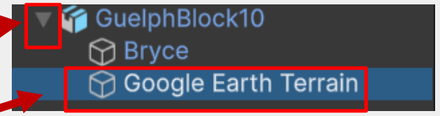
Activity: Add Different Trees to the Scene

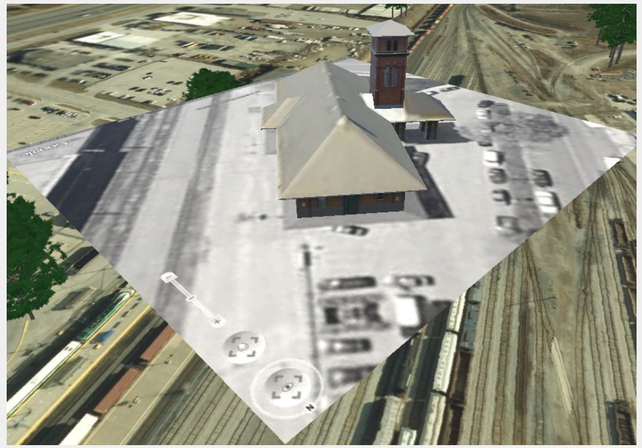
- ❑ Drag and drop different tree objects into the scene to develop the environment.
- ❑ **Use the Move, Rotate, and Scale tools to adjust each tree's position, direction, and size.**
- ❑ Use the background GIS map as a reference for tree placement.
- ❑ Do not place trees on top of railway tracks, roads, or buildings.
- ❑ This is one possible tree layout. You can create your own layout.



Adding a Station Building

- ❑ Project Folder → Models → Buildings → Station1 → Drag and Drop GuelphBlock10 into Hierarchy
- ❑ Double-click the object to zoom in on it.
- ❑ Expand the imported object in the Hierarchy.
- ❑ Remove the unnecessary child object: Google Earth Terrain
- ❑ Move the station building to the Oshawa GO Station area.
- ❑ Use the Move, Rotate, and Scale tools to align the building with the background GIS map.







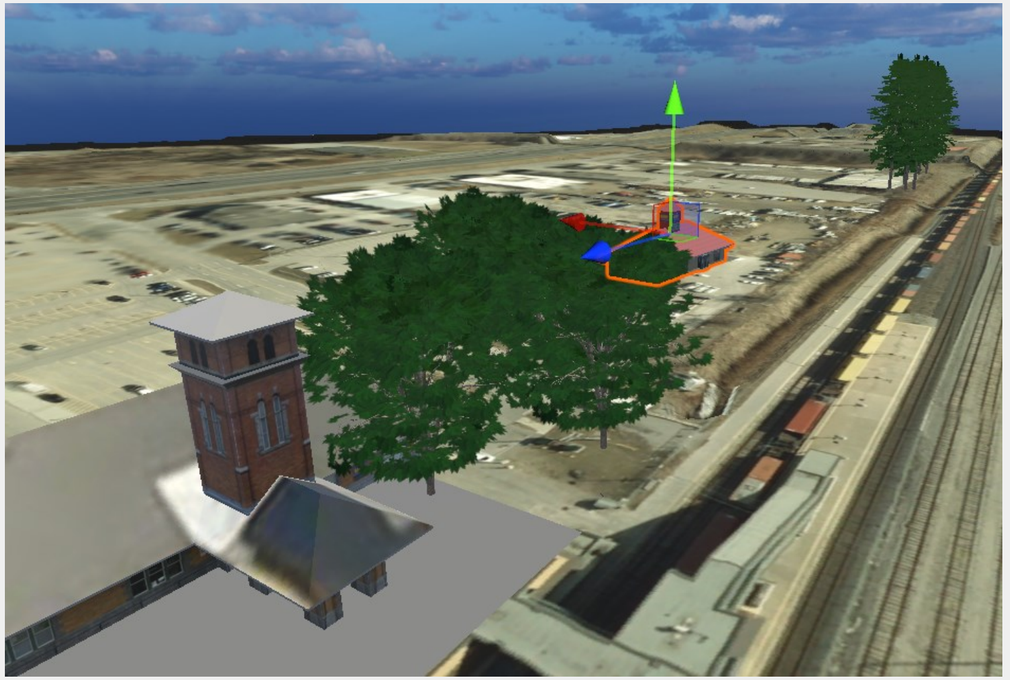
Creating a Ground Plane for the Station

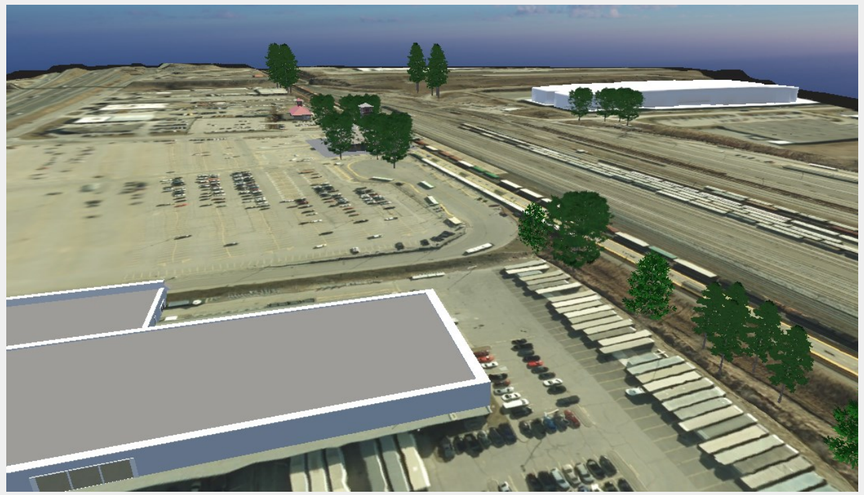
- ❑ In Hierarchy Window → Right click→ 3D Object → Plane
- ❑ Move and scale the Plane so it is located under the station building.
- ❑ View the station from a human-scale perspective to check that the plane is properly sized and positioned.
- ❑ This exercise shows how Unity can create simple built-in objects, such as a Plane.
- ❑ For complex objects such as buildings, trees, and vehicles, we usually import models created in 3D modeling software (previous lab).



Activity: Add Additional Buildings

- ❑ Add Station 2 from the Station2 folder and place it in the same location shown in the image.
- ❑ Use the Move and Scale tools to adjust its position and size.
- ❑ You can also add other building objects to complete the station environment.
- ❑ For example, I added two additional buildings to improve the scene layout.

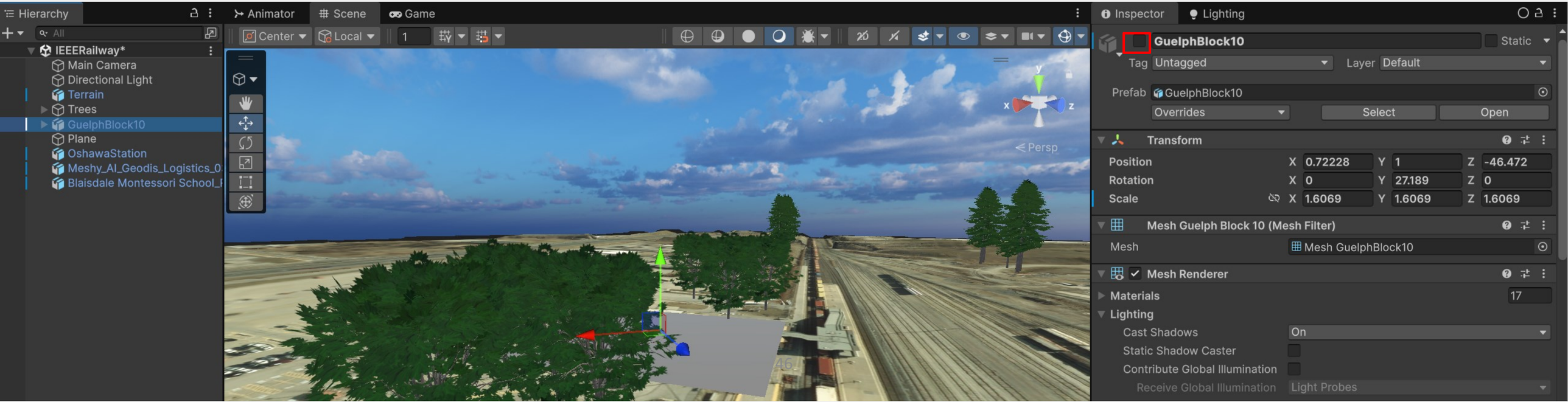




Showing and Hiding Objects in Unity

Goal: Hide all objects except GuelphBlock10.

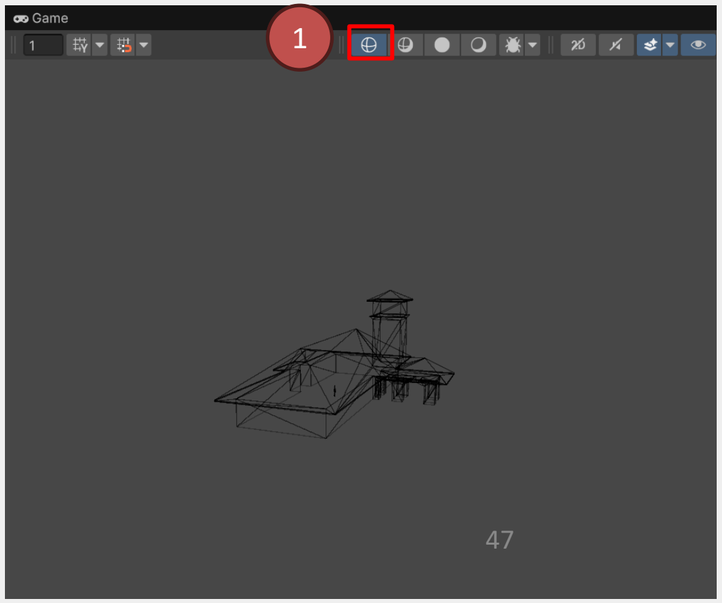
- ❑ Select GuelphBlock10 in the Hierarchy.
- ❑ In the Inspector window, uncheck the checkbox beside the object name.
- ❑ The object will be hidden in the Scene view. Check the box again to show it.
- ❑ Apply the same process to the other objects in the Hierarchy so that only the building (GuelphBlock10) remains visible.



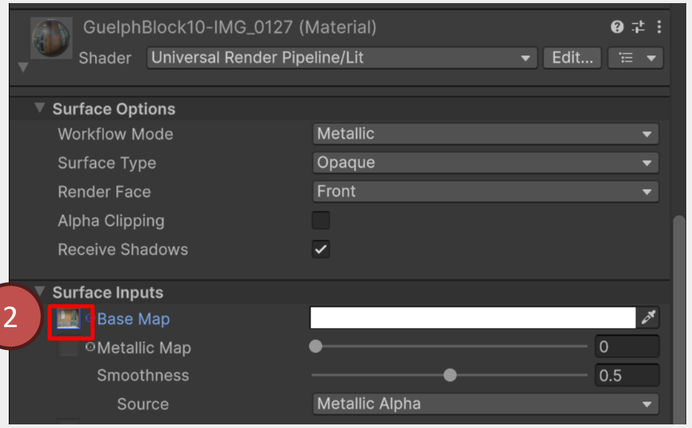
Meshes, Materials, and Textures

Goal: Understand how a 3D object is built from mesh geometry and visual materials.

- ❑ Select Wireframe mode, as shown in image 2. (1)
- ❑ The building is displayed as lines and surfaces. This is the mesh structure
- ❑ Select the building object. In the Inspector window, scroll down to the Materials section.
- ❑ Materials add color or texture images to the mesh surface. (2)
- ❑ In simple terms: A 3D object = Mesh + Materials/Textures
- ❑ Return to Shaded mode, as shown in image 1. (3)









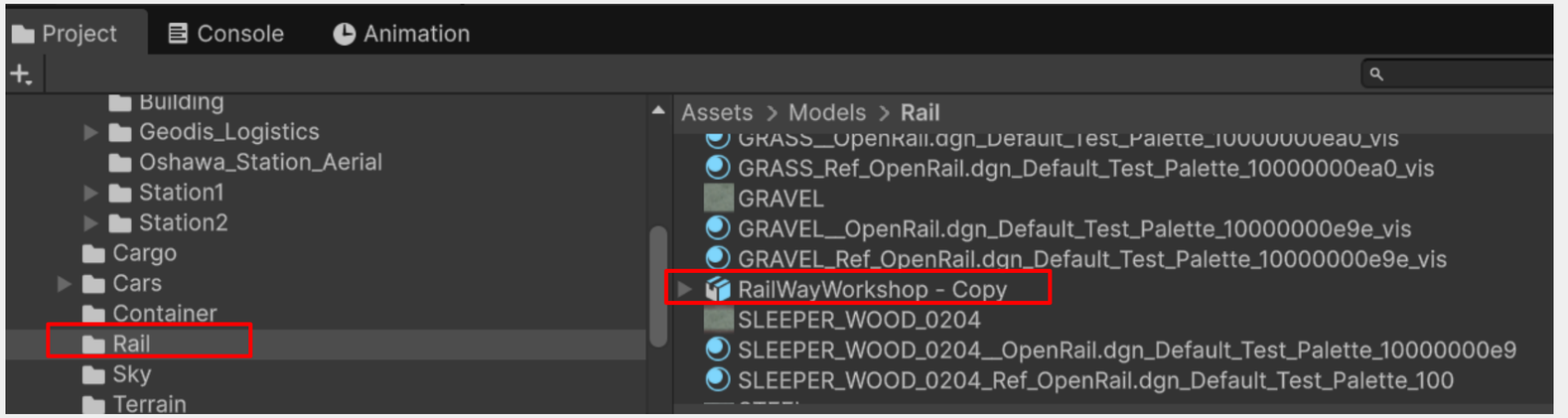


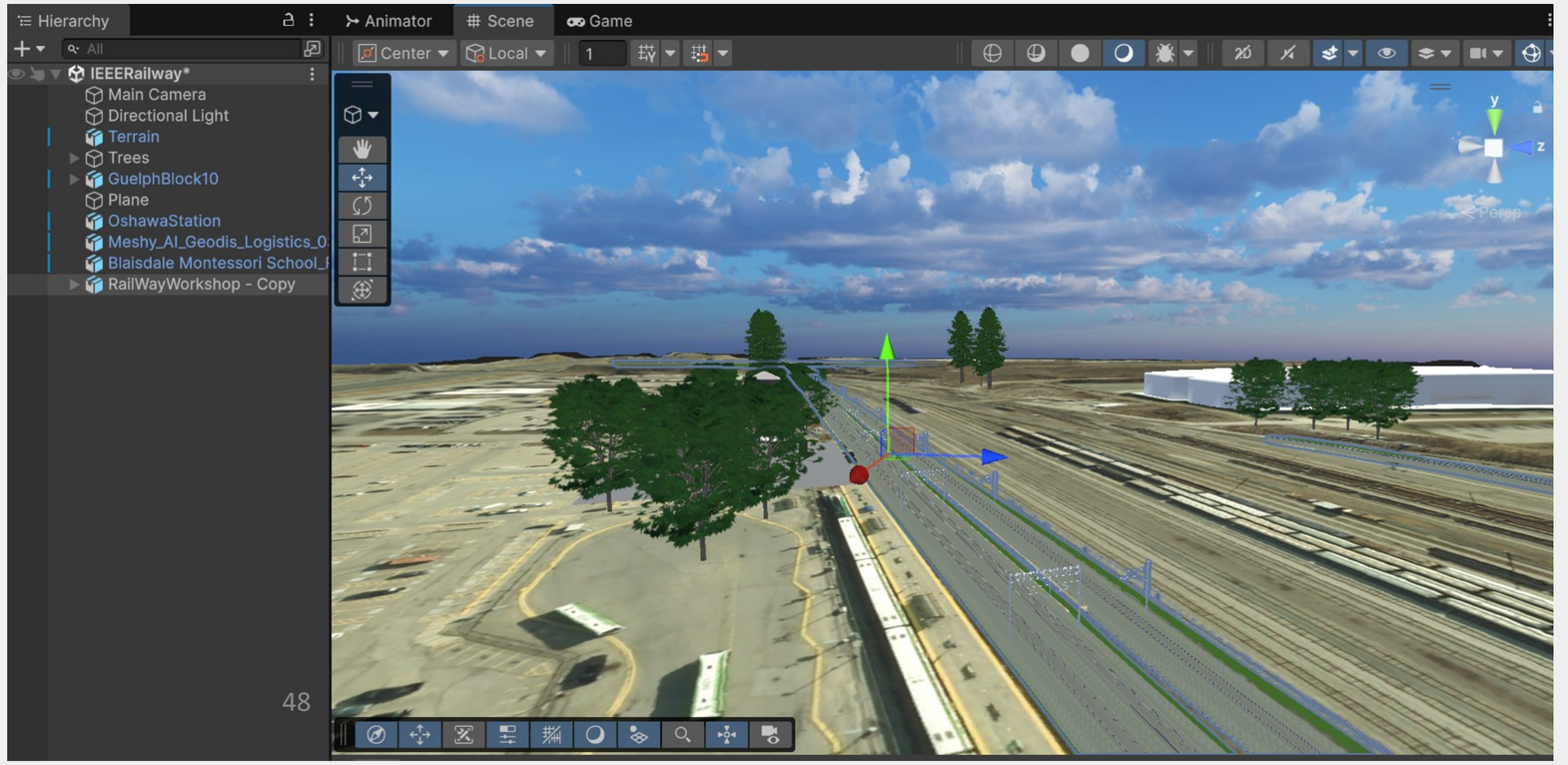
Adding the Railway Track Object

- ❑ In the Project window, go to: Models → Rail
- ❑ Drag and Drop

RailWayWorkshop – Copy.fbx into Hierarchy

- ❑ In the Inspector window, set the Position values to: X = 0, Y = 0, Z = 0
- ❑ Observe how the railway object aligns with the rail corridor on the terrain.
- ❑ This railway object was created in a highly specialized software (OpenRail) which we teach in Graduate degree (SEP 703 – Railroad Track)
- ❑ You see how different software and pieces can connect to each other





Adding Wagons in Empty Yard

- ❑ Find Wagon Models in Assets
- ❑ Drag and Drop into Hierarchy and adjust the positions and rotations

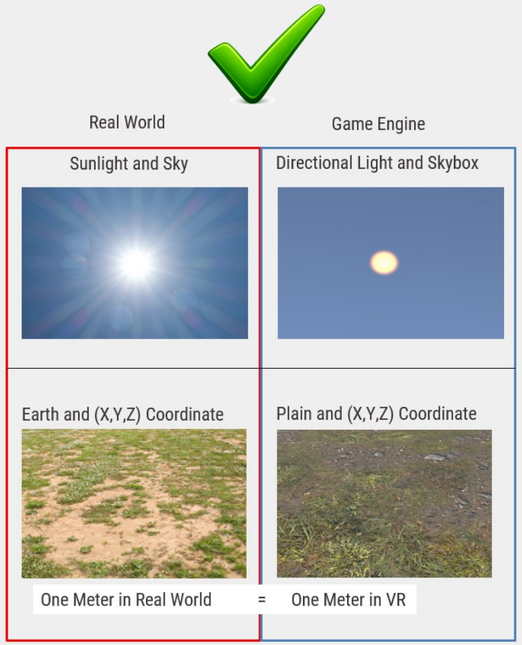
Deliverable 1

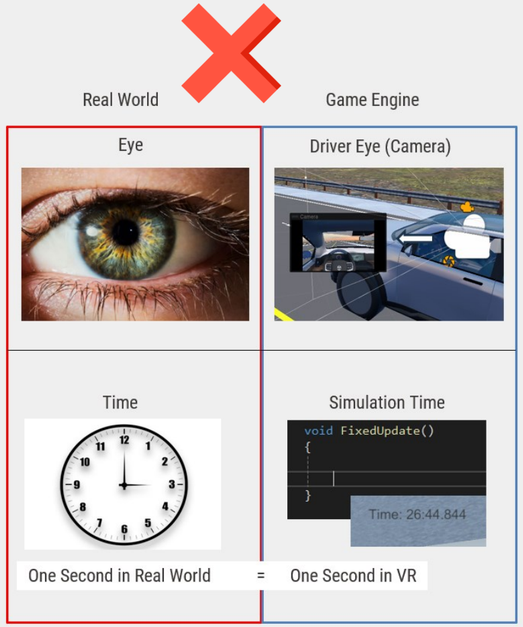
- ❑ Take a Screenshot from Your Built Environment and Save it in a word file
- ❑ Your environment should be different than below image (layout of trees, wagons, buildings, positions of clouds, time of day, etc)

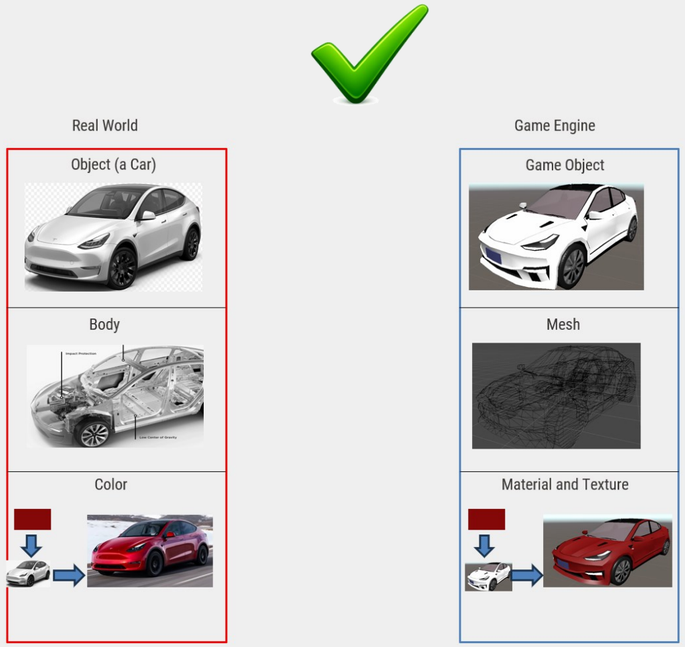


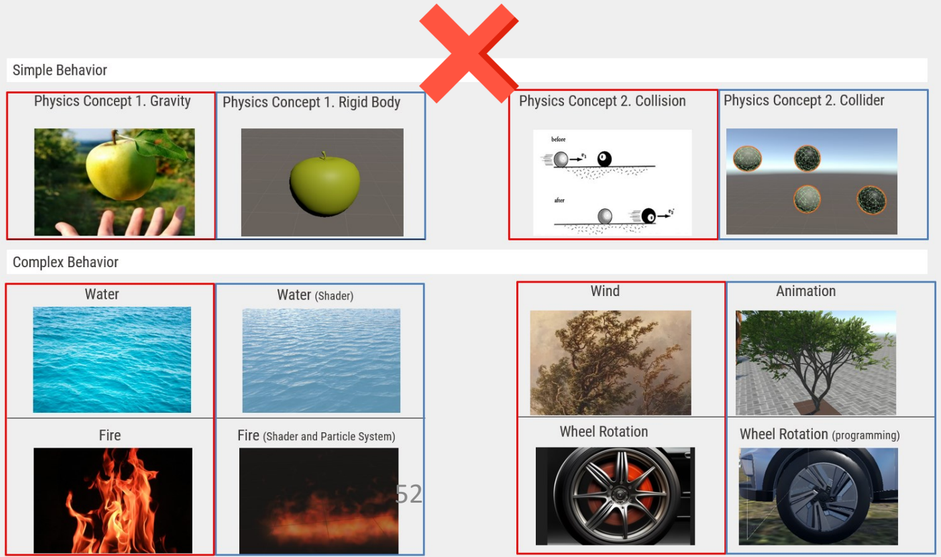
Beginning of Milestones 2: Camera and Simple Behavior

Real World vs. Unity Game Engine









✅ **Covered so far**

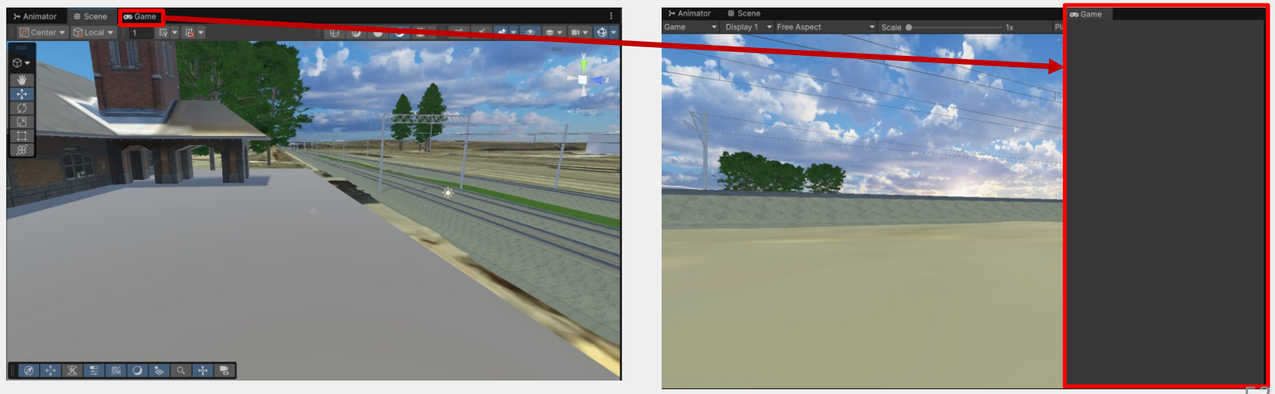
❌ **Coming next / not covered yet**

Camera and Game View in Unity

Goal: View the Scene view and Game view side by side.

❑ Drag and drop the Game tab to the right side of the Scene view.

Drag and Drop

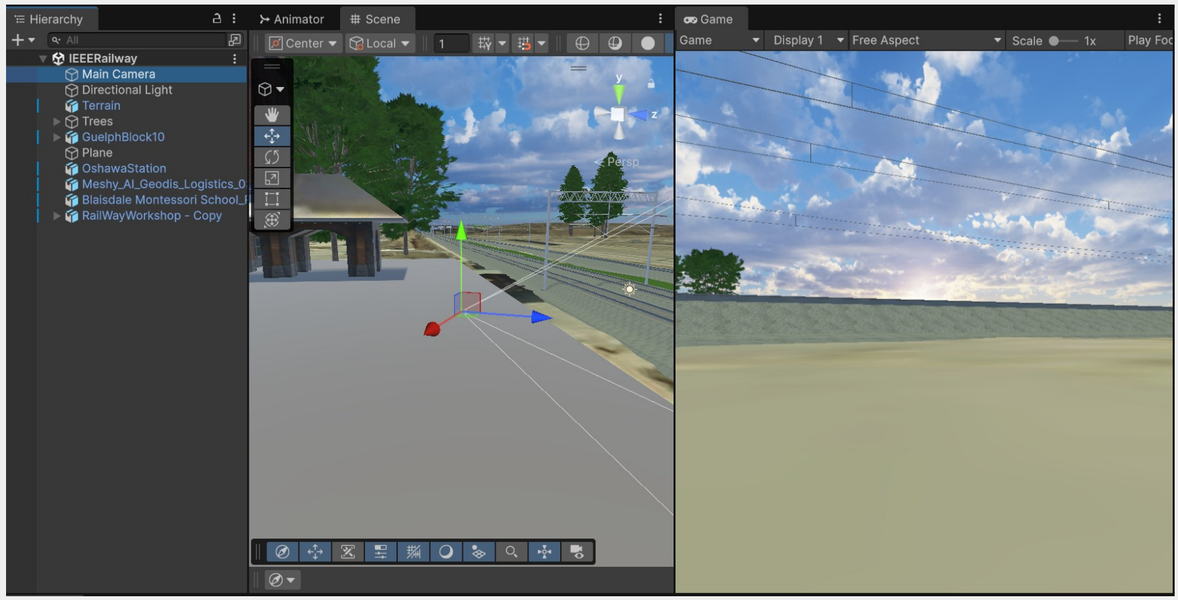


❑ Select Main Camera in the Hierarchy.

❑ The Game view shows what the camera sees.

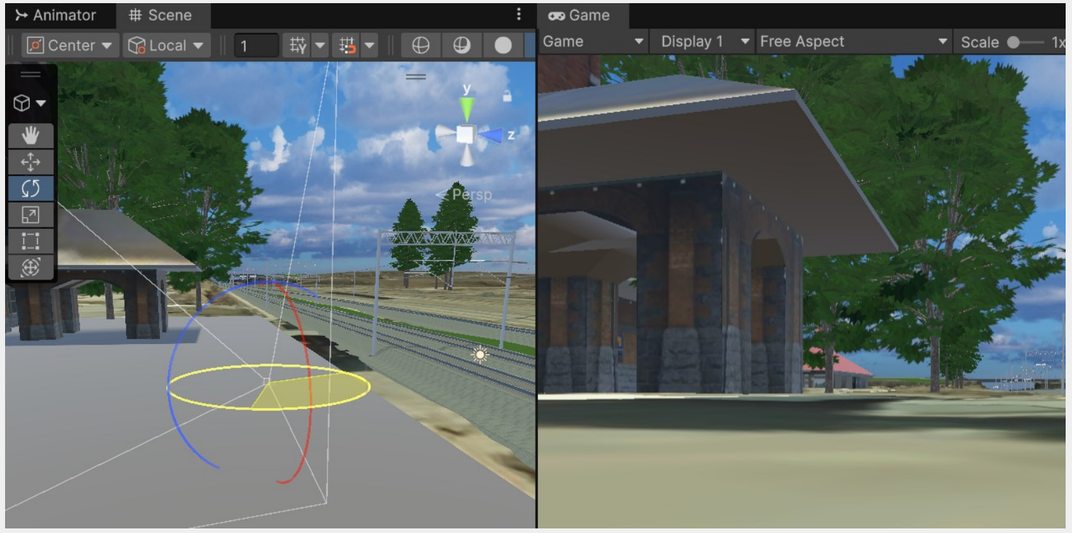
❑ In a driving or VR scene, the camera represents the viewer's or driver's eyes.

❑ Next, we will adjust the camera position and view direction.

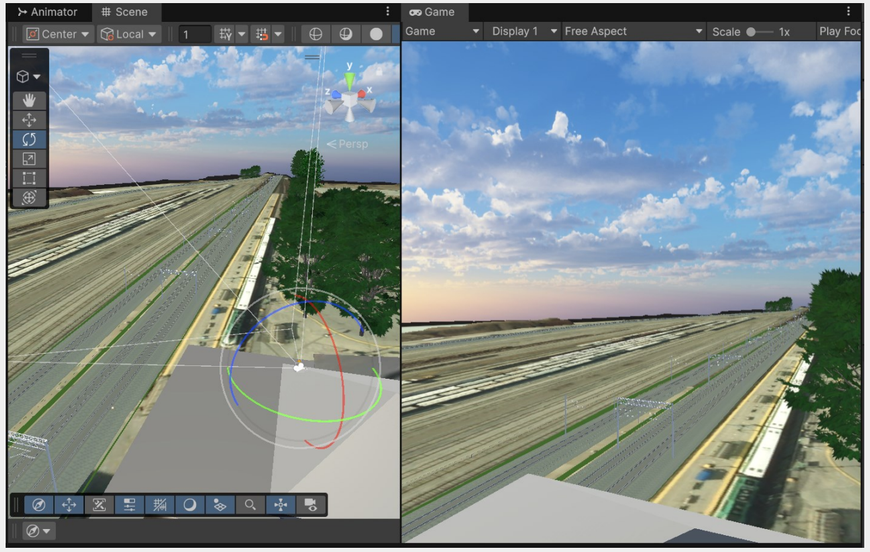


Camera View and Movement in Unity

- ❑ Select Main Camera in the Hierarchy → Press F
- ❑ Rotate the camera toward the station building.
- ❑ The Game view shows what the camera sees.



- ❑ In a driving or VR scene, the camera represents the viewer's or driver's eyes.
- ❑ In games, movement is often created by moving or rotating the camera over time.
- ❑ Exercise: Place the camera like a security camera overlooking the railway station.
- ❑ Save and Press the Play button. What do you notice?



Simple Behavior:

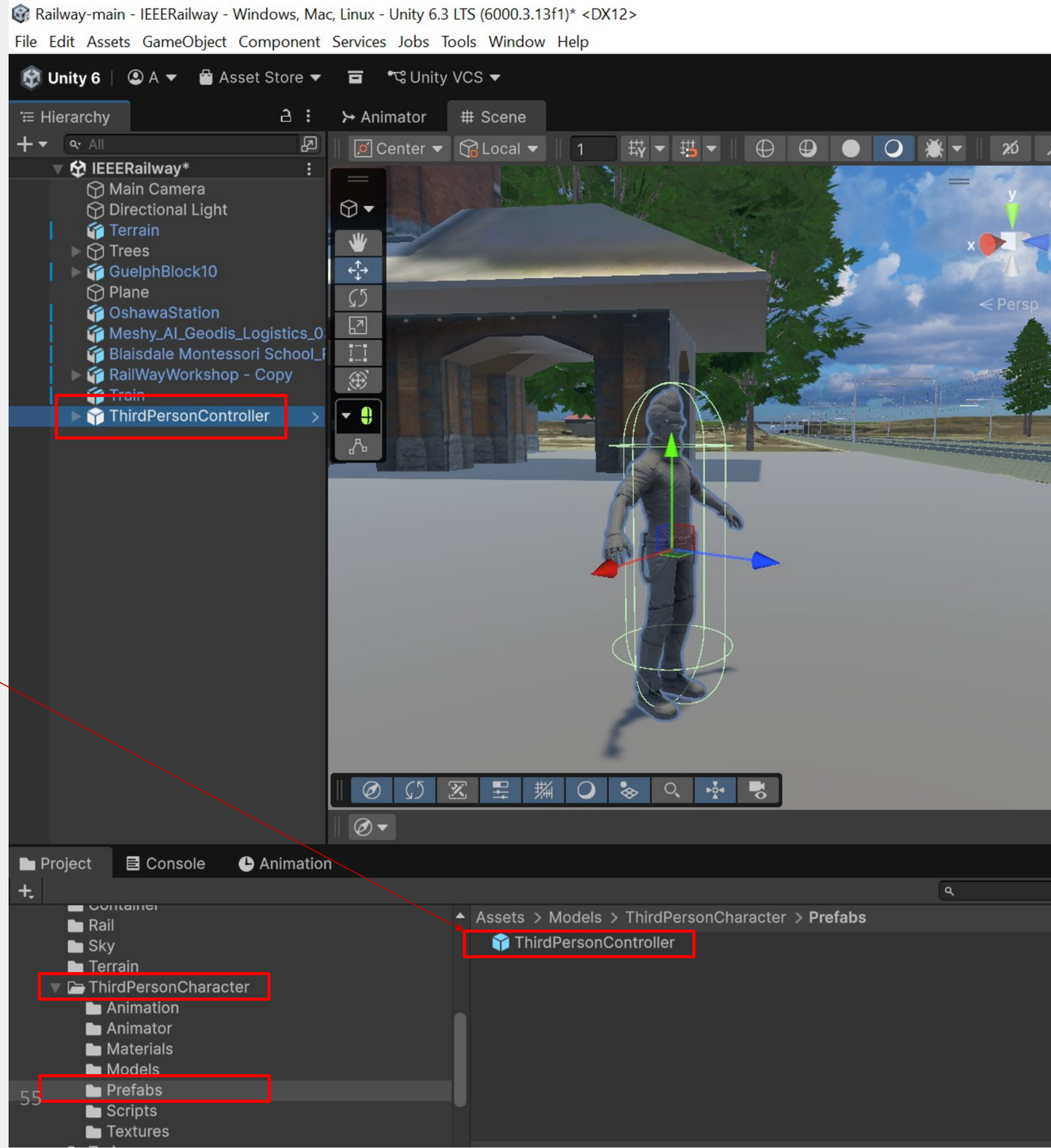
Third-Person Controller

Goal: Add a Third-Person Controller character to the scene.

- ❑ Find “ThirdPersonController.fbx”
- ❑ Drag and drop into hierarchy
- ❑ Adjust position and rotation as image
- ❑ Click the Play button.

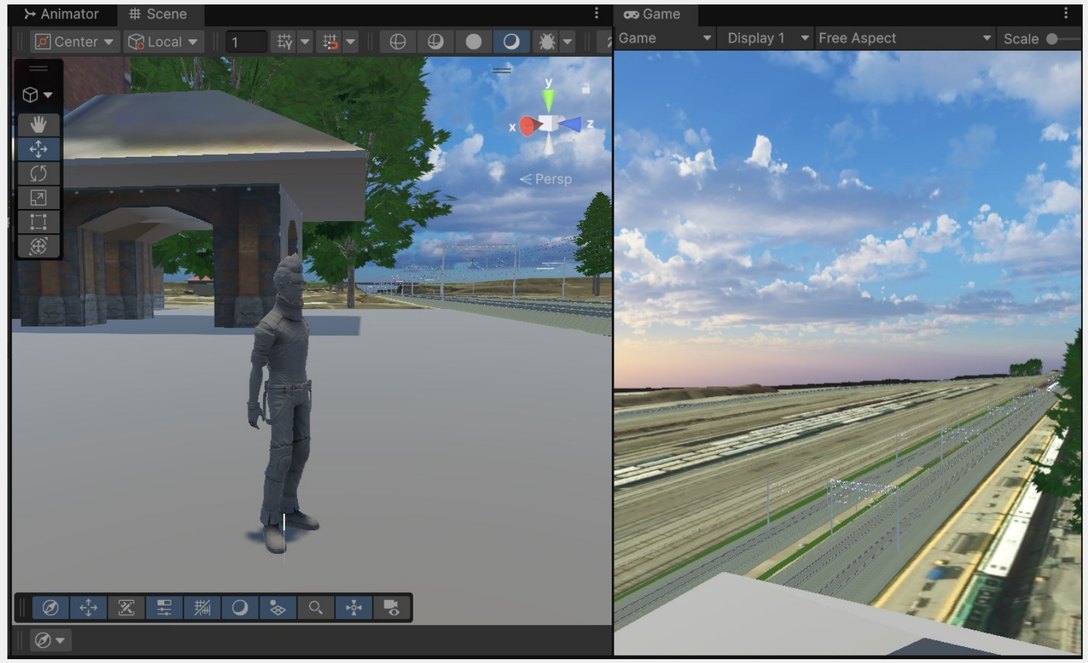
Think & Observe - No submission required

- ❑ What do you see in the Game view?

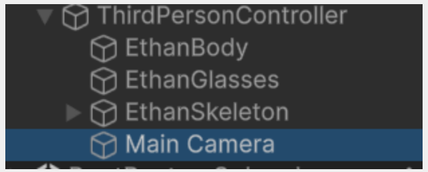


Simple Behavior:
Character Movement and Camera Follow

- ❑ While still in play mode → Click inside the Game view → Move the character using the W key on the keyboard.
- ❑ You can also try A, S, and D for movement.

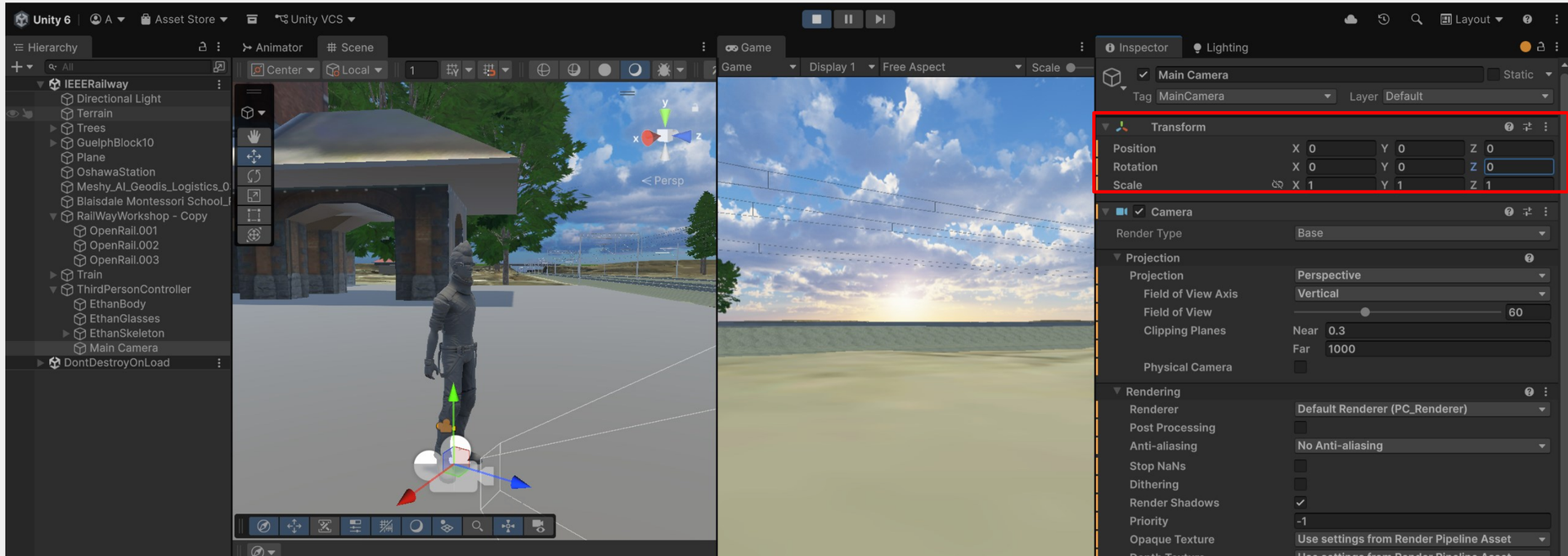


- ❑ Now Stop Play Mode
- ❑ Drag and drop Main Camera under ThirdPersonController in the Hierarchy.



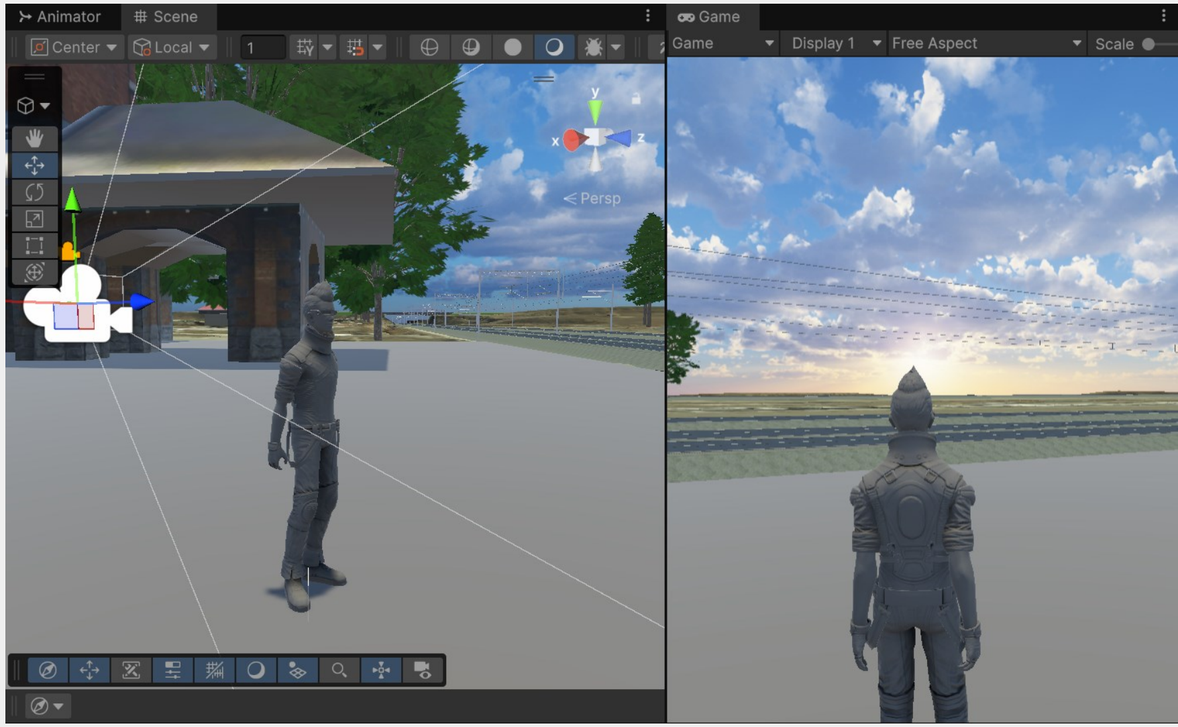
Simple Behavior:
Character Movement and Camera Follow

- ❑ Select Main Camera under ThirdPersonController.
- ❑ In the Inspector window, set Position and Rotation values to:X = 0, Y = 0, Z = 0
- ❑ Observe that the Main Camera moves to the character's origin point, near the feet.
- ❑ Move the camera behind the character.
- ❑ Adjust the camera height and rotation so it looks forward from behind the character.
- ❑ Press Play and observe the Game view → See Next Slide



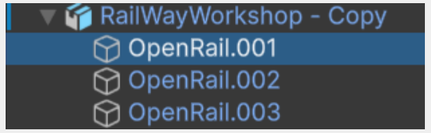
Simple Behavior:
Character Movement and Camera Follow

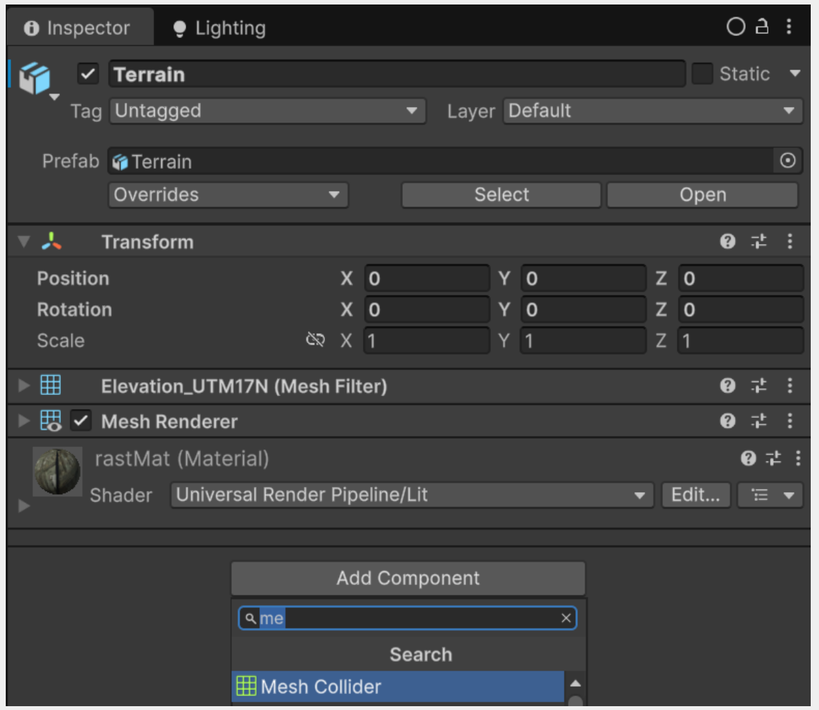
- ❑ Click inside the Game view.
- ❑ Move the character using W, A, S, and D.
- ❑ Press Space to jump.
- ❑ Exit Play Mode.
- ❑ Press Play again.
- ❑ **Remember: Always do the changes in Edit Mode -** Changes made during Play Mode are temporary and are not saved in the scene.



Simple Behavior: Adding Colliders

- ❑ In Play Mode, move the character onto the railway/terrain.
- ❑ Why does the character fall through the platform or rails?
- ❑ Because the terrain and rail objects need physics collision behavior.
- ❑ Without a Collider, they are only visible 3D objects and the character cannot stand on them.
- ❑ Select Terrain in the Hierarchy→ in inspector → Add Component → Mesh Collider
- ❑ Add a Mesh Collider to each rail object as well:
- ❑ To all OpenRail.001, OpenRail.002, OpenRail.003





- ❑ **Mesh Collider:** A Mesh Collider allows Unity to detect collision between the character and the 3D object, so the character can stand on the surface instead of falling through it.

Simple Behavior:

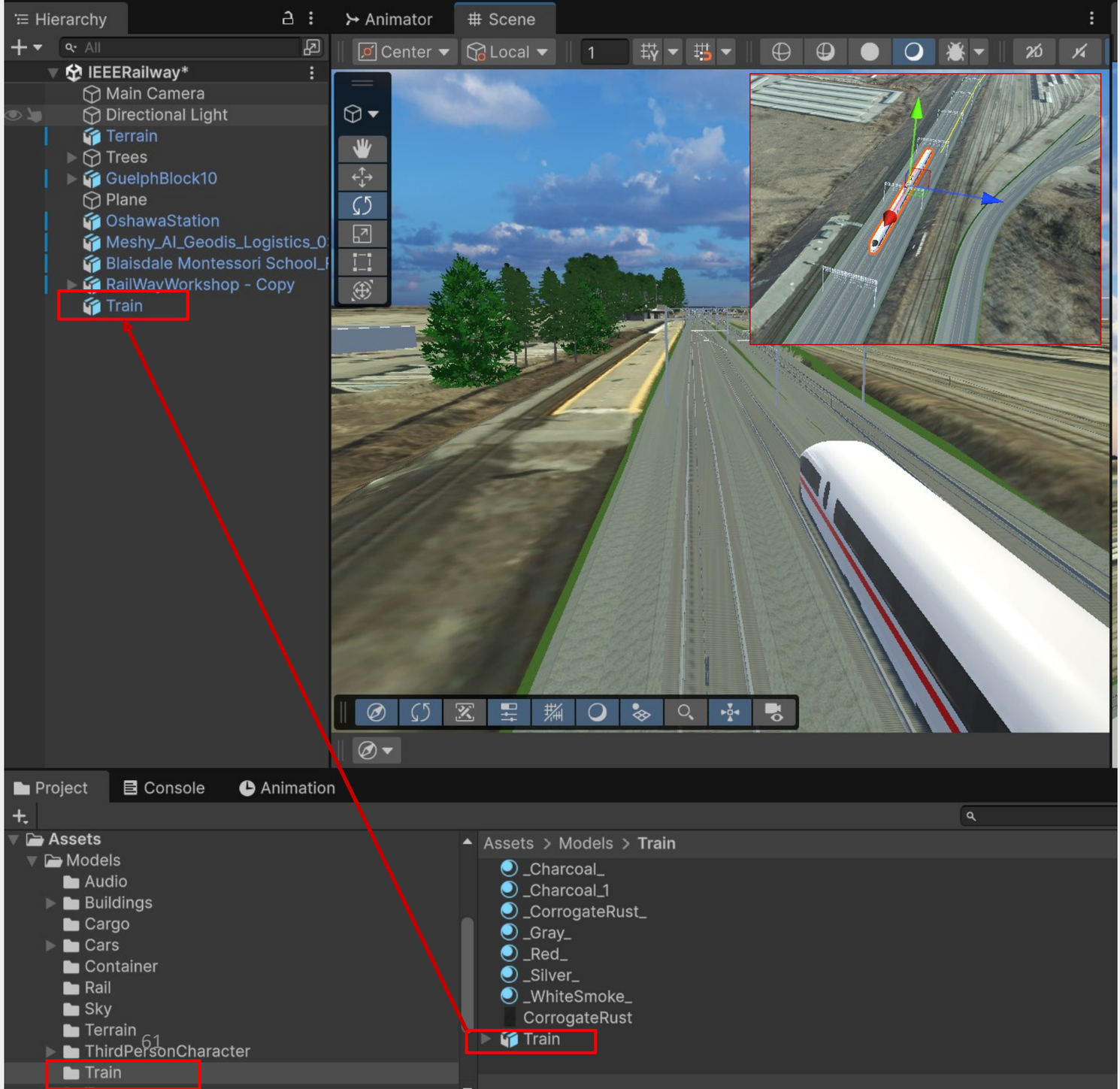
Remove the Character and Position the Camera

- **Detach the Main Camera:** Drag and drop the Main Camera from ThirdPersonController into the root of the Hierarchy.
- **Remove the Character:** Delete the ThirdPersonController GameObject.
- **Position the Camera:** Adjust the Main Camera's position and rotation so that it faces the railway tracks.

Simple Behavior:

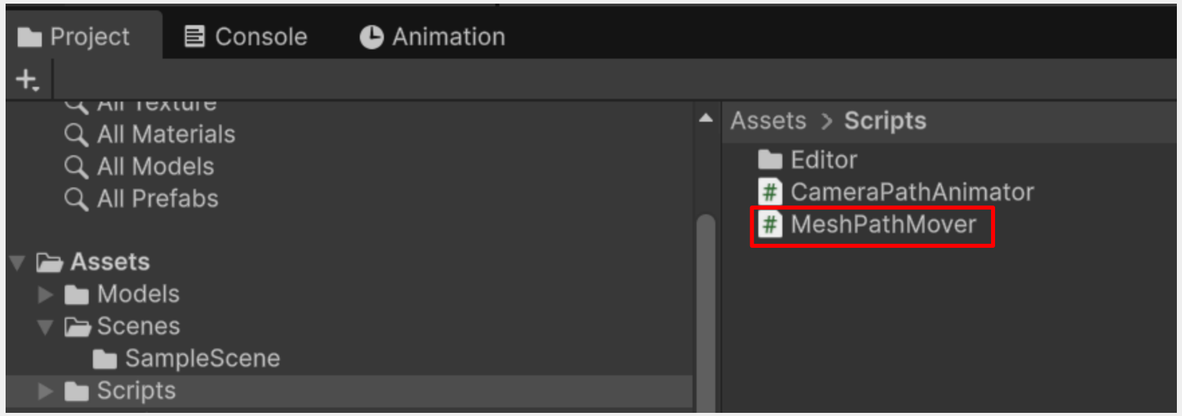
Adding a Train Object

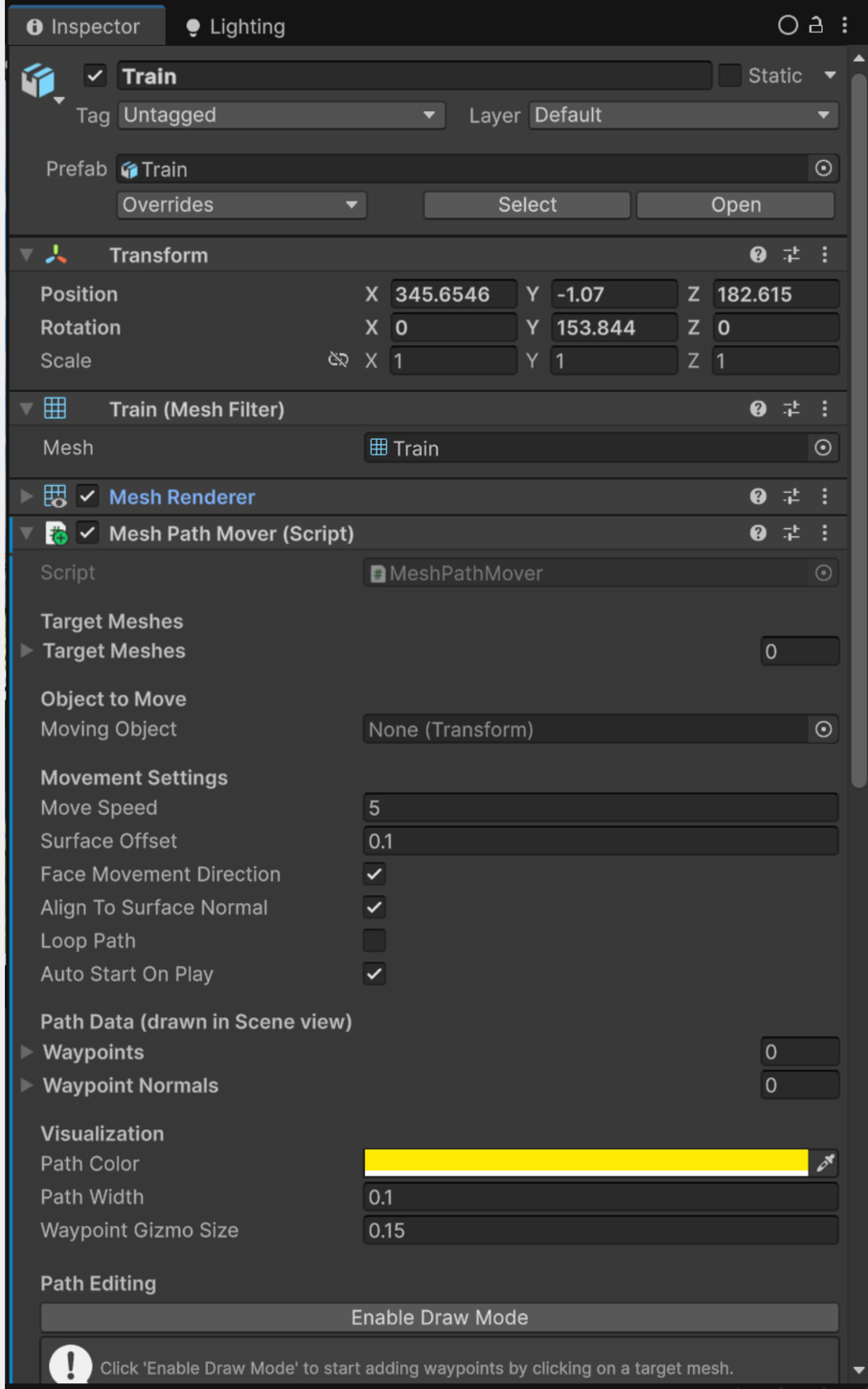
- ❑ In the Project window, go to: Models → Train
- ❑ Drag and drop Train.fbx into hierarchy
- ❑ Adjust the position and rotation of Train accurately on top of second rail at the beginning of the corridor exactly where I added in the red box
- ❑ Check the Scene view to confirm that the train is aligned with the rails.



Simple Behavior: Adding Train Movement Script

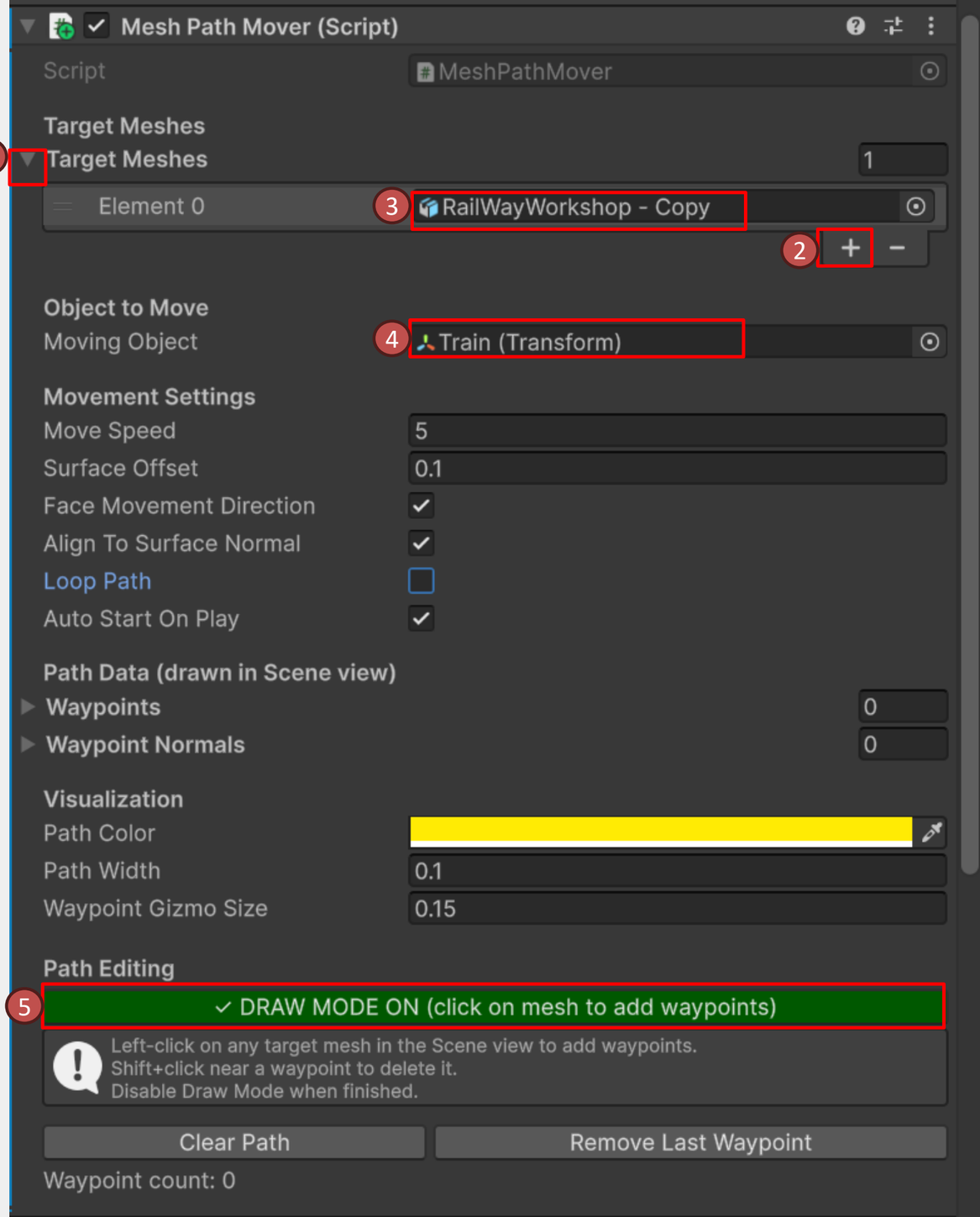
- Select the Train object in the Hierarchy
- Drag and drop the MeshPathMover.cs script into the Inspector window of the Train object
- This C# script gives movement behavior to the train.
- After adding the script, new movement settings appear in the Inspector.
- Next, we will define the train path and movement settings.



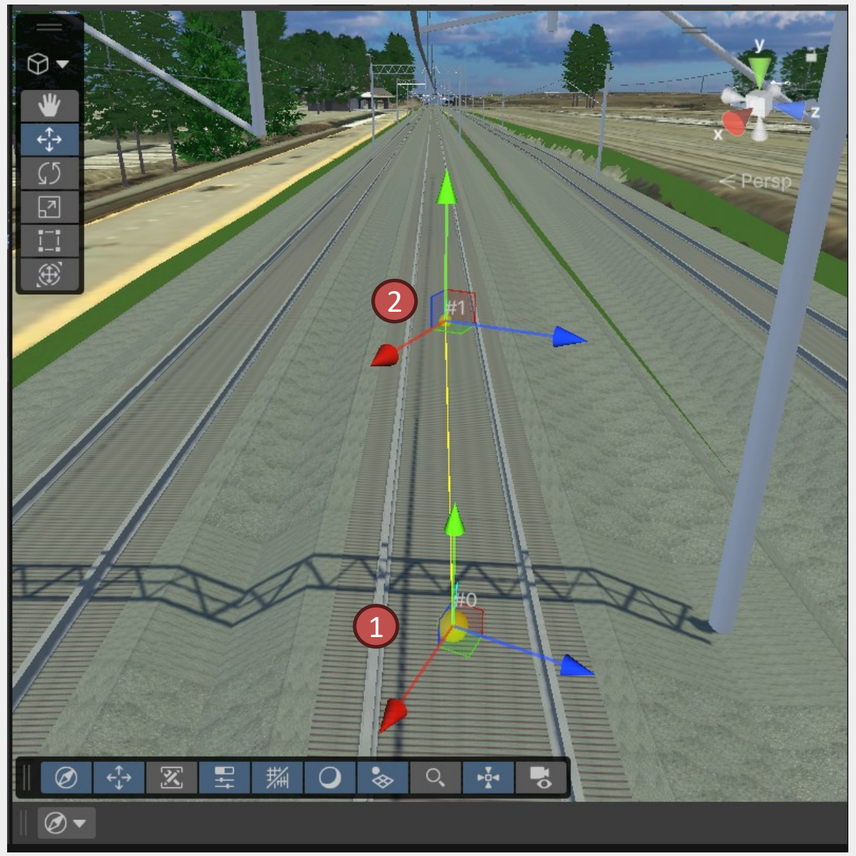


Simple Behavior: Setting the Train Movement Path

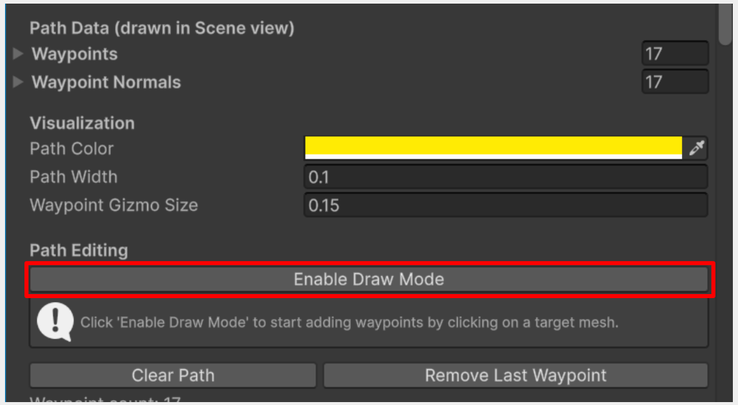
- ❑ Select the small drop-down arrow near target mesh
- ❑ Click the “+” button under Target Meshes
- ❑ **Question:** On which object should the train move?
- ❑ **Answer:** Drag and drop RailwayWorkshop - Copy from the Hierarchy into the Target Meshes slot.
- ❑ **Question:** Which object should move?
- ❑ **Answer:** Drag and drop the Train object from the Hierarchy into the Moving Object slot.
- ❑ **Question:** From where to where should the train move?
- ❑ **Answer**: Click Enable Draw Mode and add waypoints on the rail.



Simple Behavior: Drawing the Train Path



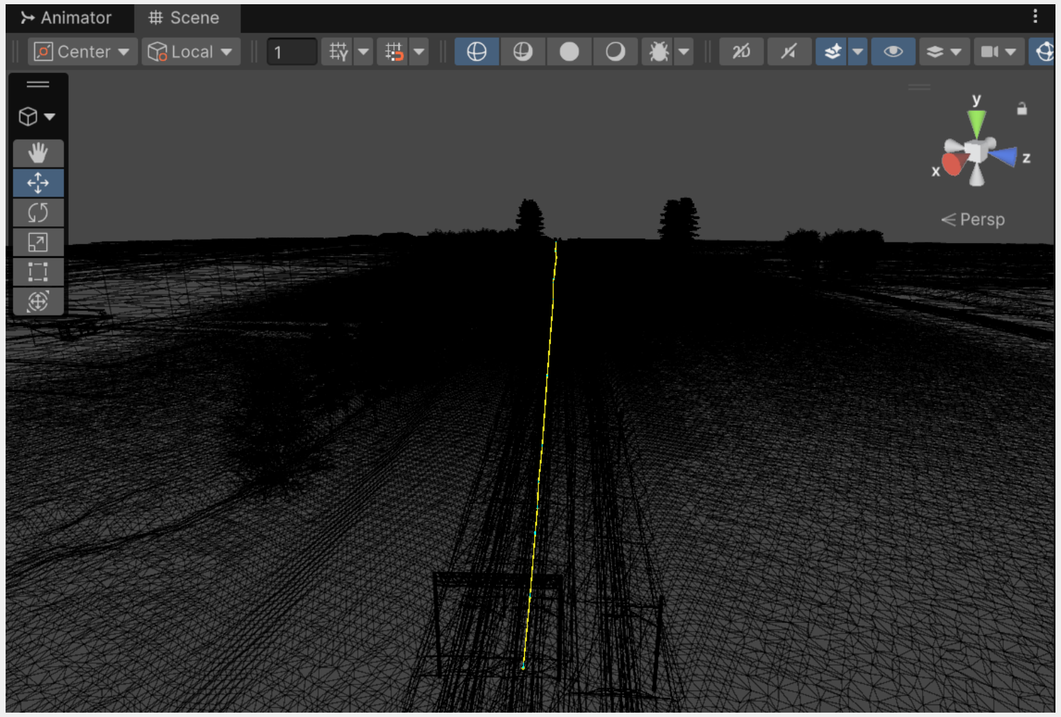
- In Scene view → click the first point where the train should start.
- Click the second point, then the third point, to continue the path.
- Repeat this process along the rail track.
- Continue adding waypoints after the bridge.
- In this example, we added 17 waypoints.
- When finished, click Enable Draw Mode again to turn it off.
- File → Save
- Press Play → The train will move along the rail path.
- You learned how a script can create simple behavior: train movement.



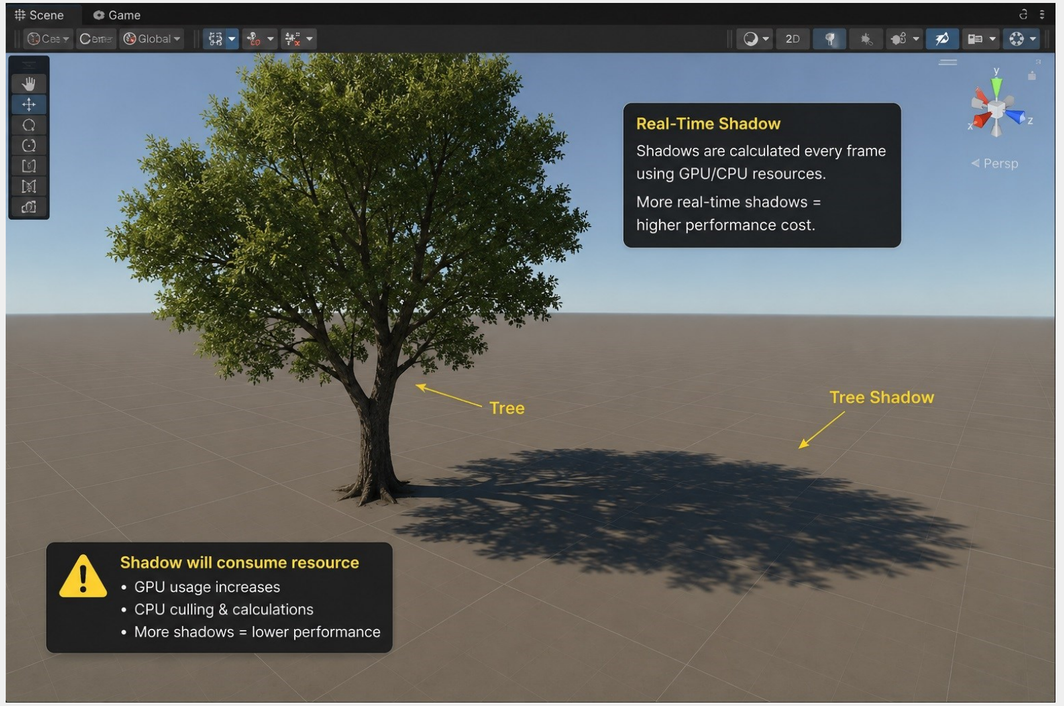
Optimization in Unity

❑ Optimization in a game engine depends on several factors:

❑ Number of Meshes:
More meshes and polygons require more computational resources.



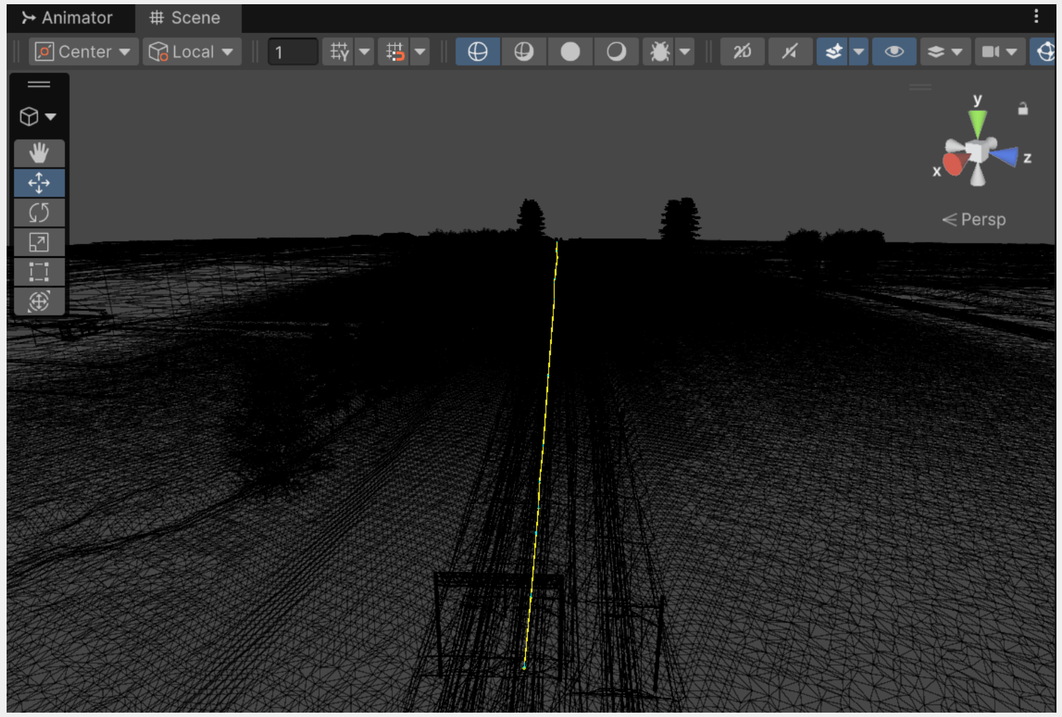
❑ Lighting and Shadows:
Real-time shadows are calculated during rendering and can reduce performance.



Optimization in Unity

❑You already learn how to optimize the meshes from the start using meshy.ai

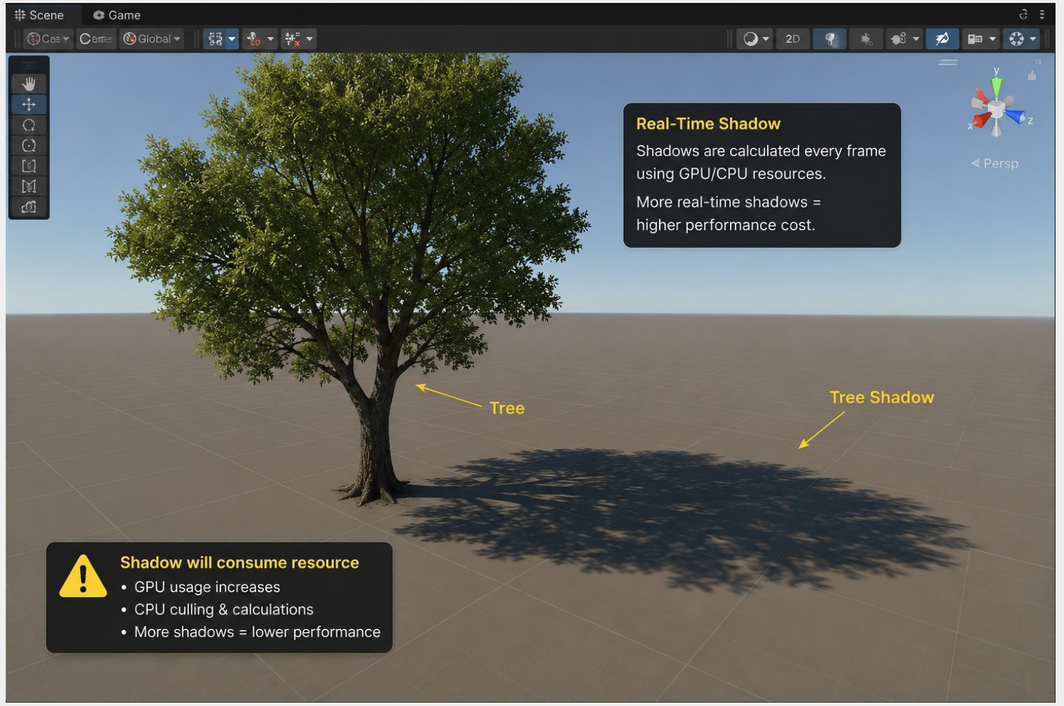
❑ Number of Meshes:
More meshes and polygons require more computational resources.



Optimization in Unity

❑Now you learn this in Unity in next slide

❑ Lighting and Shadows:
Real-time shadows are calculated during rendering and can reduce performance.



Optimization: Checking Scene Statistics

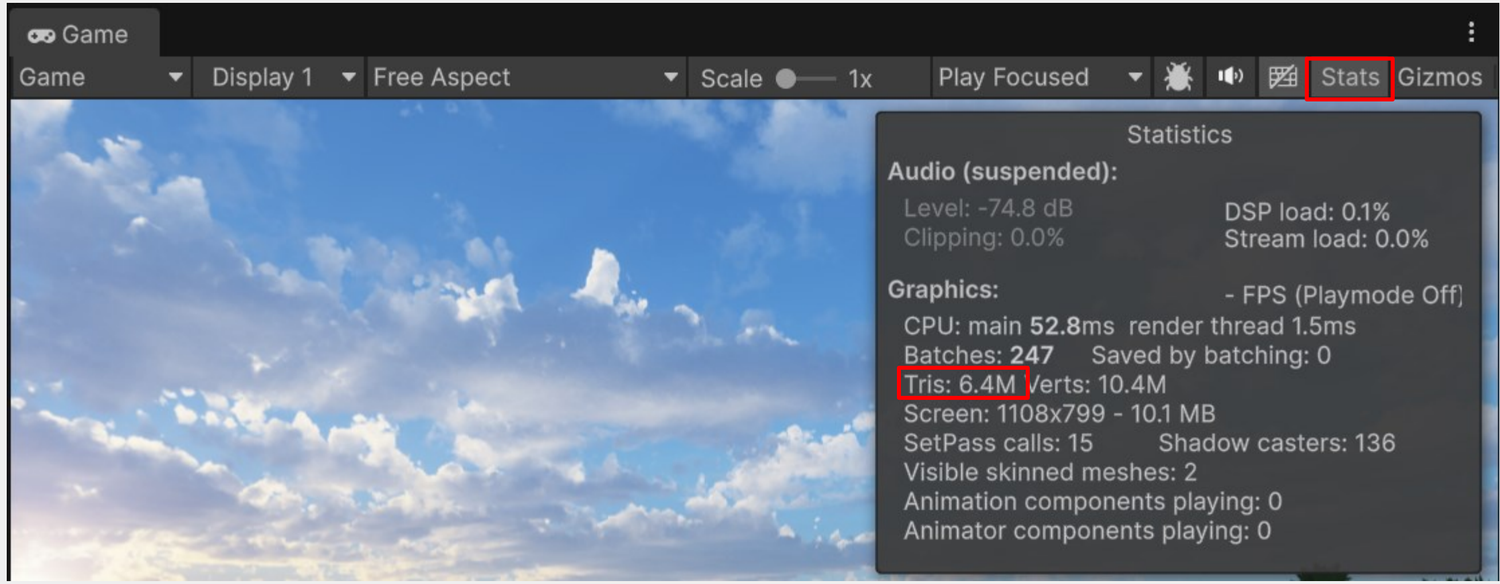
- ❑ Open the Game view.
- ❑ Click Stats to view rendering statistics.
- ❑ Check the number of Tris.

Tris means triangles used to render the scene.

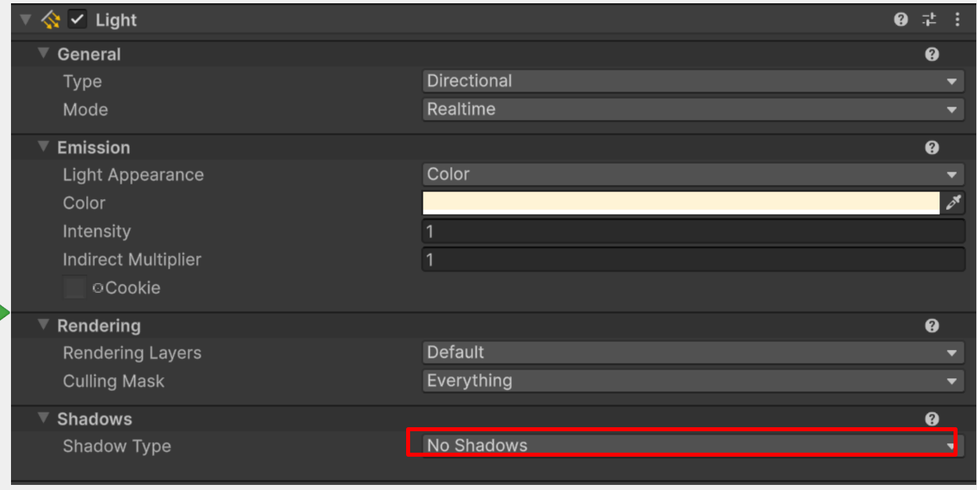
- ❑ For VR/DT, the number of triangles should be kept low to maintain smooth performance.
- ❑ In this example: Tris = 6.4M
- ❑ This is high for VR/DT and should be optimized.

One simple optimization:

- ❑ Select Directional Light → Inspector → Shadows → Shadow Type → No Shadows

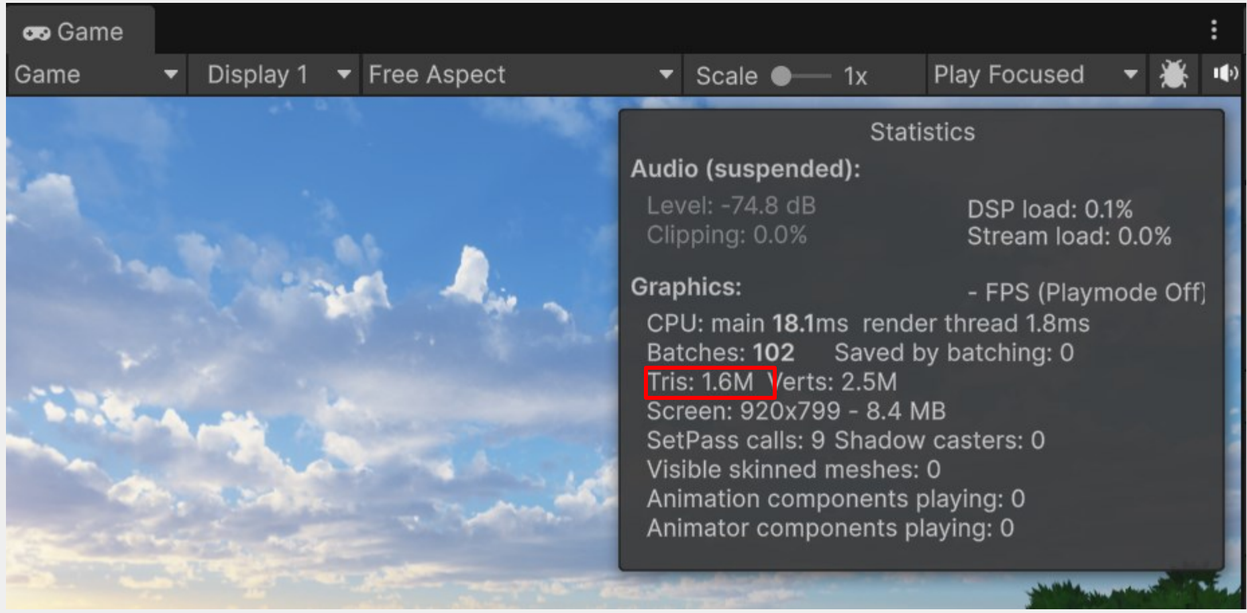






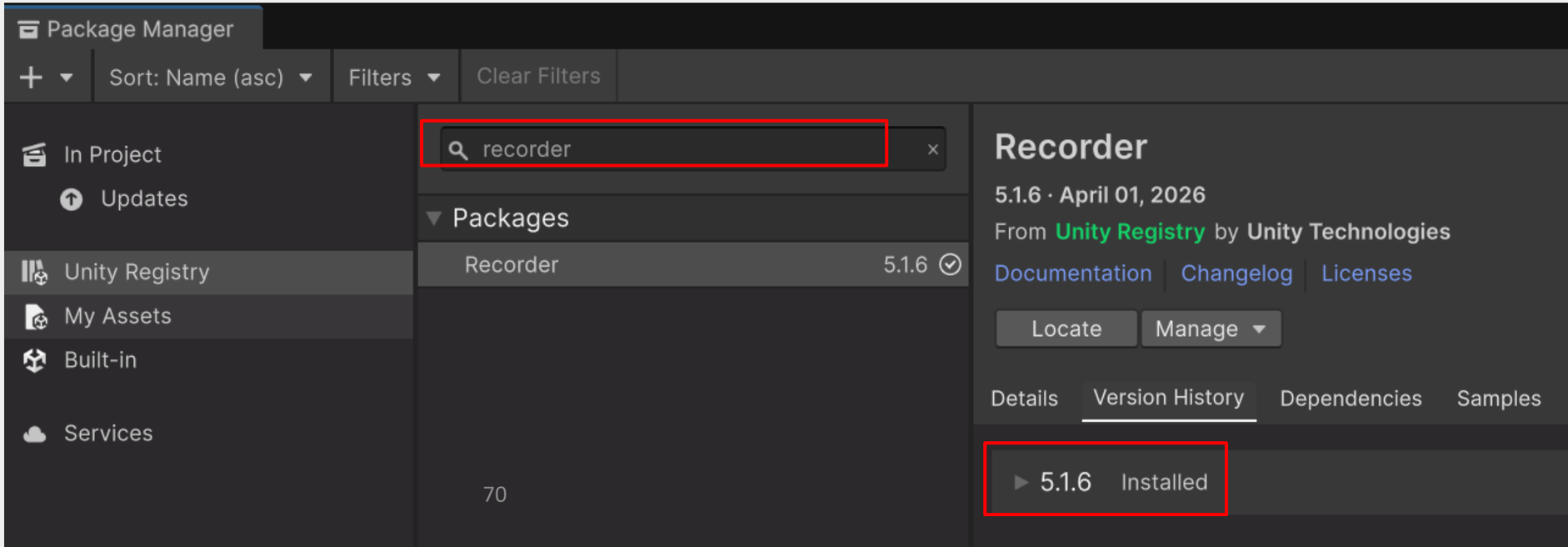
Optimization: Checking the Result

- After optimization, the triangle count is reduced.
- Before: Tris = 6.4M
- After: Tris = 1.6M
- This is a major improvement, but it may still be high for VR/DT.



Installing Unity Recorder

- ❑ Open Package Manager: Window → Package Manager
- ❑ In Package Manager, select Unity Registry.
- ❑ Search for “Recorder”.
- ❑ Select Recorder and click Install: **In this project, it is already installed**.
- ❑ Unity Recorder is an extra package used to record videos, images, or animation outputs from Unity.
- ❑ Package Manager is where we install and manage extra Unity tools and packages.



Recording a 360° Video with Unity Recorder

- ❑ Open the Recorder window:

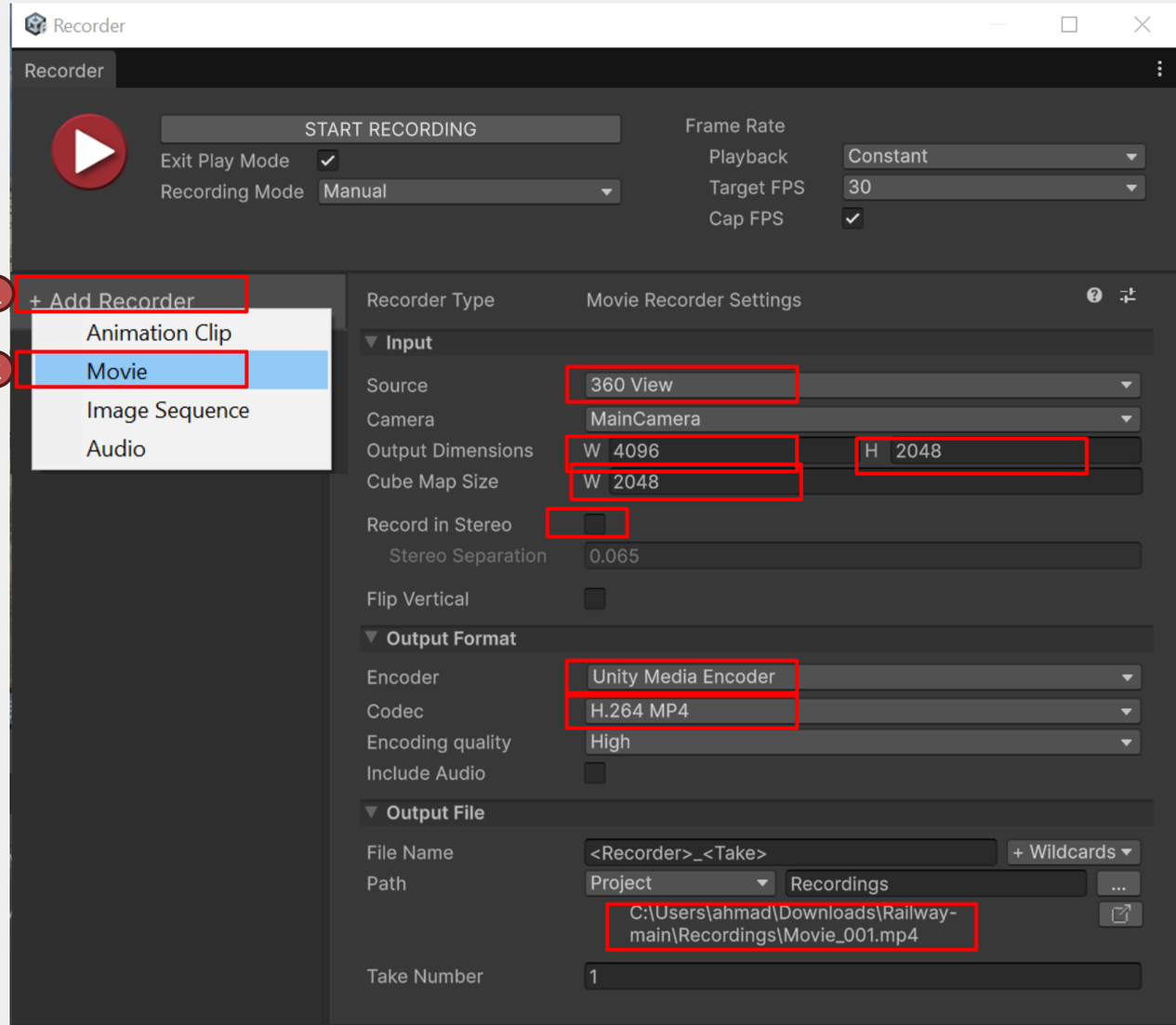
Window → General→ Recorder → Recorder Window

- ❑ Click Add Recorder ⟶ Movie
- ❑ Set the recording settings as showing in image
- ❑ Click Start Recording.
- ❑ Stop the recording after about 30 seconds (900 Frames Per Second).
- ❑ Find the video in the project folder:

…\RailwayVR-main\Recordings\

- ❑ Open the video using VLC Player.

What do you see?



Why Do We Use Spatial Media Metadata Injector?

For 360 videos exported from Unity

1. Export from Unity



360_video.mp4

You export a 360° video from Unity as a standard video file (e.g., MP4).

2. Problem



Before

Looks like a 360 video, but many platforms treat it like a normal flat video.



3. Add Metadata

Spatial Media Metadata Injector



360_video.mp4



Adds metadata = instructions that tell the player this is a spherical 360° video.

4. Result

YouTube VR Player 360 Player

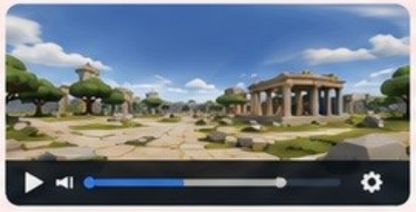
After



Now the platform recognizes it as an interactive 360 video.

The injector does not change the video content. It adds the missing 360° metadata so platforms know how to display it correctly.

Without metadata:
flat stretched video



VS

With metadata:
interactive 360° video



Spatial Meta Data Injector

- ❑ Download Spatial Media Metadata Injector from Google's GitHub release page.
- ❑ Open the release page:

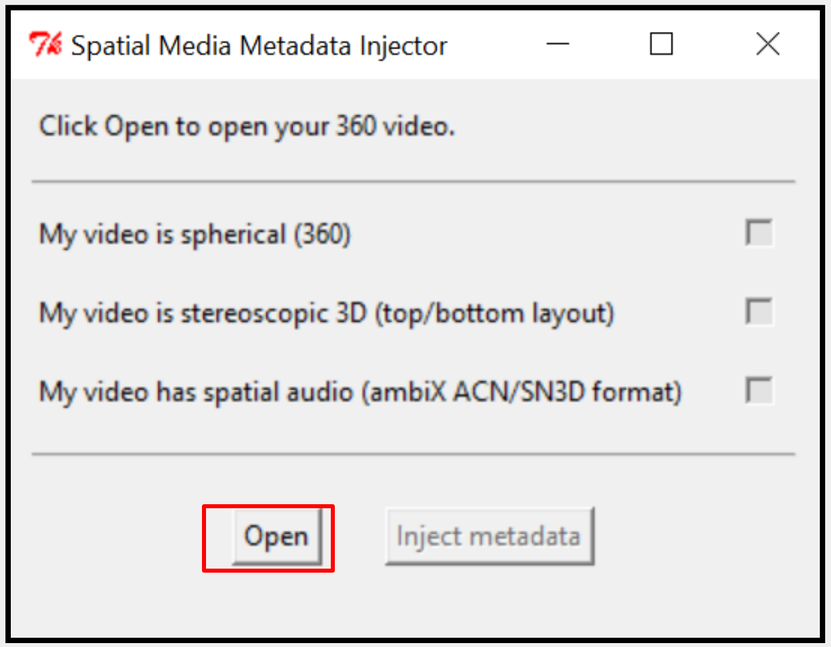
https://github.com/google/spatial-media/releases

- ❑ Download the Windows version: 360.Video.Metadata.Tool.win.zip (You can also download it from Avenue)
- ❑ This tool adds 360° metadata to the exported video.
- ❑ Without this metadata, YouTube or other platforms may treat the video as a normal flat video instead of a 360° video.

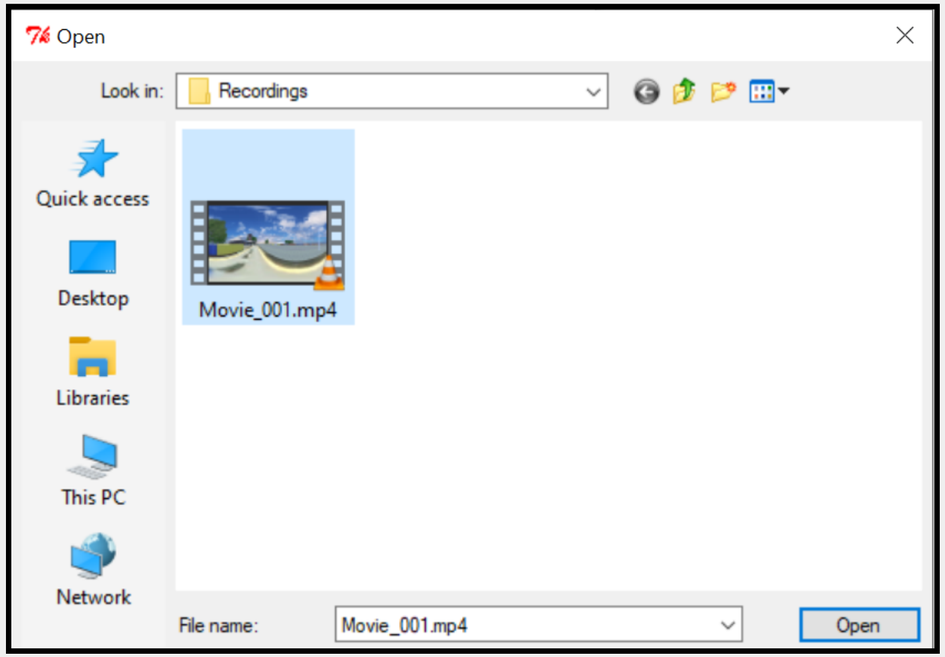


Spatial Meta Data Injector

- ❑ Extract the ZIP file.
- ❑ Open the extracted folder and run the .exe file.
- ❑ Click Open.

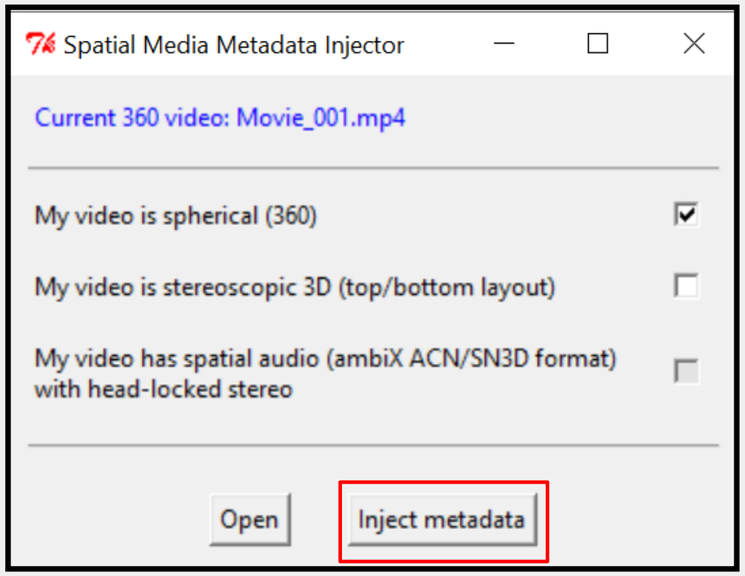


- ❑ Select the recorded 360° video file.
- ❑ Click Open.

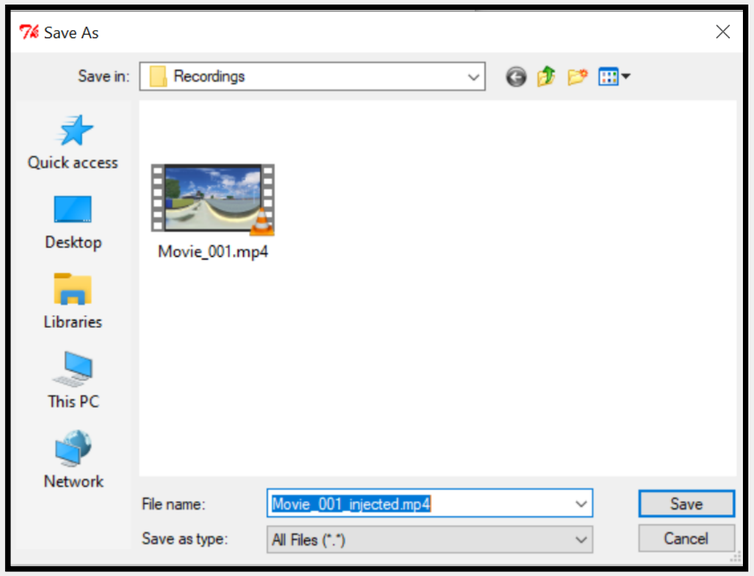


Spatial Meta Data Injector

- ❑ Make sure “My video is spherical (360)” is checked.
- ❑ Click Inject Metadata.

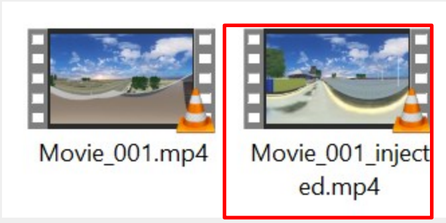


- ❑ Save the new metadata-injected video.



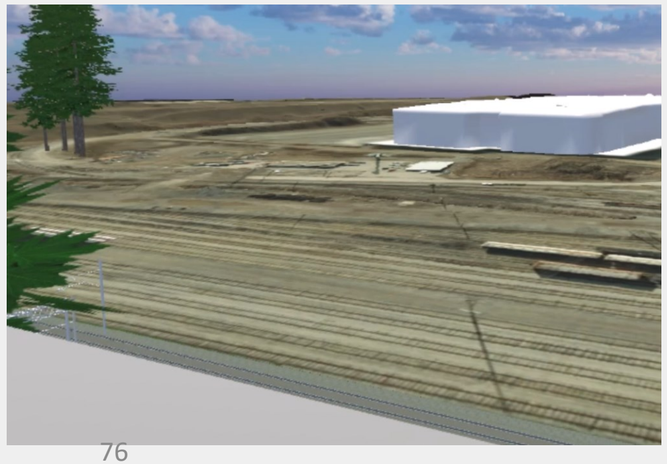
Test the 360° Video

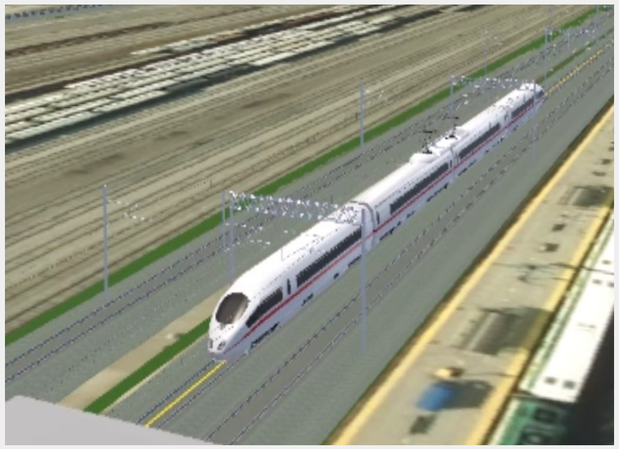
- ❑ Open the metadata-injected video with VLC Player.
- ❑ Select the file ending with:_injected.mp4



- ❑ The video should open as a 360° video.
- ❑ Click and drag with the mouse to look around the scene.
- ❑ **Think & Observe - No submission required**
- ❑ Can you rotate the view and see different directions?

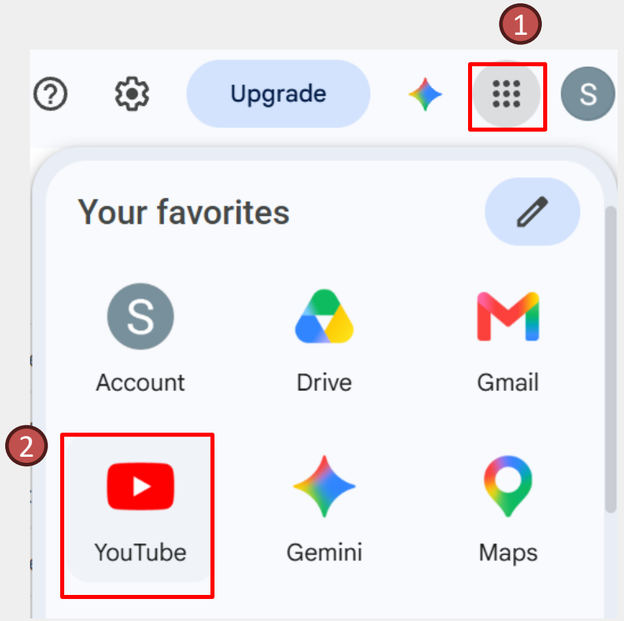
- ❑ **Optional:** Open the injected 360° video in a VR headset to experience the scene immersivity.

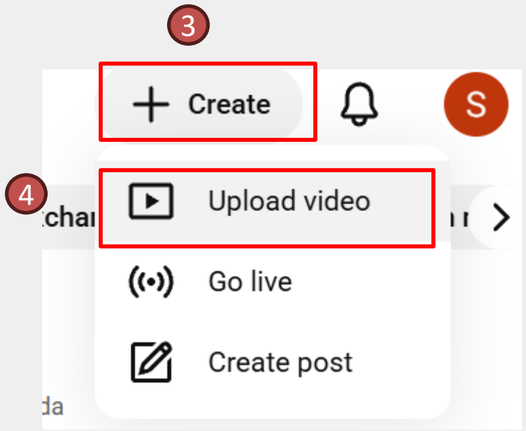




Deliverable 2

- Sing in into your Gmail or create a Gmail account
- Follow the steps as shown
- Upload the Video in YouTube
- Check if you can freely rotate your head in the uploaded video
- **Store the link in the same word file and Submit it**





Deliverable 3

- ❑ Download your developed Train in the previous lab or the trains you developed for the group project
- ❑ Replace the existing train with your developed train
- ❑ Add wagons to it and move it
- ❑ Record the 360-Degree Virtual Reality Video
- ❑ Upload in Youtube and Store the link in the same word file

Unity Installation

Unity Installation

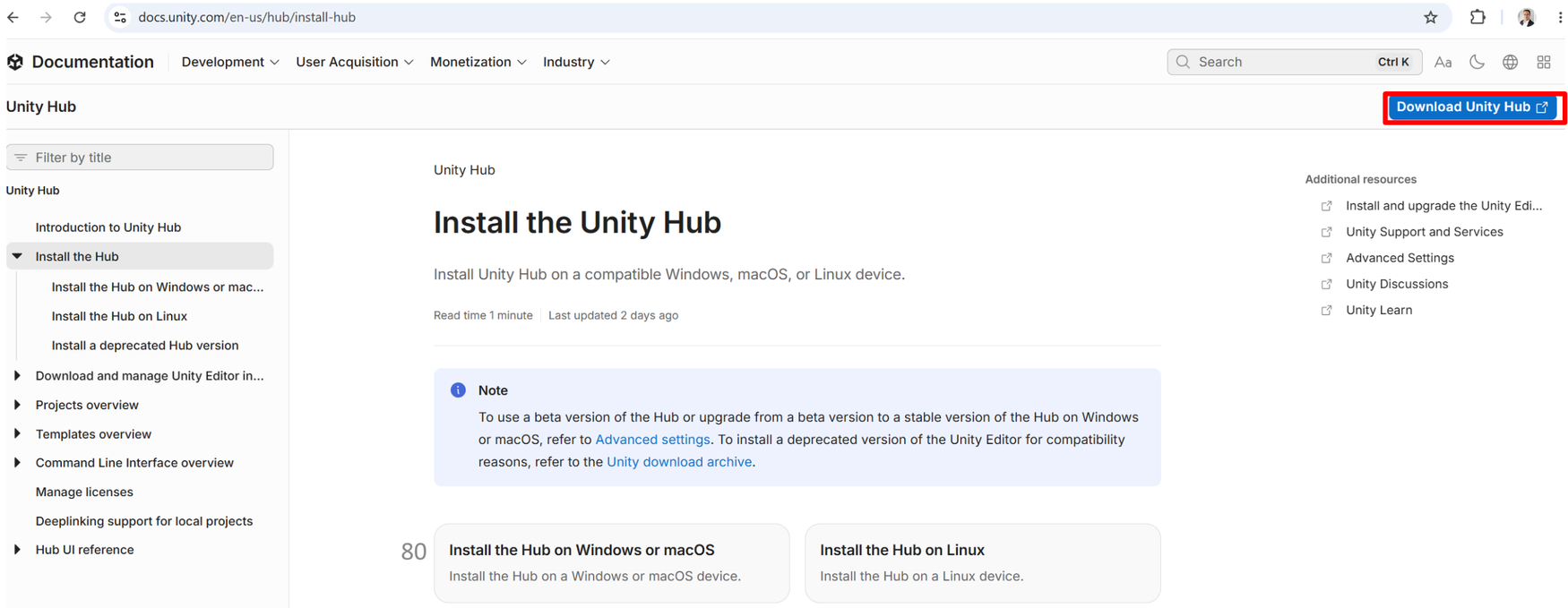
Task: Install Unity Hub

❑ Open The Following Link from Unity official website

https://docs.unity.com/en-us/hub/install-hub

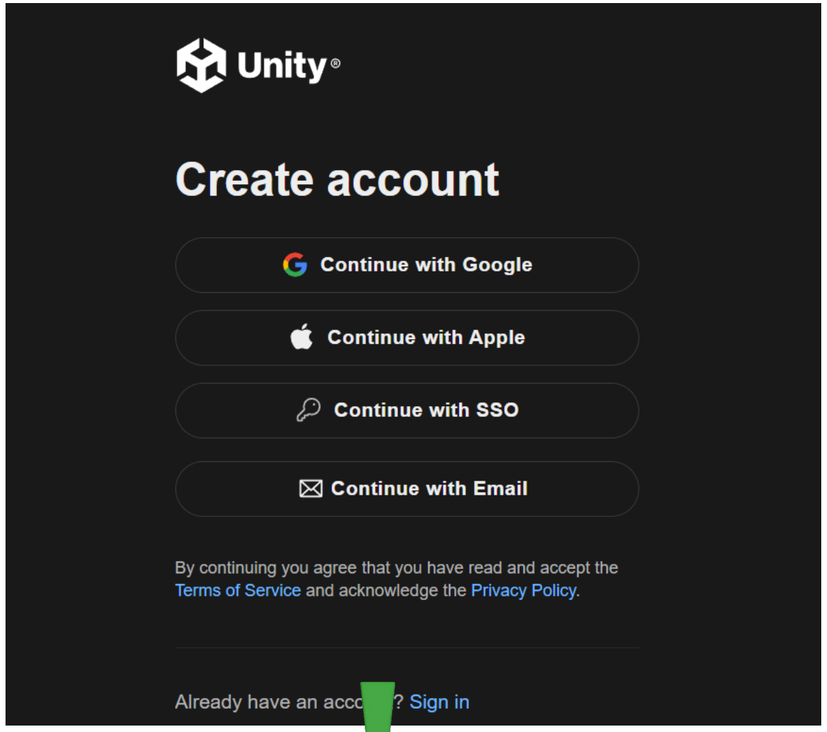
❑ Download Unity Hub

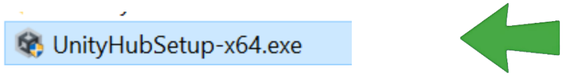
❑ Install Version “6000.3.13f1

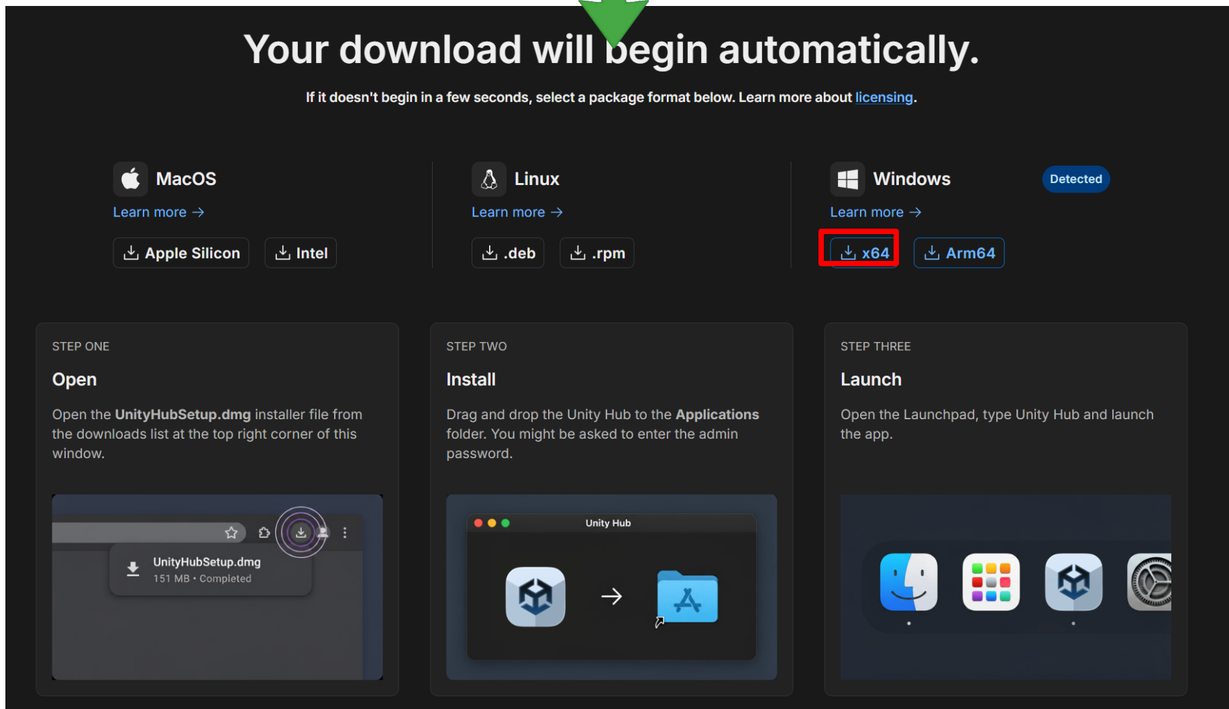


Unity Installation

- ❑ Create an account using your email
- ❑ Download Unity Hub for Windows
- ❑ Run .exe and Install Unity Hub



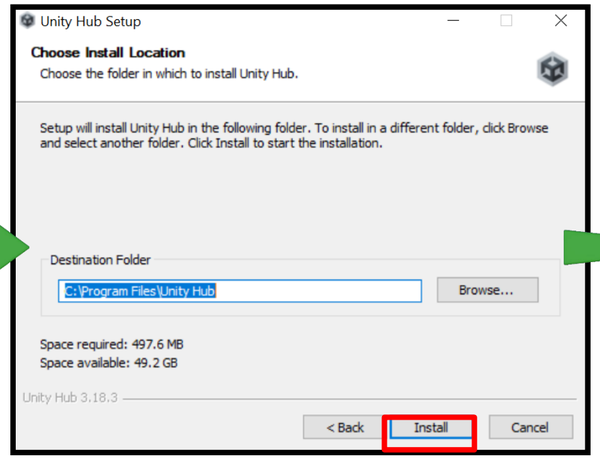


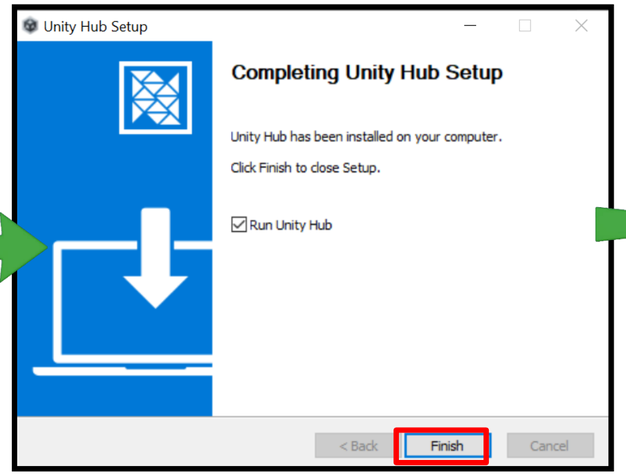


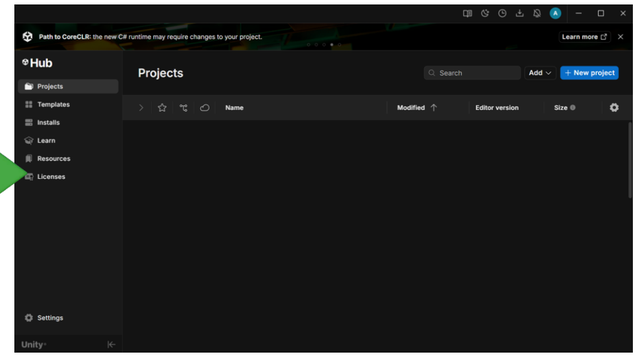
Unity Installation

❑ Install Unity Hub by following below steps









Unity Installation

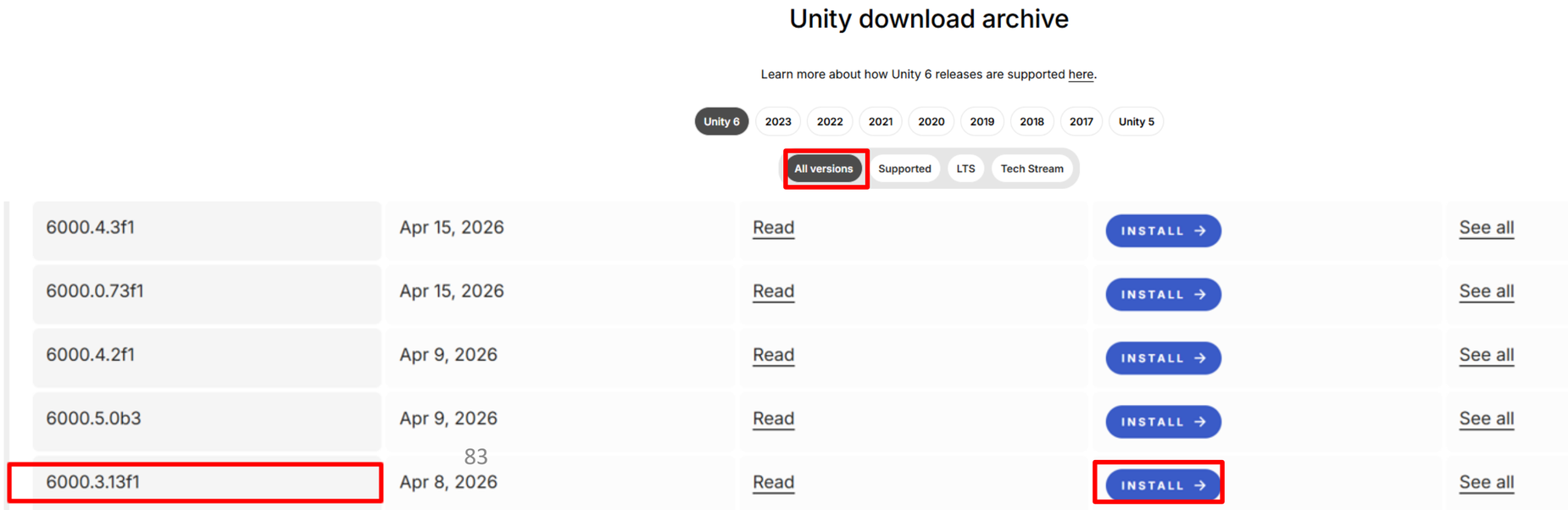
Task: Install Unity version 6000.3.13f1

❑ Open The Following Link from Unity official website

https://unity.com/releases/editor/archive

❑ Change it to “All Versions”

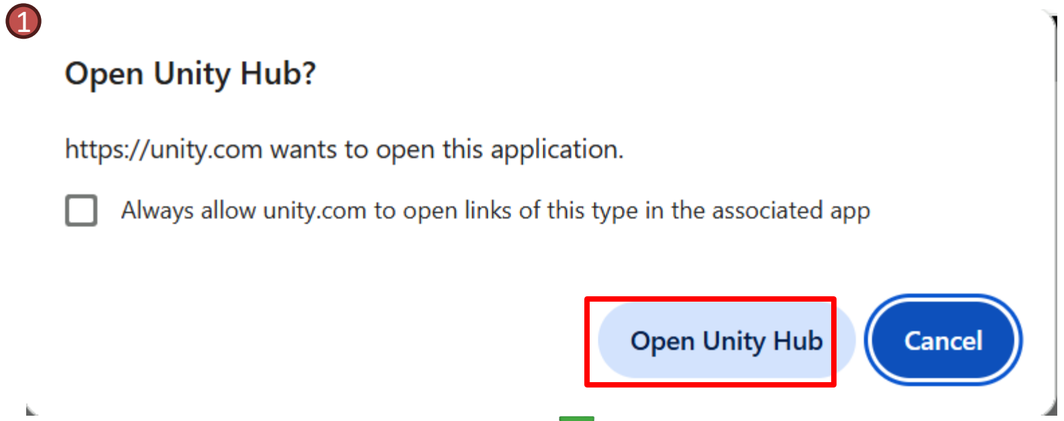
❑ Install Version “6000.3.13f1



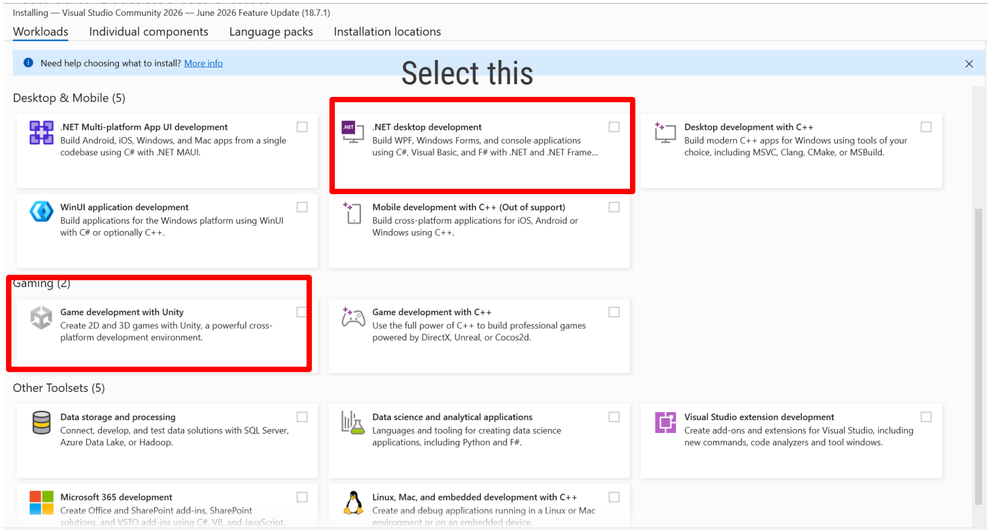
Unity Installation

❑ Follow the processes to install Unity Version “6000.3.13f1

1



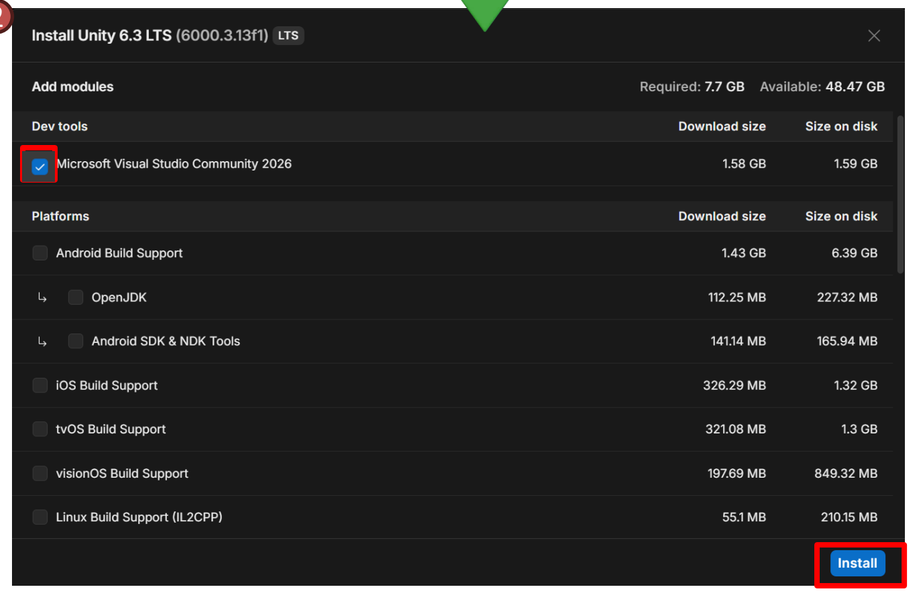
Select this



Select this

2

check this



3

